STOCHASTIC CALCULUS

FIRST PART (FQ307, OMI302) and SECOND PART (FQ308) of a global Course in Stochastic Calculus (OMI302b)

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Abstract. The course includes the following chapters of stochastic calculus.

(i) Motivations: stochastic modeling, probabilistic representations of linear PDEs, stochastic control, filtering, mathematical finance.

(ii) Stochastic processes in continuous time: Gaussian processes, Brownian motion, (local) martingales, semimartingales, Itô processes.

(iii) Stochastic integrals: forward and Itô integrals.

(iv) Itô and chain rule formulae, a first approach to stochastic differential equations.

(v) Girsanov formulae. Novikov and Benês coondition. Predictable representation of Brownian martingales.

(vi) Stochastic differential equations with Lipschitz coefficients. Markov flows.

(vii) Stochastic differential equations without Lipschitz coefficients: strong existence, pathwise uniqueness, existence and uniqueness in law. Engelbert-Schmidt criterion. Non-explosion conditions

(viii) Bessel processes and Cox-Ingersoll-Ross model in mathematical finance.

(ix) Backward stochastic differential equations and connections with semilinear PDEs.

1 Introduction

The present document constitutes a second-level course of *stochastic calculus*. We would like to illustrate the richness of stochastic calculus in itself and at the level of applications. It is often related to a microscopic way to observe the world but it also helps to show important links to the macroscopic vision. A preliminary list of complementary monographs is the following: [18, 23, 31, 27, 30].

The range of applications goes from mathematical biology (neurosciences, population growth, oncology, epidemiology), engineering sciences, physics, economics and finance. More recently, applications to very recent fields have appeared, including *big data*.

We start with some possible problems.

(A) Stochastic analogue of ordinary differential equations

Allowing randomness in some coefficients of an ordinary differential equation (ODE), produces a more realistic mathematical model of the situation.

PROBLEM 1

Consider a simple model of population growth

$$\frac{dN}{dt}(t) = a(t)\, N(t), \quad N(0) = A, \tag{1.1}$$

where $N(t)$ is the size of the population at time t, $a(t)$ is the growth rate at time t. It might happen that $a(t)$ is not completely known, but subject to some random environmental effects, so that we have

$$a(t) = r(t) + \text{“noise”}, \tag{1.2}$$

where we do not know the exact behaviour of the noise term, only its probability distribution. The function $r(t)$ is assumed to be nonrandom. How do we solve (1.1) in this case?

(B) Stochastic approach to a deterministic problem with boundary or initial condition

PROBLEM 2 The most celebrated example is the stochastic solution of the **Dirichlet problem**.
Let U be a (reasonable) domain in $\mathbb{R}^n$ and a continuous function $f : \partial U \to \mathbb{R}$. Find a continuous function $\tilde{f} : \bar{U} \to \mathbb{R}$ such that

i) $\tilde{f} = f$ on ∂U,

ii) $\tilde{f}$ is harmonic on U i.e.

$$\Delta \tilde{f} = \sum_{i=1}^{n} \frac{\partial^2 \tilde{f}}{\partial x_i^2} = 0 \text{ sur } U.$$

In 1944 Kakutani proved that the solution could be expressed in terms of **Brownian motion** (which will be introduced later): $\tilde{f}(x) = E(f(B_\tau^x))$ where $(B_t^x, t \geq 0)$ is a Brownian motion starting from x and τ is the first exit point from the domain.

It turned out that this was just the top of an iceberg. The Dirichlet problem could be solved for a large class of elliptic (hypoelliptic) second order PDE operator L more general than the laplacian. In that case the Brownian motion can be replaced by the solution (X_t) of an associated stochastic differential equation.

PROBLEM 3

We consider the problem

$$\begin{cases} \frac{\partial u}{\partial t}(t,x) - \frac{1}{2}\Delta u(t,x) = 0 \\ u(0,x) = f(x), \end{cases}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is continuous non-negative or with polynomial growth. It is possible to show that

$$u(t,x) = E(f(B_t^x)),$$

is a solution of the problem.

(C) Optimal stopping

PROBLEM 4

Suppose that a person has an asset or resource (e.g. a house, stocks, oil...) that she is planning to sell. The price X_t at time t of her asset on the open market varies according to a stochastic differential equation of the same type as in PROBLEM 1, i.e.

$$\frac{d}{dt}X_t = X_t(r + \alpha \text{ “noise”}),$$

where r, α are known constants. The discount rate is a known constant ρ. At what time should she decide to sell?

We assume that she knows the behaviour of (X_s) up to the present time t, but because of the noise in the system, she can of course never be sure at the time of the sale if her choice of time will turn out to be the best. So what we are searching for a *stopping strategy* that gives the best result in the long run, i.e. maximizes the *expected* profit when the inflation is taken into account.

This is an *optimal stopping problem*. It turns out that the solution can be expressed in terms of the solution of a corresponding boundary value problem (PROBLEM 2), except that the boundary is unknown (free) as well.

(D) Filtering problem.

PROBLEM 5

Suppose that, in order to improve our knowledge about the solution of a *random differential equation*, say of (1.1) in PROBLEM 1, we perform observations $Z(s)$ of $N(s)$ at times $s \leq t$. However, due to inaccuracies in our measurements we do not really measure $N(s)$ but a disturbed version of it, i.e.

$$Z(s) = N(s) + \text{“noise”}, \tag{1.3}$$

So in this case, there are two sources of noise, the second coming from the error of measurement.

The *filtering problem* is the following. What is the best estimate of $N(s)$ satisfying (1.1), based on the observations (1.3) when $s \leq t$?
Intuitively, the problem is to *filter* the noise away from the observations in an optimal way. In 1960 Kalman and in 1961 Kalman and Bucy proved what is now known as the Kalman-Bucy filter. Basically the filter gives a procedure for estimating the state of a system which satisfies a "noisy" linear differential equation, based on a series of "noisy" observations. Almost immediately the discovery found applications in aerospace engineering (Ranger, Mariner, Apollo etc.) and it now has a broad range of applications but it has known a wide range of other applications.

(E) Black-Scholes-Samuelson model (Financial mathematics)

It is a *continuous time model* to describe the price of two financial assets: one is risky (say a stock with price S_t at time t) and a non-risky asset (with price S_t^0 at time t). We suppose that the evolution of S_t^0 is described by the ordinary differential equation

$$dS_t^0 = rS_t^0 dt, \tag{1.4}$$

where r is a positive constant.

Remark 1.1 *This equation is the continuous time version of the difference equations*

$$S_{n+1}^0 = S_n^0(1+r). \tag{1.5}$$

In fact (1.5) gives

$$\Delta S_n^0 = rS_n^0 \Delta t, \tag{1.6}$$

where

$$\Delta t = 1, \quad \Delta S_n^0 = S_{n+1}^0 - S_n^0.$$

Now (1.6) motivates (1.4).

In (1.4), the interest rate related to the non-risky asset is constant and equal to r.

We set

$$S_0^0 = 1,$$

such that

$$S_t^0 = e^{rt}, \quad \forall\, t \geq 0.$$

We suppose that the evolution of the risky asset price (stock) is described by SDEs of the type (1.1) and (1.2)

$$dS_t = S_t(\mu + \text{"noise"})dt.$$

At time t, we set π_t^0 (resp. π_t) the quantity of non-risky (resp. risky) detained in the portfolio.

Suppose that at time $t = 0$ the investor is offered the right (but without obligation) to buy one unity of risky asset at a specified price K at a specified future time $t = T$. Such a right is called a **European call option**. How much should the person be willing to pay for such an option? This problem was solved when Fisher Black and Myron Scholes (1973) used stochastic analysis to compute a theoretical value for the price, the now famous Black-Scholes opion price formula. They also solved the so called *hedging problem* i.e. the investment that the seller of the option has to perform in order to reproduce the sum to provide to the owner of the option in case he exercise it. In particular he has to evaluate the quantities π_t^0 and π_t to invest at each time t in order to be able to reimburse the owner of the option at the maturity T.

PROBLEME 6 Stochastic control

It constitutes a maximization (resp. minimization) problem in expectation. We consider again the previous (Samuelson) model and let π_t^0 (resp. π_t) denote the quantity of non-risky (resp. risky) asset that an investor owns in his portfolio.

We express by $u_t = \frac{\pi_t}{X_t}$ the wealth portion invested in the stock (risky asset). The total wealth X_t at time t is given by

$$X_t = S_t^0 \pi_t^0 + S_t \pi_t.$$

At each time t, the investor can decide the portion u_t; he will put $(1-u_t)X_t$ in the non-risky asset. Given a utility function U and a maturity T, the problem consists in determining the *optimal portfolio*, i.e. to find a good allocation investment, $u_t, \quad 0 \leq t \leq T$, which maximizes at the maturity time the expected utility of the wealth $X_t^{(u)}$; one tries to maximize on $0 \leq u_t \leq 1$ the expected utility

$$E\left\{U(X_T^{(u)})\right\}.$$

2 Basic recalls in probability theory

2.1 Preliminaries

In general, we will be given an underlying probability space $(\Omega, \mathcal{F}, P)$.

Let $(E, \mathcal{E})$ be a measurable set, i.e. a set E equipped with a σ-field $\mathcal{E}$. In most cases E will be of the type

$$E = \mathbb{R}^d, \mathbb{C}, \overline{\mathbb{R}}, \tag{2.1}$$

but it can also be an infinite dimensional Banach space E of real continuous functions on $[0, T]$. $\mathcal{E}$ will be then the Borel σ-field $\mathcal{B}(E)$. If E_1, E_2 are Banach spaces then $f : E_1 \rightarrow E_2$ is called **Borel** if it is measurable with respect to the corresponding Borel σ-fields. If X takes values in E, $\quad (X : \Omega \rightarrow E)$, we denominate by **law of** X (or induced probability measure through X) on $(E, \mathcal{E})$ the probability $\nu = \nu_X : \mathcal{B}(E) \rightarrow [0, 1]$ defined by

$$\nu(B) = P\{X \in B\} = P(X^{-1}(B)).$$

All the (σ-fields) included in $\mathcal{F}$ will be supposed to be completed with the P-null sets. In particular, if X is a $\mathcal{G}$-measurable r.v. and $X = Y$ a.s. then Y is also $\mathcal{G}$-mesurable.

By definition, a property Π will be said to hold ν-a.e. on E if it holds for every $x \in E$ except on a ν -negligible set. When ν is Lebesgue measure dx, we omit the mention ν and we will simply say a.e.
A property Π will be said to hold a.s. it holds P-a.e. on Ω.

If $\underline{X} = (X_1, \ldots, X_d)$ is a r.v. taking values in $\mathbb{R}^d$ (or random vector), we call **characteristic function** of $\underline{X}$ the function $\varphi_{\underline{X}} : \mathbb{R}^d \to \mathbb{C}$ defined by

$$t \longmapsto E\{\exp(it \cdot \underline{X})\}, \quad t \in \mathbb{R}^d.$$

Remark 2.1 *The characteristic function uniquely determines the law of* $\underline{X}$.

Let $\underline{X}$ be a square integrable random vector, i.e. such that $E(|\underline{X}|^2) < \infty$.

The **covariance matrix** of $\underline{X}$ is the matrix $\Gamma(\underline{X}) = (\sigma_{ij})_{1 \le i,j \le d}$ whose coefficients are given by

$$\sigma_{i,j} = \mathrm{Cov}(X_i, X_j) = E\{(X_i - \mu_i)\,(X_j - \mu_j))\},$$

where $\underline{\mu} = (\mu_1, ..., \mu_d)$ is the expectation of $\underline{X}$, i.e. $\mu_i = E(X_i), 1 \le i \le d$.

Remark 2.2 $X_1, \cdots, X_d$ *are independent if and only if*

$$\varphi_{\underline{X}}(t) = \Pi_{j=1}^d \varphi_{X_j}(t_j), \quad t = (t_1, \cdots, t_d).$$

Remark 2.3 *If the r.v.* $X_1, \ldots, X_d$ *are independent then* $\Gamma(X)$ *is diagonal. The opposite implication is generally wrong.*

2.2 Real Gaussian random variables

Let $\mu \in \mathbb{R}$, $\sigma > 0$. A real r.v. X (i.e. taking values in $E = \mathbb{R}$) is called **Gaussian r.v.** or **normal r.v.** with parameter μ and σ^2, notation $X \sim N(\mu, \sigma^2)$, its law admits a density given by

$$f_{\mu,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The law of X is equally called **Gaussian** or **normal law** with parameters μ and σ^2. In particular we also have

$$E(X) = \mu, Var(X) = \sigma^2.$$

If $\mu = 0$, $\sigma = 1$, X is called **standard**. Similarly the law of X is also called standard Gaussian law.

If $\sigma = 0$, we say that $X \sim N(\mu, \sigma^2)$ if the law of X is the (discrete) Dirac law δ_μ, i.e.

$$\delta_\mu(A) = \begin{cases} 1 & \text{if } \mu \in A \\ 0 & \text{otherwise.} \end{cases}$$

In that case $X \equiv \mu$ a.s.

Proposition 2.4 *Suppose $X \sim N(\mu, \sigma^2)$. Then*

$$E(e^{zX}) = e^{\mu z + \frac{\sigma^2 z^2}{2}}, \qquad \forall\, z \in \mathbb{C}. \tag{2.2}$$

Remark 2.5 *(i) If $Y \sim N(0, 1)$, we have*

$$E(e^{zY}) = e^{\frac{z^2}{2}}, \qquad \forall\, z \in \mathbb{C}. \tag{2.3}$$

(ii) The characteristic function of $X \sim N(\mu, \sigma^2)$ is given by

$$\varphi_X(t) = e^{i\mu t - \frac{\sigma^2 t^2}{2}}, \qquad \forall\, t \in \mathbb{R}.$$

2.3 Gaussian vectors

Definition 2.6 *A r.v. $\underline{X} = (X_1, \dots, X_d)$ taking values in $\mathbb{R}^d$ is called* **Gaussian vector** *if for every reals $a_1, \dots, a_d$, the real-valued r.v. $\sum_{i=1}^{d} a_i X_i$ is a Gaussian random variable.*

We recall that the components $X_1, \dots, X_d$ of a Gaussian vector are Gaussian random variables. However, if the coordinates of a random vector $\underline{X}$ are Gaussian, this is not enough to ensure that $\underline{X}$ is a Gaussian vector. When the r.v. are independent then the situation is different. We recall the proposition below without proof.

Proposition 2.7 *Let $X_1, \dots, X_d$ be* **independent** *Gaussian r.v. Then the vector $\underline{X} = (X_1, \dots, X_d)$ is Gaussian.*

Theorem 2.8 *Let $\underline{X} = (X_1, \dots, X_d)$ an $\mathbb{R}^d$-valued Gaussian vector. The r.v. $X_1, \dots, X_d$ are independent if and only if the matrix $\Gamma(\underline{X})$ is diagonal.*

Proof. Exercise. Indication: use Remark 2.2. ■

Exercise 2.9 *Let $\underline{X}$ be a square integrable random vector. Prove that the covariance matrix Γ is a positive definite symmetric matrix.*

We need to characterize the notion of multi-dimensional Gaussian law. Let $\underline{\mu} = (\mu_1, ..., \mu_d)$ in $\mathbb{R}^d$ and Γ a positive-definite real matrix. A probability ν on $\mathcal{B}(\mathbb{R}^d)$ is called **Gaussian** or **normal** with parameter $\underline{\mu}$ and Γ if a r.v. X distributed according to ν verifies

$$\varphi_X(t) = \exp\left(it \cdot \underline{\mu} - \frac{t\Gamma t^\top}{2}\right).$$

We will use the notation $X \sim N(\underline{\mu}, \Gamma)$.

This leads naturally to the following proposition whose proof constitutes again an exercise, at least when Γ is diagonal.

Proposition 2.10 *Let $\underline{X}$ be a Gaussian vector such that $E(\underline{X}) = \underline{\mu}$ and covariance matrix Γ. With the first statement we mean $E(X_i) = \mu_i$ with $1 \leq i \leq d$.*

- *Then $\underline{X} \sim N(\underline{\mu}, \Gamma)$.*
- *If Γ is invertible then the Gaussian law is absolutely continuous with density*

$$p(x) = \frac{1}{(2\pi)^{\frac{d}{2}} \sqrt{\det \Gamma}} \exp\{-\frac{1}{2}(x - \underline{\mu})'\Gamma^{-1}(x - \underline{\mu})\}.$$

2.4 Complements on conditional expectation

Here we only recall some well-known results that we have non necessarily in mind.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{G}$ a sub σ-field of $\mathcal{F}$.

Remark 2.11 *If X is an integrable r.v., independent of $\mathcal{G}$, then $E(X|\mathcal{G}) = E(X)$ a.s.*

This property has no converse implication. However the interesting result below holds.

Proposition 2.12 *Let X be a given r.v. X is independent on a σ-field $\mathcal{G}$ if and only if*

$$\forall\, u \in \mathbb{R} : \quad E(e^{iuX}|\mathcal{G}) = E(e^{iuX}) \quad a.s. \tag{2.4}$$

A useful result in the sequel is the following.

Theorem 2.13 *Let $(E_1, \mathcal{E}_1)$, $(E_2, \mathcal{E}_2)$ two sets equipped with a σ-field. Let X be a $\mathcal{G}$-measurable r.v. taking values in $(E_1, \mathcal{E}_1)$ and Y be a r.v. with values in $(E_2, \mathcal{E}_2)$. We suppose that Y is independent of $\mathcal{G}$.*

Let $\Phi : (E_1 \times E_2, \mathcal{E}_1 \otimes \mathcal{E}_2) \longrightarrow \mathbb{C}$ measurable.

We define $\varphi : E_1 \to \mathbb{C}$ by

$$\varphi(x) = E(\Phi(x, Y)), x \in E_1.$$

Then

$$E(\Phi(X, Y)|\mathcal{G}) = \varphi(X) \quad a.s.$$

provided that the left-hand side exists.

Remark 2.14 *i) Under the stated hypotheses, we can calculate $E(\Phi(X, Y)|\mathcal{G})$ as if X were constant.*

ii) If $\Phi(x, y) = f(x)g(y)$ the result follows by the fondamental properties of conditional expectation.

If $(\xi_\alpha)_{\alpha \in I}$ is a family of integrable r.v. then $E(X|\xi_\alpha, \alpha \in I)$ will be by definition $E(X|\sigma(\xi_\alpha, \alpha \in I))$. Let $(\xi_1, \ldots, \xi_{n+1})$ be a square integrable r.v. In general $E(\xi_{n+1}|\xi_1, \ldots, \xi_n)$ is the orthogonal projection of ξ_{n+1} on $\mathcal{L}^2(\Omega, \sigma(\xi_1, \ldots, \xi_n), P)$.

Let $\hat{\xi}_{n+1}$ denote the orthogonal projection of ξ_{n+1} on the finite-dimensional subspace generated by $1, \xi_1, \ldots, \xi_n$ i.e. the linear regression of ξ_{n+1} on $\xi_1, \ldots, \xi_n$. In general if $\eta_{n+1} = E(\xi_{n+1}|\xi_1, \ldots, \xi_n)$, then

$$E(\eta_{n+1} - \xi_{n+1})^2 \leq E(\hat{\xi}_{n+1} - \xi_{n+1})^2.$$

Proposition 2.15 *Suppose that the covariance matrix $\Gamma(n)$ of $(\xi_1, \ldots, \xi_n)$ is invertible. Then the linear regression $\hat{\xi}_{n+1}$ of ξ_{n+1} on $\xi_1, ..., \xi_n$ is given by*

$$\mu_{n+1} + \sum_{j=1}^{n} (\xi_j - \mu_j)\, b_j,$$

where $\mu_\ell = E(\xi_\ell)$ and $1 \leq \ell \leq n+1$. The coefficients (b_j) are solutions of the system

$$\begin{pmatrix} \sigma_{1,n+1} \\ \vdots \\ \sigma_{n,n+1} \end{pmatrix} = \Gamma(n) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix},$$

$\sigma_{ij} = \mathrm{Cov}(\xi_i, \xi_j)$.

Remark 2.16 *When $(\xi_1, \ldots, \xi_{n+1})$ is* **Gaussian**, *then the linear regression $\hat{\xi}_{n+1}$ coincides with $E(\xi_{n+1} | \xi_1, \ldots, \xi_n)$.*

3 Stochastic processes, Brownian motion

We introduce now basic material about the topic, which will be widely used in the sequel. For more specific properties, the reader can however consult for instance [18, 31].

3.1 Preliminaries in analysis

We go on with some reminders about measure theory on the real line, see [19] (Chapter 9) and [32] (Chapter 8) for more details. See also [26].

Definition 3.1 *(i) A* **subdivision** *of an interval $[a, b]$ is a finite sequence $\mathcal{T} := (t_i)_{0 \leq i \leq n}$ of ordered real numbers in $[a, b]$, i.e. $a = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n = b$. The mesh of this subdivision is $|\mathcal{T}| := \sup_{1 \leq i \leq n-1}(t_{i+1} - t_i)$.*

(ii) If f is a function defined on an interval of $\mathbb{R}, f(x_+)$ (resp. $f(x_-)$) denotes the right-limit (resp. left-limit) of f at point x. We set $\Delta f(x) = f(x_+) - f(x_-)$.

f is said **right-continuous** *or* **càd** *(resp.* **to have a left-limit** *or* **làg***) if for any $x, f(x_+) = f(x)$ (resp. $f(x_-)$ exists).*

f is said **left-continuous** *or* **càg** *(resp.* **to have a right-limit** *or* **làd***) if for any $x, f(x_-) = f(x)$ (resp. $f(x_+)$ exists).*

Let $f : [a, b] \to \mathbb{R}$ be a real valued function. The **total variation** $\|f\|_{[a,b]}$ of f over $[a, b]$ is the supremum of

$$\sum_{i=0}^{n} |f(t_{i+1}) - f(t_i)|,$$

over all the subdivisions $(t_i)_{i=0}^{N}$ of $[a, b]$.

A function $f : [0, T] \to \mathbb{R}$ is said to be of **bounded (or finite) variation** if the total variation $\|f\|_{[0,T]}$ is finite. We denote by BV the set of Borel real functions with bounded variation.

An important property of bounded variation functions is the following.

Proposition 3.2 *A function $f : [0, T] \to \mathbb{R}$ is of bounded variation if and only if f is a difference of two non-decreasing functions.*

Remark 3.3 *Let $f : [0, T] \to \mathbb{R}$ with finite variation.*

(i) Then f is bounded and the function $t \to \|f\|_{[0,t]}$ from $[0, T]$ to $\mathbb{R}_+$ is a non-decreasing function.

(ii) If $f : [0, T] \to \mathbb{R}$ is differentiable and $f' \in L^1([0, T])$, then $f \in BV$ and $\|f\|_{[0,t]} = \int_0^t |f'|(s)ds$.

(iii) f is almost everywhere differentiable.

(iv) If $f : [0, T] \to \mathbb{R}$ is monotone then f is in BV, and $\|f\|_{[a,b]} = |f(b) - f(a)|$.

$T > 0$ will be a fixed maturity in the whole course.

3.2 Generalities on continuous time processes

Let $(\Omega, \mathcal{F}, P)$ be a probability space and E be a set equipped with a σ-algebra $\mathcal{E}$.

We denominate **continuous time stochastic process** (or simpler **process**) taking values in $(E, \mathcal{E})$, a family $(X_t, t \geq 0)$ of r.v. defined on $(\Omega, \mathcal{F}, P)$ with values in $(E, \mathcal{E})$. The latter space will be omitted if self-explanatory.

Remark 3.4 *In practice the index t represents the time.*

a) *A process can be considered as an application $\mathbb{R}_+ \times \Omega \to E$. When that application is measurable with $\mathbb{R}_+ \times \Omega$ being equipped with the product σ-field $\mathcal{B}(\mathbb{R}_+) \otimes \mathcal{F}$ and E of the σ-field $\mathcal{E}$, then X is said to be a measurable process.*

b) *To each ω in Ω, we associate the function $\mathbb{R}_+ \to E$, $t \longmapsto X_t(\omega)$, called* **path** *of the process.*

c) *When $E = \mathbb{R}^d$, $\mathcal{E}$ will be the Borel σ-field.*

d) *We will also consider processes indexed by the time interval $[0, T]$. Most of the notions introduced for $\mathbb{R}_+$ can be expressed for $[0, T]$. We will maintain some ambiguity with respect to this.*

$\square$

We provide now some general definitions about processes.

Definition 3.5 *(i) A process $(X_t)_{t \geq 0}$ is said to have* **independent increments** *if for every $0 \leq t_0 < t_1 < \ldots < t_n < \infty$, the family $\{X_{t_1} - X_{t_0}, \ldots, X_{t_n} - X_{t_{n-1}}\}$ is independent.*

(ii) It is said to be **stationary** *if for every $0 \leq t_0 < t_1 < \ldots < t_n < \infty$, the law of $(X_{t_0+h}, \ldots, X_{t_n+h})$ does not depend on h.*

(iii) A process $(X_t)_{t \geq 0}$ is said to have **stationary increments** *if for every $0 \leq t_0 < t_1 < \ldots < t_n < \infty$, the law of $(X_{t_1+h} - X_{t_0+h}, \ldots, X_{t_n+h} - X_{t_{n-1}+h})$ does not depend on h.*

(iv) $(X_t)_{t\geq 0}$ *is said to be* **square integrable** *if* $\mathbb{E}(|X_t|^2) < \infty, \forall t \geq 0$.

(v) *Suppose that* $E = \mathbb{R}$. (X_t) *is said to be* **Gaussian**, *if for every* $0 \leq t_0 < t_1 < \ldots < t_n < \infty, (X_{t_0}, \ldots, X_{t_n})$ *is a Gaussian vector.*

We state now some basic definitions related to the measurability of processes.

A process (Y_t) is called a **modification** of (X_t) if $X_t = Y_t$ a.s. for every positive t.
Two processes (X_t) and (Y_t) are said **indistinguishable** if there is a P-negligible set N such that

$$X_t(\omega) = Y_t(\omega), \quad \forall t \geq 0, \forall \omega \in N^c.$$

If (X_t) and (Y_t) are indistinguishable, then (X_t) is a modification of (Y_t).

The concept of process may be extended to the case when the parameter belongs to a subset of $\mathbb{R}^n$. A family of r.v. $(X_t, t \in A)$, where $A \subset \mathbb{R}^n$, is called **random field** (indexed by A). Similarly as for processes, we can speak about **indistinguishable random fields** and **modifications**.

When $E = \mathbb{R}^d$, a process $(X_t)_{t\geq 0}$ is said to be **continuous**, if a.s. all its paths are continuous.
More generally, if a.s. all the paths of a process X verify a given property Π, we will say that X has the property Π. For instance, X is **right-continuous** (or **càd**), **right-continuous with left limits** (or **càdlàg**), increasing, having bounded variation, if a.s. all the paths of X fulfill the corresponding property.

Proposition 3.6 *There is at most one left-continuous (resp. right-continuous) modification of a given process in the sense of indistinguishability. In other words, if* $(Y_t^1)_{t\geq 0}$ *and* $(Y_t^2)_{t\geq 0}$ *are two left-continuous (resp. right-continuous) modifications of* $(X_t)_{t\geq 0}$*, then they are indistinguishable.*

Proof.

It is enough to show that two left-continuous (resp. right-continuous) processes (X_t) and (Y_t), modification of each other are indistinguishable. We only consider the case when the paths are left-continuous, the other cases

being similar. Let N be a negligible set such that for $\omega \notin N$ the paths $t \mapsto X_t(\omega)$ and $t \mapsto Y_t(\omega)$ are left-continuous.
On the other hand, for each $n \in \mathbb{N}^*, k \in \mathbb{N}$, there is a null set $N_{k,n}$ such that

$$X_{\frac{k}{2^n}}(\omega) = Y_{\frac{k}{2^n}}(\omega), \quad \forall \omega \notin N_{k,n}.$$

We set $\tilde{N} = \bigcup_{k,n} N_{k,n} \cup N$ which is still a null set.
By left-continuity of paths, for each $\omega \notin \tilde{N}, t \geq 0$, we have

$$\begin{aligned} X_t(\omega) &= \lim_{n\to\infty} \sum_{k\geq 0} \mathbf{1}_{[\frac{k}{2^n}, \frac{k+1}{2^n}[}(t) X_{\frac{k}{2^n}}(\omega) \\ &= \lim_{n\to\infty} \sum_{k\geq 0} \mathbf{1}_{[\frac{k}{2^n}, \frac{k+1}{2^n}[}(t) Y_{\frac{k}{2^n}}(\omega) = Y_t(\omega). \end{aligned} \tag{3.1}$$

Finally, $(X_t)_{t\geq 0}$ and $(Y_t)_{t\geq 0}$ are indistinguishable. ■

A classical criterion to establish if a process is continuous is Garsia-Rudemich-Rumsey criterion, see for instance [3].

Proposition 3.7 *Let $(X_t, 0 \leq t \leq T)$ be a process such that*

$$E\big[|X_t - X_s|^\alpha\big] \leq C|t-s|^{1+\beta}, \quad 0 \leq s, t \leq T, \tag{3.2}$$

where $C, \alpha, \beta > 0$.

Then, there is a modification of X (still denoted by the same letter) such that for any $\gamma \in]0, \frac{\beta}{\alpha}[$, a.s. we have

$$\frac{1}{h^\gamma} \sup_{|t-s|\leq h} |X_t - X_s| \to 0, \quad (h \downarrow 0). \tag{3.3}$$

A simplified version of this result is the so-called Kolmogorov theorem.

Corollary 3.8 *Let $(X_t, 0 \leq t \leq T)$ be a process satisfying (3.2). Then it admits a modification which is Hölder continuous with parameter $\gamma < \frac{\beta}{\alpha}$. In particular for ω a.s. there is a constant $c(\omega)$ such that*

$$|X_t - X_s| \leq c(\omega)|t-s|^\gamma.$$

We now introduce the concept of filtration and related definitions.

Definition 3.9 *(i) A* **filtration** $(\mathcal{F}_t, t \geq 0)$ *on* $(\Omega, \mathcal{F}, P)$ *will be an increasing family of sub-σ-fields of* $\mathcal{F}$. *Namely* $\mathcal{F}_t$ *will be a σ-algebra included in* $\mathcal{F}$, *and* $\mathcal{F}_s \subset \mathcal{F}_t$, *for any* $0 \leq s \leq t$. *The σ-field* $\mathcal{F}_t$ *generally represents the state of the information available at time* t.

(ii) A filtration $(\mathcal{F}_t)$ *is supposed to satisfy* **the usual conditions**, *if* $\mathcal{F}_0$ *contains all the P-negligible sets in* $\mathcal{F}$ *and is* **right-continuous** *in the sense*

$$\mathcal{F}_{t+} := \bigcap_{s>t} \mathcal{F}_s = \mathcal{F}_t. \tag{3.4}$$

In the sequel all the filtrations will verify the usual conditions.

(iii) A process (X_t) *is said to be* **adapted** *to a filtration* $(\mathcal{F}_t, t \geq 0)$, *if for each* t, X_t *is* $\mathcal{F}_t$*-measurable.*

Remark 3.10 *In particular* $\mathcal{F}_t$ *contains the family* $\mathcal{N}$ *of the P-negligible sets of* $\mathcal{F}$. *The aim of this technical assumption is to allow the validity of the following property: if* ξ, η *are two random variables such that* $\xi = \eta$ *a.s., and* ξ *is* $\mathcal{F}_t$*-measurable then* η *is* $\mathcal{F}_t$*-measurable.*

(In fact, the set $N = \{\omega | \xi(\omega) \neq \eta(\omega)\}$ *is negligible. For any* $B \in \mathcal{B}(\mathbb{R})$,

$$\eta^{-1}(B) = (\eta^{-1}(B) \cap N^c) \cup \tilde{N} = (\xi^{-1}(B) \cap N^c) \cup \tilde{N} \in \mathcal{F}_t,$$

where $\tilde{N} = \eta^{-1}(B) \cap N$ *is negligible).*

Given a process $(X_t)_{t\geq 0}$, let $(\mathcal{F}_t)$ be the filtration $\mathcal{F}_t = \sigma(X_s, s \leq t)$. Obviously $(\mathcal{F}_t)$ is the smallest filtration for which $(X_t)_{t\geq 0}$ is adapted. A priori the above filtration does not verify the usual conditions.

Remark 3.11 *Given the (natural) filtration* $(\mathcal{F}_t)$ *it is no difficult to show the existence of a smallest filtration* $(\mathcal{F}_t^X)$ *fulfilling the usual conditions such that* $\mathcal{F}_t \subset \mathcal{F}_t^X$ *for every* $t \in [0, T]$. *In particular,* $\mathcal{F}_0^X$ *contains all the P-negligible subsets in* $\mathcal{F}$. *This filtration is called the* **canonical filtration** *of the process* $(X_t)_{t\geq 0}$. *When we speak about a filtration associated with a process* $(X_t)_{t\geq 0}$ *without other comments, we intend its canonical filtration.*

Remark 3.12 *(i) If $(X_t), (Y_t)$ are two modifications of each other, they will have the same canonical filtration.*

(ii) If (X_t) is $(\mathcal{F}_t)$-adapted , then every modification remains $(\mathcal{F}_t)$-adapted.

Sometimes it is not sufficient to only suppose that a process $(X_t)_{t\geq 0}$ is $(\mathcal{F}_t)$-adapted, we need a little bit more.

Definition 3.13 *(i) A process $(X_t)_{t\geq 0}$ is said to be* **progressively measurable** *(with respect to $(\mathcal{F}_t)$) if for any $t \geq 0$, the map $(\omega, s) \mapsto X_s(\omega)$, from $(\Omega \times [0,t]\ ;\ \mathcal{F}_t \otimes \mathcal{B}([0,t]))$ is measurable.*

(ii) Let $\mathcal{P}_{prog}$ be the σ-field generated by subsets $A \subset \Omega \times [0,T]$ in $\mathcal{F}$ such that 1_A is a progressively measurable process. It can be checked that a process is progressively measurable if and only if it is $\mathcal{P}_{prog}$-measurable.

Remark 3.14 *(i) It is obvious that any $(\mathcal{F}_t)$-progressively measurable process is $(\mathcal{F}_t)$-adapted.*

(ii) If (X_t) is $(\mathcal{F}_t)$-adapted, and càd (or càg), then (X_t) is progressively measurable, see [18] Proposition 1.13, chapter 1.

(iii) An $(\mathcal{F}_t)$-adapted process is nearly progressively measurable. More precisely, it admits a progressively measurable modification, [18] Proposition 1.12, chapter 1. This result will not be used in the sequel.

We introduce now the notion of random time and stopping time.
A r.v. τ with values in $[0,T] \cup \{\infty\}$ (or in $\mathbb{R}_+ \cup \{\infty\}$) if the process is indexed by the whole half line) is called **random time**. If $(X_t)_{t\geq 0}$ is a stochastic process, (X_t^τ) symbolizes the **stopped process** and is defined by

$$X_t^\tau = X_{t\wedge\tau}, t \geq 0.$$

A **stopping time** is a particular random time. We call **stopping time** with respect to a filtration $(\mathcal{F}_t)_{t\geq 0}$ a random time τ such that

$$\{\tau \leq t\} \in \mathcal{F}_t, \quad \forall t \geq 0.$$

Let τ be a stopping time. We introduce

$$\mathcal{F}_\tau = \{A \in \mathcal{F} | A \cap \{\tau \leq t\} \in \mathcal{F}_t;\ \forall t \geq 0\}.$$

$\mathcal{F}_\tau$ represents the available information up to τ.

Exercise 3.15 *(i) Prove that $\mathcal{F}_\tau$ is a σ-field. Prove that it coincides with $\mathcal{F}_t$ when $\tau = t$.*

Proposition 3.16 *Let $\mathcal{S}, \tau$ be two stopping times.*

a) τ is $\mathcal{F}_\tau$-measurable.

b) If $\mathcal{S} \leq \tau$ a.s., then $\mathcal{F}_\mathcal{S} \subset \mathcal{F}_\tau$.

c) $\mathcal{S} \wedge \tau = \inf(\mathcal{S} \wedge \tau)$ is a stopping time. In particular, if t is a deterministic time, then $\mathcal{S} \wedge t$ is again a stopping time.

Proof. Left to the reader. ■

3.3 Brownian motion

A particularly important example of stochastic process is **Brownian motion**. It constitutes the basis of a great quantity of stochastic models, in physics, engineering and finance.

Definition 3.17 *A real valued process $(B_t)_{t \geq 0}$ will be called* **Brownian motion (or Wiener process) with variance** σ^2 *, if*

(i) it has independent increments,

(ii) it is such that $B_t - B_s \sim N(0, \sigma^2(t-s))$, for any $t > s$.

Remark 3.18 *a) A Brownian motion admits a continuous modification, see Proposition 3.21 below. So in general (without restriction of generality) we will also suppose*
(iii) B is continuous.

b) Assumptions (i) and (ii) imply that that process has stationary increments.

Theorem 3.19 *Let (B_t) be a Brownian motion. If*

$$B_0 \equiv x,$$

for some $x \in \mathbb{R}$ then it is a Gaussian process.

Proof. It will be enough to show that for all $0 \leq t_0 < t_1 < \ldots < t_n, (B_{t_1} - B_{t_0}, \ldots, B_{t_n} - B_{t_{n-1}})$ is a Gaussian vector. This is a consequence of independence and of the Gaussian character of each component. ■

Definition 3.20 *A Brownian motion (Wiener process) (W_t) is called* **standard** *or* **classical** *if its variance is 1 and $W_0 = 0$ a.s.*

Proposition 3.21 *Let $W = (W_t)$ be a standard Brownian motion. Then its paths are Hölder continuous with parameter $\gamma < \frac{1}{2}$.*

Proof. Let n be a positive integer. If Z is a standard Gaussian r.v. we have

$$E(W_t - W_s)^{2n} = E(\sqrt{t-s}Z)^{2n} = c|t-s|^n,$$

where $c = E(|Z|^{2n})$. According to Corollary 3.8 the consequence follows taking $\alpha = 2n, 1 + \beta = n$, so that $\frac{\beta}{\alpha} = \frac{1}{2} - \frac{1}{2n}$. ■

In the sequel, when we will mention a Brownian motion, without other precision, this will be a classical Brownian motion.

In this case the law of W_t is $N(0,t)$, i.e. its density with respect to Lebesgue measure is

$$\frac{1}{\sqrt{2\pi t}} \exp\left(\frac{-x^2}{2t}\right).$$

We will also need the notion of Brownian motion related to a particular filtration $(\mathcal{F}_t)_{t\geq 0}$.

Definition 3.22 *Let $(\mathcal{F}_t)$ be a σ-field fulfilling the usual conditions. We will call $(\mathcal{F}_t)$-***Brownian motion** *(with variance σ^2), a (continuous) real process $(B_t)_{t\geq 0}$ which verifies the following properties.*

(i) $\forall t \geq 0, B_t$ *is* $\mathcal{F}_t$*-measurable;*

(ii) If $s < t, B_t - B_s$ *is independent of* $\mathcal{F}_s$*;*

(iii) If $s < t, B_t - B_s \sim N(0, \sigma^2(t-s))$.

Also, we will call **classical** $(\mathcal{F}_t)$**-Brownian motion** *an $(\mathcal{F}_t)$-Brownian motion (W_t) if $\sigma = 1$ and $W_0 = 0$ a.s.*

.

Remark 3.23 *Let $(B_t)_{t\geq 0}$ be an* $(\mathcal{F}_t)$**-Brownian motion** *Then, the filtration $\sigma(B_s, s \leq t) \vee \mathcal{N}$ where $\mathcal{N}$ is the family of P-null sets is the canonical filtration $(\mathcal{F}_t^B)$ of B, see Theorem 31, section 4, Chapter 1 of [30].*

Moreover we have the following.

a) $(B_t)_{t\geq 0}$ is $(\mathcal{F}_t)$-adapted. Point (i) shows that $\mathcal{F}_t^B \subset \mathcal{F}_t$

b) A Brownian motion (B_t) is an $(\mathcal{F}_t^B)$-Brownian motion.

c) An $(\mathcal{F}_t)$-Brownian motion is always an $(\mathcal{F}_t^B)$-Brownian motion.

3.4 White noise

We consider a simple model of population growth of the type

$$\frac{dN}{dt} = a(t)N(t), \quad N(0) = A, \tag{3.5}$$

where $N(t)$ is the population tail at time t, $a(t)$ is the increasing growth rate at time t. It can happen that $a(t)$ is not completely determined, because it is influenced by some environmental effects; this drives us to

$$a(t) = r(t) + \alpha\text{“noise”}, \tag{3.6}$$

where we do not know the exact behaviour of the noise, but only its probability distribution and its "mean" $r(t)$ is known.

A particular case of (3.5) and (3.6) can be

$$\frac{dN}{dt} = (r + \alpha\text{"noise"})N(t), \quad N(0) = A. \tag{3.7}$$

$N(t)$ can represent for instance the price of a resource, as oil, apartments, currencies, stocks and so on. α could represent the *volatility* of the resource. If $\alpha = 0$, one would have $X_t = X_0 e^{rt}$.

More generally, we will be interested in equations of the type

$$\frac{dX_t}{dt} = b(t, X_t) + a(t, X_t).\text{"noise"},$$

where b, a are given functions.

It seems reasonnable to look for a noise process (ξ_t) such that

$$\frac{dX_t}{dt} = b(t, X_t) + a(t, X_t), \xi_t, t \geq 0. \tag{3.8}$$

Starting from different situations arising for instance in engineering, one could suppose that $(\xi_t)_{t\geq 0}$ has, at least approximatively the following properties.

(i) the r.v. $(\xi_t)_{t\geq 0}$ are independent;

(ii) the process $(\xi_t)_{t\geq 0}$ is **stationary**, i.e. the law of $(\xi_{t_1+h}, \xi_{t_2+h}, \ldots, \xi_{t_n+h})$ does not depend on h for every $(t_1, \ldots, t_n)$;

(iii) $E(\xi_t) = 0, \quad \forall t \geq 0$.

Unfortunately, there is **no** reasonnable stochastic process satisfying (i) and (ii), see for instance, [27]. For instance, such a (ξ_t) cannot be continuous. If moreover one requests $E(\xi_t^2)$ to be finite, then function $(t, \omega) \mapsto \xi_t(\omega)$ will even not be measurable with respect to the σ-algebra $\mathcal{B}([0, T]) \otimes \mathcal{F}$ where $\mathcal{B}(I)$ is the Borel σ-algebra of I where $I \subset \mathbb{R}$.
Nevertheless, it is possible to represent $(\xi_t)_{t\geq 0}$ as a generalized process, see e.g. [15] (with the help of tempered Schwartz distributions $\mathcal{S}'$). Here we will avoid such difficulties, so formally we write $B_t = \int_0^t \xi_s ds$ ou $B_t(\omega) = \int_0^t \xi_s(\omega) ds$. Properties (i), (ii), (iii) can be translated into

(i') (B_t) has independent increments,

(ii') (B_t) has stationary increments.

On the other hand, logically we have

(iii') $B_0 = 0$ a.s.

(iv') (B_t) is continuous.

Theorem 3.24 *There are $\mu \in \mathbb{R}, \sigma \geq 0$ such that $(B_t - \mu t)_{t \geq 0}$ is a Brownian motion with variance σ^2.*
Moreover, if $\sigma > 0$, $W_t = \frac{1}{\sigma}(B_t - \mu t)$ is a classical Brownian motion.

Proof. See [14]. ■

Equation (3.8) can be formally rewritten as

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t a(s, X_s) dB_s. \tag{3.9}$$

Difficulty. It is possible to show that a.s., every path of the Brownian motion is nowhere differentiable, so that it is necessary to give a meaning to the "stochastic integral" with respect to dB_s appearing in (3.9). It is indeed also possible to show that the paths of B are a.s. never Hölder-continuous with parameter $\gamma \geq \frac{1}{2}$.

4 Martingales and semimartingales.

4.1 Continuous time martingales

The following definition is similar to the case of discrete time processes.
Let $(\mathcal{F}_t)_{t \geq 0}$ be a filtration on the probability space $(\Omega, \mathcal{F}, P)$. An adapted process (M_t) of integrable r.v., i.e. verifying $E(|M_t|) < \infty, \quad \forall t \geq 0$ is

- an $(\mathcal{F}_t)$-martingale if $E(M_t|\mathcal{F}_s) = M_s, \quad \forall t \geq s$;
- an $(\mathcal{F}_t)$-supermartingale if $E(M_t|\mathcal{F}_s) \leq M_s, \quad \forall t \geq s$;

- an $(\mathcal{F}_t)$-submartingale if $E(M_t|\mathcal{F}_s) \geq M_s, \quad \forall t \geq s$.

Remark 4.1 *From that definition, we can deduce that if $(M_t)_{t\geq 0}$ is a $\mathcal{F}_t$-martingale, then $E(M_t) = E(M_0), \quad \forall t \geq 0$. If $(M_t)_{t\geq 0}$ is an $\mathcal{F}_t$-supermartingale (resp. $\mathcal{F}_t$-submartingale) then $t \mapsto E(M_t)$ is decreasing (resp. increasing).*

Definition 4.2 *An $(\mathcal{F}_t)$-martingale (resp. $(\mathcal{F}_t)$-submartingale, $(\mathcal{F}_t)$-supermartingale), is said* **square integrable** *if $E(M_t^2) < \infty$, for any $t \geq 0$.*

When one speaks about a martingale, without σ-field specification, one refers to the canonical filtration.

Remark 4.3 *An $(\mathcal{F}_t)$-martingale (resp. supermartingale, submartingale) is also an $(\mathcal{F}_t^M)$-martingale (resp. supermartingale, submartingale).*

Exercise 4.4 *Let $(M_t; 0 \leq t \leq T)$ be a continuous $(\mathcal{F}_t)$-martingale such that $E[M_T^2] < \infty$. Prove the following.*

(i) For any $t \in [0,T], E[M_t^2] < \infty$.

(ii) The function $t \mapsto E[M_t^2]$ is non-decreasing.

(iii) For any s, t in $[0,T]$,

$$E[(M_t - M_s)^2|\mathcal{F}_s] = E[M_t^2|\mathcal{F}_s] - M_s^2, \quad \text{if} \quad 0 \leq s \leq t \leq T.$$

Exercise 4.5 *Let (M_t) be a submartingale, $\varphi : \mathbb{R} \to \mathbb{R}$ a convex function such that $\varphi(M_t)$ is integrable for every t. We suppose that one of the two following properties are realized.*

- *(M_t) is a martingale,*
- *φ is non-decreasing.*

Then $(\varphi(M_t))$ is a submartingale.

At this stage we can provide some examples coming from Brownian motion.

Proposition 4.6 *Let $(W_t)_{t\geq 0}$ be $(\mathcal{F}_t)$-classical Brownian motion. Then, the following processes are $(\mathcal{F}_t)$-martingales:*

$$\begin{array}{l} 1) W_t \\ 2) W_t^2 - t \\ 3) \exp\left(\sigma W_t - \frac{\sigma^2}{2} t\right). \end{array}$$

Proof. If $s \leq t$ then $W_t - W_s$ is independent from the $\sigma-$field $\mathcal{F}_s$. So $E(W_t - W_s|\mathcal{F}_s) = E(W_t - W_s)$; since a classical Brownian motion is mean zero, then $E(W_t - W_s) = 0$. Therefore (i) follows.
In order to prove (ii), we remark that

$$\begin{aligned} E(W_t^2 - W_s^2|\mathcal{F}_s) &= E\left((W_t - W_s)^2 + 2W_s(W_t - W_s)|\mathcal{F}_s\right) \\ &= E\left((W_t - W_s)^2|\mathcal{F}_s\right) + 2W_s \underbrace{E(W_t - W_s|\mathcal{F}_s)}_{0}; \end{aligned}$$

$E(W_t - W_s|\mathcal{F}_s) = 0$ since (W_t) is a martingale. Therefore

$$E(W_t^2 - W_s^2|\mathcal{F}_s) = E((W_t - W_s)^2|\mathcal{F}_s) = E(W_t - W_s)^2 = t - s,$$

because $W_t - W_s$ is independent of $\mathcal{F}_t$ and $W_t - W_s \sim N(0, t-s)$. We can deduce that $E(W_t^2 - t|\mathcal{F}_s) = W_s^2 - s$, when $s < t$.
We finally prove (iii). If $s < t$

$$E\left(e^{\sigma W_t - \frac{\sigma^2 t}{2}}|\mathcal{F}_s\right) = e^{\sigma W_s - \frac{\sigma^2 t}{2}} E\left(e^{\sigma(W_t - W_s)}|\mathcal{F}_s\right),$$

because W_s is $\mathcal{F}_s$-measurable. Since $W_t - W_s$ is independent of $\mathcal{F}_s$ we have

$$E\left(e^{\sigma(W_t - W_s)}|\mathcal{F}_s\right) = E\left(e^{\sigma(W_t - W_s)}\right) = E\left(e^{\sigma G\sqrt{t-s}}\right),$$

where $G \sim N(0,1)$. At this stage (2.3) implies that the previous quantity equals $e^{\frac{\sigma^2(t-s)}{2}}$. This yields the announced result. ∎

Previous processes provide in fact example of *continuous* martingales, since they are based on Brownian motion which is a continuous process.

The reader may ask if the definition of martingale guarantees a minimum of path regularity. This can be ensured since our "usual" filtration are essentially right-continuous, see [30], Corollary Chapter I, section 2.

Lemma 4.7 *Any $(\mathcal{F}_t)$-martingale admits a càdlàg modification.*

Exercise 4.8 *Let A be an integrable random variable and (M_t) a process such that $M_t = E(A|\mathcal{F}_t)$.*

(i) Show that M is an $(\mathcal{F}_t)$-martingale.

(ii) Conclude that M has a càdlàg modification.

If $(M_t)_{t\geq 0}$ is a martingale, the relation $E(M_t|\mathcal{F}_s) = M_s, \quad t > s$, can be generalized to random times, if those are *bounded stopping times.* Next theorem, namely **Doob's stopping time** theorem, will be stated without proof.

Theorem 4.9 *Let $(M_t)_{t\geq 0}$ be a càdlàg martingale with respect to a filtration $(\mathcal{F}_t)_{t\geq 0}$ and τ_1, τ_2 are two stopping times such that $\tau_1 \leq \tau_2 \leq K, \quad K$ being a real positive constant. Then M_{τ_2} is integrable and*

$$E(M_{\tau_2}|\mathcal{F}_{\tau_1}) = M_{\tau_1} \quad \text{Pa.s.} \tag{4.10}$$

Remark 4.10 *(i) Let τ be a bounded stopping time. Previous result implies that $E(M_\tau) = E(M_0)$. To observe this, it is enough to apply Doob's stopping time to $\tau_1 = 0, \tau_2 = \tau$ and to take the expectation of both members.*

(ii) If (M_t) is a submartingale, the same result holds replacing (4.10) with

$$E(M_{\tau_2}|\mathcal{F}_{\tau_1}) \geq M_{\tau_1} \quad \text{P} \quad \text{a.s.} \tag{4.11}$$

We will provide an application example of this result to the evaluation of the law of the hitting time of a barrier by classical Brownian motion.

Given a continuous process $(X_t)_{t\geq 0}$ adapted to a filtration $(\mathcal{F}_t)$. Let $a \in \mathbb{R}$. We consider the random time

$$T_a = \begin{cases} \inf\{s \geq 0 | X_s \geq a\} & \text{if} \quad \{s \geq 0 | X_s \geq a\} \neq \emptyset \\ \infty & \text{if} \quad \{s \geq 0 | X_s \geq a\} = \infty. \end{cases}$$

Proposition 4.11 *T_a is a stopping time with respect to $(\mathcal{F}_t)_{t\geq 0}$.*

Proof. We can replace (X_t) with an indistinguishable process whose paths are continuous, see Remark 3.18 a).

Remark 4.12 *The path continuity allows to write* $T_a(\omega) = \min\{s \geq 0 | X_s(\omega) \geq a\}$, *if* $\{s \geq 0 | X_s(\omega) \geq a\} \neq \emptyset$. *In particular* $X_{T_a}(\omega) = a$ *if* $\{s \geq 0 | X_s(\omega) \geq a\} \neq \emptyset$.

Since
$$\{T_a \leq t\} = \{\max_{s \leq t} X_s \geq a\},$$
it remains to show that $\max_{s \leq t} X_s$ is $(\mathcal{F}_t)$-measurable to conclude that T_a is a stopping time. This is true because it is the limit of

$$\max_{s = tk2^{-n}, k=0\ldots 2^n} X_s,$$

which is obviously $(\mathcal{F}_t)$-measurable. ∎

Proposition 4.13 *Let* $(W_t)_{t \geq 0}$ *be a* $(\mathcal{F}_t)$ *classical Brownian motion. If* $a \in \mathbb{R}$, *we set*

$$\tau_a = \begin{cases} \inf\{s \geq 0 | W_s = a\} & \text{if} \quad \{s \geq 0 | W_s = a\} \neq \emptyset \\ \infty & \text{if} \quad \{s \geq 0 | X_s = a\} = \emptyset. \end{cases}$$

Then, τ_a *is a.s. a finite stopping time and*

$$E\left(e^{-\lambda \tau_a}\right) = e^{-\sqrt{2\lambda}|a|}, \quad \lambda \geq 0. \tag{4.12}$$

Remark 4.14 *(4.12) provides the Laplace transform of* τ_a. *As the characteristic function, it uniquely determines the law.*
In fact, $\lambda \longrightarrow E\left(e^{-\lambda \tau_a}\right)$ *is analytical (being holomorphic) on* $\{\mathrm{Re}\lambda > 0\}$. *The same holds for the right member* $\lambda \mapsto e^{-\sqrt{2\lambda}|a|}$.
Equality (4.12) means that the two previous complex functions are identical on $\mathbb{R}_+$. *By analytical continuation, they also coincide on* $\{\mathrm{Re}\lambda > 0\}$. *By continuity, they are equal on* $\{\mathrm{Re}\lambda \geq 0\}$; *this determines the characteristic function which uniquely determines the law.*

Proof (of Proposition 4.13).
There is no restriction of generality to suppose that all paths of $(W_t)_{t \geq 0}$ are

continuous. The obtained indistinguishable process is still a $(\mathcal{F}_t)$ -Brownian motion.
We suppose $a \geq 0$. Since the paths are continuous, $\tau_a = T_a$, so that τ_a is again a stopping time.
We will apply Doob's stopping time theorem to the martingale

$$M_t = \exp\left(\sigma W_t - \frac{\sigma^2 t}{2}\right).$$

Unfortunately, we cannot apply directly that stopping time to τ_a, which is not a priori bounded. However, if n is a strictly positive integer, Proposition 3.16 says that $\tau_a \wedge n$ is still a stopping time, which is this time bounded. Applying Remark 4.10 (i), we obtain $E(M_{\tau_a \wedge n}) = 1$. Since $W_r(\omega) \leq a$, if $r \leq \tau_a(\omega)$,

$$M_{\tau_a \wedge n} = e^{\sigma W_{\tau_a \wedge n} - \frac{\sigma^2}{2}(\tau_a \wedge n)} \leq e^{\sigma a}.$$

Moreover, if $\tau_a < \infty$, $\quad \lim_{n\to\infty} M_{\tau_a \wedge n} = M_{\tau_a}$. If $\tau_a = +\infty$, for every $t \geq 0$, $\quad W_t \leq a$, so that

$$M_{\tau_a \wedge n} \leq \exp\left(\sigma a - \frac{\sigma^2 n}{2}\right) \longrightarrow_{n\to\infty} 0.$$

In fact, $W_{\tau_a \wedge n} \leq a$ and $\tau_a \wedge n = n$. Therefore we have $\lim_{n\to\infty} M_{\tau_a \wedge n} = \mathbf{1}_{\{\tau_a < \infty\}} M_{\tau_a}$.
By the dominated convergence theorem, we have

$$1 = \lim_{n\to\infty} \underbrace{E\left(M_{\tau_a \wedge n}\right)}_{1} = E\left(\mathbf{1}_{\{\tau_a<\infty\}} M_{\tau_a}\right).$$

Since $W_{\tau_a} = a$, if $\tau_a < \infty$,

$$\begin{aligned} E\left(e^{-\sigma^2 \frac{\tau_a}{2}} \mathbf{1}_{\{\tau_a<\infty\}}\right) &= E\left(M_{\tau_a} e^{-\sigma W_{\tau_a}} \mathbf{1}_{\{\tau_a<\infty\}}\right) = e^{-\sigma a} \underbrace{E\left(M_{\tau_a} \mathbf{1}_{\{\tau_a<\infty\}}\right)}_{1} \\ &= e^{-\sigma a}. \end{aligned}$$

Letting σ go to 0, we obtain

$$P\{\tau_a < \infty\} = 1.$$

This means that the classical Brownian motion reaches the barrier a with probability one. Then

$$E\left(e^{-\frac{\sigma^2 \tau_a}{2}}\right) = e^{-\sigma a}. \tag{4.13}$$

Let us suppose now that $-a > 0$. We remark that

$$\tau_a = \inf\{s \geq 0, -W_s = -a\}.$$

Now $(-W_t)_{t\geq 0}$ is an $(\mathcal{F}_t)$-Brownian motion and the result follows. ■

Remark 4.15 *Doob's stopping theorem cannot be applied to τ_a. In particular τ_a is not bounded. In fact, if it were the case, we would have $E\left(W_{\tau_a}\right) = 0$. Since $W_{\tau_a} = a$, we would obtain a contradiction.*

Another consequence of Doob's stopping theorem concerning stopping of a martingale.

The result below is stated without proof.

Proposition 4.16 *Let $(M_t)_{0\leq t\leq T}$ be a càdlàg $(\mathcal{F}_t)$-martingale (resp. submartingale, supermartingale). Then M^τ is still an $(\mathcal{F}_t)$-martingale (resp. submartingale, supermartingale).*

Remark 4.17 *It is immediate to see that $(M_t)_{0\leq t\leq T}$ is $(\mathcal{F}_{t\wedge\tau})$-martingale. The statement of the proposition is stronger.*

4.2 Local martingales, finite variation processes and uniform convergence in probability

We denote by $\mathcal{C}([0,T])$ the vector algebra of continuous processes indexed by $[0,T])$. $\mathcal{C}([0,T])$ is an F-space (metrizable, with homogeneous distance, complete) equipped with the topology of the **uniform convergence in probability (ucp)**. We recall that a sequence of processes X_n in $\mathcal{C}([0,T])$ converges ucp to a process X (necessarily in $\mathcal{C}([0,T])$ if

$$\sup_{t\leq T} |X_n(t) - X(t)| \xrightarrow{P} 0. \tag{4.14}$$

Remark 4.18 *$X_n \to X$ ucp if and only if $\rho(X_n, X) \to 0$ where ρ is the distance defined by*

$$\rho(X,Y) = \mathbb{E}\left(\frac{\sup\limits_{t\leq T}|X_t - Y_t|}{1 + \sup\limits_{t\leq T}|X_t - Y_t|}\right).$$

Remark 4.19 *As mentioned earlier, the space $\mathcal{C}([0,T])$ equipped with the distance ρ is a complete metric space.*

Indeed the space of random elements with values in the Banach space $E := C([0,T])$ (equipped with the sup-norm $\|\cdot\|$) is a complete metric space if equipped with the metric

$$\rho^C(X,Y) = \frac{\|X-Y\|}{1+\|X-Y\|},$$

describing the convergence in probability.

So if (X^n) is a Cauchy sequence in $\mathcal{C}([0,T])$, then by definition of ρ, (X^n) is Cauchy as sequence of random elements with values in E with respect to the metric ρ^C. Therefore there is a random element X with values in E such that X^n converges in probability to X.

Finally this shows that $\mathcal{C}([0,T])$ is complete.

We denote by $\mathcal{C}$ the vector algebra of continuous processes indexed by $\mathbb{R}_+$. We say again that a sequence (X_n) in $\mathcal{C}$ converges ucp to X if (4.14) holds for every $T > 0$. $\mathcal{C}$ is again an F-space. We denote by $\mathcal{C}_{\mathcal{F}}$ the subalgebra of $\mathcal{C}$ constituted by $(\mathcal{F}_t)$- adapted continuous processes.

We provide now some recalls concerning the theory of martingales.

Doob's stopping time theorem allows to obtain estimates for the maximum of a martingale. If (M_t) is a martingale, we can get a bound for the moments of $\sup_{0\le t\le T}|M_t|$. First, we state the so called **Doob inequality**.

Proposition 4.20 *Let $p>1$, $0\le s<T$. If $(M_t)_{s\le t\le T}$ is a right-continuous non-negative $(\mathcal{F}_t)$-submartingale, we have*

$$E\left(\sup_{s\le t\le T} M_t^p\right) \le \left(\frac{p}{p-1}\right)^p E\left(M_T^p\right).$$

Proof. See [18], Theorem 3.8, ch.1. ■

Corollary 4.21 *Let $p>1$, $0\le s<T$. Let $(M_t)_{s\le t\le T}$ be a right-continuous $(\mathcal{F}_t)$-martingale. We have*

$$E\left(\sup_{s\le t\le T} |M_t|^p\right) \le \left(\frac{p}{p-1}\right)^p E\left(|M_T|^p\right).$$

Proof. $|M_t|_{s\leq t\leq T}$ is a submartingale and $x \to |x|$ is a convex function. ∎

Example 4.22 *If $p = 2$, we have*

$$E\left(\sup_{0\leq t\leq T} |M_t|^2\right) \leq 4E\left(|M_T|^2\right).$$

Definition 4.23 *Let $I = \mathbb{R}_+$ or $[0, T]$. A process $(X_t)_{t\in I}$, is called $(\mathcal{F}_t)$-* **local martingale** *if there is an increasing sequence (τ_n) of stopping times such that*

- *X^{τ_n} is an $(\mathcal{F}_t)$-martingale;*
- *$\lim_{n\to\infty}\tau_n = +\infty$ a.s.*

Remark 4.24

- *An $(\mathcal{F}_t)$-martingale is an $(\mathcal{F}_t)$-local martingale.*
- *The set of $(\mathcal{F}_t)$-local martingales is a linear space.*
- *If M is a càdlàg $(\mathcal{F}_t)$-local martingale, τ is a stopping time, then M^τ is again an $(\mathcal{F}_t)$- local martingale.*
- *A similar notion can be defined for* **local submartingale** *and* **local supermartingale**.

Definition 4.25 *A process S is called (continuous) $(\mathcal{F}_t)$-***semimartingale** *if it is the sum of a (continuous) $(\mathcal{F}_t)$-local martingale and an $(\mathcal{F}_t)$-adapted (continuous) finite variation process.*

Example 4.26 *We give two very simple examples of semimartingales arising from standard Brownian motion W.*

- *W itself since it is a martingale.*
- *$S_t := \sup_{s\leq t} W_s$. Why?*

Remark 4.27 *When the filtration is not mentioned, again the canonical filtration will be underlying.*

Proposition 4.28 *(Stricker theorem).*

Let S be an $(\mathcal{F}_t)$- semimartingale and $(\mathcal{G}_t)$ be a subfiltration of $(\mathcal{F}_t)$ such that S is $(\mathcal{G}_t)$- adapted. Then S is also a $(\mathcal{G}_t)$- semimartingale.

Proof. See [30], Th. II 2.4 and [35]. ■

We may ask under which conditions a local martingale is a true martingale. Next lemma provides a partial answer.

Lemma 4.29 *Let $(M_t)_{t\in[0,T]}$ be an $(\mathcal{F}_t)$-càdlàg local martingale such that*

$$\sup_{t\in[0,T]} |M_t| \in L^1.$$

Then $(M_t)_{t\in[0,T]}$ is an $(\mathcal{F}_t)$-martingale.

Proof. Suppose that $M_T^\star = \sup_{t\in[0,T]} |M_t|$ is an integrable r.v. In particular $M_t \in L^1(\Omega)$ for all $t \in [0,T]$. Let $(\tau_n)_{n\geq 1}$ be a non-decreasing sequence of stopping times such that $\tau_n \uparrow +\infty$, and $(M_{t\wedge\tau_n}; 0 \leq t \leq T)$ is an $(\mathcal{F}_t)$-martingale.

Obviously, for any $t \in [0,T]$, $M_{t\wedge\tau_n}$ converges a.s. as $n \to \infty$ to M_t, and $\sup_{t\in[0,T]} |M_{t\wedge\tau_n}| \leq M_T^\star \in L^1(\Omega)$. Hence by Lebesgue dominated convergence theorem, for every $t \in [0,T]$, $M_{t\wedge\tau_n}$ converges to M_t, in $L^1(\Omega)$.

Let us check the martingale property. Let $0 \leq s \leq t \leq T$ and $\Lambda \in \mathcal{F}_s$. Since $(M_{t\wedge\tau_n}; 0 \leq t \leq T)$ is a martingale

$$E[M_{s\wedge\tau_n} 1_\Lambda] = E[M_{t\wedge\tau_n} 1_\Lambda].$$

By the dominated convergence theorem may be applied, n going to ∞, then $E[M_s 1_\Lambda] = E[M_t 1_\Lambda]$. This proves that (M_t) is a martingale. ■

A basic decomposition in stochastic analysis is the following. It will be a particular case among many theorems in the literature. We only state a particular case. It is a consequence of Theorem 4.10 and Remark 4.16 of chapter 1. of [18].

Theorem 4.30 *(***Doob-Meyer decomposition of a continuous submartingale***)*

Let X be a continuous $(\mathcal{F}_t)$-submartingale non-negative or such there is $\gamma > 1$ such that $E(|X_T|^\gamma) < \infty$. Then, there is a continuous $(\mathcal{F}_t)$- martingale M and an adapted, continuous, and increasing process V (such that $V_0 = 0$) with $X = M + V$. The decomposition is unique.

The *moment* condition above can be replaced by a *uniform integrability condition.*

Remark 4.31

- *In particular, an $(\mathcal{F}_t)$-submartingale is an $(\mathcal{F}_t)$-semimartingale.*
- *The uniqueness can also be a consequence of the uniqueness of the decomposition for Dirichlet processes.*

By localization arguments (that we will omit) it is possible to establish the following.

Proposition 4.32 *Let M be a continuous $(\mathcal{F}_t)$-local martingale. There is a continuous $(\mathcal{F}_t)$-local martingale and an increasing process V such that $M^2 = N + V$.*
The decomposition is unique if we set $V_0 = 0$.

Exercise 4.33

- *Prove the proposition if M is an $(\mathcal{F}_t)$-martingale.*
- *Provide the explicit Doob-Meyer decomposition for the submartingale (W_t^2) if W is a classical Wiener process.*

Definition 4.34 *Let M be a continuous $(\mathcal{F}_t)$-local martingale. We denote by $< M >$ the increasing process vanishing at zero; which appears in the Doob-Meyer decomposition of the local submartingale M^2. As a result, $M^2 - < M >$ is an $(\mathcal{F}_t)$-local martingale. It will be called* **oblique bracket** *or* **angle bracket***.*

Remark 4.35 *The angle bracket of a classical Brownian motion W equals $\langle W \rangle_t = t$.*

Remark 4.36 *Let $K = (K_s)$ be process such that $\int_0^T |K_s| ds < \infty$ a.s.*

(i) $V_t = \int_0^t K_s ds$ defines a finite variation process and its total variation process is given by

$$\|V\|_{[0,t]} = \int_0^t |K_s| ds.$$

(ii) Exercise: deduce that V is a semimartingale with respect to some filtration.

5 Covariation and Stochastic integrals via discretization

5.1 Definitions and fundamental properties

We introduce here the notion of covariation of two processes and associated integral close to a similar definition of Hans Föllmer, see [12].

Let $(X_t)_{t\in[0,T]}$ be a càdlàg processes. Let $(Y_t)_{t\geq 0}$ be a càdlàg or càglàd process.

Remark 5.1 *The process $Y_t^- := Y_{t-} := \lim_{s\uparrow t} Y_s$ is well-defined and it is càglàd.*
A càglàd process is locally bounded in the sense that there exists a sequence (τ_n) of stopping times such that Y^{τ_n} is bounded. This happens setting $\tau^n := \inf\{s \in [0,T] | |Y_s^{\tau_n}| > n\}$. See [30] Theorem 15, Chapter IV.

In the sequel, if we do not mention differently, processes $(X_t)_{t\in[0,T]}$ will be prolonged by continuity to the whole line, using the same notation. In particular, we have

$$X_t = \begin{cases} X_0 & \text{if} \quad t \leq 0 \\ X_T & \text{if} \quad t > T. \end{cases}$$

In the sequel we will often have to deal with processes depending on a parameter ε which converge to zero when $\varepsilon \to 0$. For this reason, the generic notation $(R(\varepsilon, s), s \in [0,T])$ indicate a family of processes such that

$$\sup_{s\leq T} |R(\varepsilon, s)| \to_{\epsilon \to 0} 0,$$

in probability. In the definition below we come back to the notion of *subdivision* introduced in Definition 3.1.

Definition 5.2 *(i) Let us consider a* **subdivision** *of* $[0,T]$*, which is, by definition, a finite subset of the type*

$$\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}. \tag{5.1}$$

For a sequence (Π^m) *of subdivisions of* $[0,T]$ *of the type* $\Pi = \{0 = t_0^m < t_1^m < \cdots < t_n^m = T\}$*, we say that its* **mesh converges to zero** *if* $|\Pi^m| := \sup_{0\le i\le n-1}(t_{i+1}^m - t_i^m)$ *converges to* 0 *when* $m \to \infty$*.*

(ii) We consider a subdivision Π *of the type* (5.1)*. For* $t \in [0,T]$*, we set*

$$\begin{aligned}
S_t^-(\Pi, Y, X) &= \sum_{i=0}^{n-1} Y_{t_i\wedge t}(X_{t_{i+1}\wedge t} - X_{t_i\wedge t})\\
S_t^+(\Pi, Y, X) &= \sum_{i=0}^{n-1} Y_{t_{i+1}\wedge t}(X_{t_{i+1}\wedge t} - X_{t_i\wedge t})\\
S_t^\circ(\Pi, Y, X) &= \sum_{i=0}^{n-1} \frac{Y_{t_{i+1}\wedge t} + Y_{t_i\wedge t}}{2}(X_{t_{i+1}\wedge t} - X_{t_i\wedge t})\\
&= \frac{S_t^-(\Pi, Y, X) + S_t^+(\Pi, Y, X)}{2}\\
C_t(\Pi, Y, X) &= \sum_{i=0}^{n-1}(X_{t_{i+1}\wedge t} - X_{t_i\wedge t})(Y_{t_{i+1}\wedge t} - Y_{t_i\wedge t}).
\end{aligned}$$

Provided that the corresponding limits exist in probability, and do not depend on the underlying sequence of subdvisions, we introduce the following definite integrals and covariations.

a) We denote

$$\int_0^t Y d^- X := \lim_{|\Pi|\to 0} S_t^-(\Pi, Y, X).$$

b) We set

$$\int_0^t Y d^\circ X := \lim_{|\Pi|\to 0} S_t^\circ(\Pi, Y, X).$$

We introduce now the **(indefinite) integrals** *as processes. We say that the* **forward (resp. symmetric) integral** *of* Y *with respect to* X *exists if, for*

any $t \in [0,T]$ $\int_0^t Y d^- X$ (resp. $\int_0^t Y d^\circ X$) exists and the process $(\int_0^t Y d^- X)_t$ (resp. $(\int_0^t Y d^\circ X)_t$) admits a càdlàg version. In that case that version will be chosen by default.
If $S_t^-(\Pi, Y, X)$ (resp. $S_t^\circ(\Pi, Y, X)$) converge ucp (without dependence from the chosen sequence of subdivisions), then we say that the forward (resp. symmetric) integral exists **in the ucp sense**.

If the forward integral of Y with respect to X exists, for $a < b$ in $[0,T]$ we also set $\int_a^b Y d^- X := \int_0^b Y d^- X - \int_0^a Y d^- X$. Similar considerations can be done for the symmetric integrals.

We define now the notions of covariation and quadratic variation.

c) (Covariation)
We set

$$[Y,X]_t := \lim_{|\Pi| \to 0} C(\Pi, Y, X)_t \tag{5.2}$$

if the limit holds in the ucp sense. $[Y,X]$ is called the covariation of Y and X. If $Y = X$, then $[X] := [X,X]$ is called quadratic variation of X.

d) α-variation) *Let $\alpha \geq 1$. We set*

$$[X]_t^\alpha = \lim_{|\Pi| \to 0} \sum_{i=0}^{n-1} |X_{s_{i+1} \wedge t} - X_{s_i \wedge t}|^\alpha.$$

In reality, previous objects, if they exist, depend on the chosen family of subdivisions. We will suppose moreover that they will be in fact independent.

Remark 5.3 *i) The covariation of continuous processes is a symmetric operation.*

ii) Let τ a random time. If $[X,Y]$ exists then

$$[Y^\tau, X^\tau]_t = [Y,X]_{t \wedge \tau}, \tag{5.3}$$

where X^τ is the process X stopped at time τ defined by $X_t^\tau = X_{t \wedge \tau}, t \geq$ 0.

iii) If the forward (resp. symmetric) integral exists in the ucp sense then the (indefinite) forward (resp. symmetric) integral exists.

iv) As covariations, also stochastic integrals, if they exist in the ucp sense, allow a stopping property:

$$\int_0^t Y^\tau d^\star X = \int_0^t Y d^\star X^\tau = \int_0^t Y^\tau d^\star X^\tau = \int_0^{t\wedge\tau} Y d^\star X,$$

where $\star$ denotes $-, \circ$.

v) $[X,X]$ coincides with the 2-variation $[X]^2$.

Remark 5.4 *When it exists, $[X]^\alpha$ is an increasing process. If H is measurable (pathwise bounded or positive) one can always define*

$$\int_0^\cdot H_s d[X]_s^\alpha.$$

In particular, if $\alpha = 2$, the integral $\int_0 H_s d[X,X]_s$ is meaningful.

Definition 5.5

- *If $[X,X]$ exists, then X will be said* **finite quadratic variation process**. *$[X,X]$ is called* **quadratic variation** *of X. We also use for covariation the terminology* **square bracket**.
- *If $[X,X] = 0$ (resp. $[X]^\alpha = 0$), X is called* **zero quadratic variation process** *(resp.* **zero α- variation process***).*

Proposition 5.6 *Suppose that X is a continuous process. If $[X]^\alpha$ exists then $[X]^\beta \equiv 0$ for any $\beta > \alpha$. In particular, if $[X,X]$ exists, then $[X]^\alpha = 0$ for every $\alpha > 2$.*

Proof. Let $\varepsilon > 0$. Let Π such that $|\Pi| < \varepsilon$. Setting, $s_i = t_i \wedge t, 1 \le i \le n$, we have

$$\begin{aligned}\sum_{i=0}^{n-1} |X_{s_{i+1}} - X_{s_i}|^\beta &\le \sum_{i=0}^{n-1} |X_{s_{i+1}} - X_{s_i}|^\alpha \sup_{i=0}^{n-1} |X_{t_{i+1}} - X_{t_i}|^{\beta-\alpha} \\ &\le \sup_{i=0}^{n-1} \delta(X,\varepsilon)^{\beta-\alpha} \sum_{i=0}^{n-1} |X_{s_{i+1}} - X_{s_i}|^\alpha,\end{aligned}$$

where $\delta(X,\varepsilon)$ is the modulus of continuity of X.

Since the paths of X are uniformly continuous, the previous expression converges a.s. (and therefore in probability) to zero. ∎

Definition 5.7 *A vector* $(X^1, \dots, X^n)$ *of càdlàg processes is said to have all its* **mutual brackets (covariations)** *if* $[X^i, X^j]$ *exist for every* $1 \le i, j \le n$.

Remark 5.8 *If* $(X^1, \dots, X^n)$ *has all its mutual covariations, then we have*

$$[X^i + X^j, X^i + X^j] = [X^i, X^i] + 2[X^i, X^j] + [X^j, X^j]; \tag{5.4}$$

from previous equality, it follows that $[X^i, X^j]$ *is the difference of increasing processes therefore it has bounded variation; so, that bracket is a classical integrator in the Lebesgue-Stieltjes sense.*

Remark 5.9 *a) Relation (5.4) holds as soon as three of the appearing brackets exist. More generally an identity of the type* $I_1 + \cdots + I_n = 0$ *has the following meaning: if* $n-1$ *terms among the* I_j *exist, the remaining one also makes sense and the identity holds true.*

b) We will see later, studying Dirichlet processes, that a covariation process $[X, Y]$ *will not always be a finite variation process; in particular* (X, Y) *may not have all its mutual brackets and* $[X, Y]$ *will however exist.*

The properties below can be established in a elementary way working out the definition of the integral via discretization.

Proposition 5.10 *Let* $X = (X_t)_{t \in [0,T]}$ *be a continuous process and* $Y = (Y_t)_{t \in [0,T]}$ *be a càdlàg (or càglàd) process.*

(i) We have $\displaystyle\int_0^\cdot Y d^\circ X = \int_0^\cdot Y d^- X + \frac{1}{2}[Y, X]$.

(ii) (**Integration by parts**).

We have

$$X_t Y_t = X_0 Y_0 + \int_0^t X d^- Y + \int_0^t Y d^- X + [X, Y]_t, t \in [0, T],$$

*(iii) (*__Kunita-Watanabe__*).*

If X *and* Y *are such that* (X, Y) *has all its mutual brackets, we have*

$$|[X, Y]| \le \{[X, X][Y, Y]\}^{\frac{1}{2}}.$$

(iv) If X is a finite quadratic variation process and Y is a zero quadratic variation process then (X, Y) has all its mutual brackets and $[X, Y] = 0$.

(v) Let X be a bounded variation continuous process. Then for any càdlàg (or càglàd) process Y we have

a) $\int_0^t Y d^- X = \int_0^t Y dX = \int_0^t Y d^\circ X$,

b) $[X, Y] = 0$. In particular a bounded variation continuous process is a zero quadratic variation process.

Proof (of Proposition 5.10).

(i) Let

$$\Pi = \{0 = t_0 < ... < t_n = T\}, \tag{5.5}$$

be an element of a sequence of subdivisions of $[0, T]$ whose mesh converges to zero. For fixed $t \in [0, T]$,

$$S_t^\circ(\Pi, Y, X) = \sum_i \frac{Y_{t_{i+1}\wedge t} + Y_{t_i \wedge t}}{2}(X_{t_{i+1}\wedge t} - X_{t_i\wedge t}) = S_t^-(\Pi, Y, X) + \frac{1}{2}C(\Pi, X, Y)_t.$$

Taking the limit when $|\Pi|$ goes to zero gives the result (i).

(ii) Let Π of the type (5.5). For $t \in [0, T]$, we write

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \sum_{i=0}^{n-1}(X_{t_{i+1}\wedge t}Y_{t_{i+1}\wedge t} - X_{t_i\wedge t}Y_{t_i\wedge t}) \\ &= S_t^-(\Pi, Y, X)_t + S_t^-(\Pi, X, Y) + C_t(\Pi, X, Y). \end{aligned}$$

Letting $|\Pi|$ go to zero, the integration by parts is then established.

(iii) If X is a finite quadratic variation process, Cauchy-Schwarz inequality

$$C_t(\Pi, Y, X) \leq \sqrt{C_t(\Pi, Y, Y) C_t(\Pi, X, X)}.$$

(iv) is a direct consequence of (iii).

(v) The first equality of a) follows through Lebesgue dominated convergence theorem. In fact $S^-(\Pi, Y, X)_t = \int_0^t Y^\Pi dX$, where

$$Y_s^\Pi = \sum_{i=0}^{n-1} Y_{t_i} 1_{[t_i, t_{i+1}[}(s).$$

Clearly for almost all ω

$$\sup_{t \le T} \left| \int_0^t (Y^\Pi - Y) dX \right| \le \int_0^T |Y_s^\Pi - Y_s| d\|X\|_{[0,s]}$$

converges to zero since, for each underlying ω, the quantity of jumps of Y is countable and the total variation measure of $d\|X\|$ of each singleton is zero.

Concerning b), we also use Lebesgue dominated convergence theorem. Now

$$C(\Pi, Y, X)_t = \int_0^t (Y^{+,\Pi} - Y^\Pi) dX + R(|\Pi|, t),$$

where

$$Y_s^{+,\Pi} = \sum_{i=1}^n Y_{t_{i+1}} 1_{]t_i, t_{i+1}]}(s).$$

By similar arguments as for a) $\sup_{t \le T} |C(\Pi, Y, X)_t|$ converges to zero a.s.

This together with the first equality of a) together with (i) gives the second equality of a).

■

We need at this point a Dini type lemma in the stochastics framework. The deterministic version of this lemma is well-known.

Lemma 5.11 *Let $(Z_\ell(t), t \in [0,T])$ be a sequence of continuous processes verifying the following properties.*

(i) $\forall \ell \in \mathbb{N}^*, t \to Z_\ell(t)$ is non-decreasing.

(ii) There is a continuous process $\big(Z(t)\big)_{0 \le t \le T}$ such that $Z_\ell(t) \to Z(t)$ in probability when ℓ goes to infinity. Then Z_ℓ converges to Z ucp.

Given a process X, the following corollary shows that the convergence in probability of expression (5.2) with $X = Y$, guarantees the existence of $[X, X]$.

Corollary 5.12 *Let $(X_t)_{0\le t\le T}$ a process. Suppose there is a continuous process (A_t)*

$$\lim_{|\Pi|\to 0} C(\Pi, X, X)_t$$

converges in probability to A_t for any $t \ge 0$. Then X is a finite quadratic variation process and $[X] = A$.

Proof (of the Lemma 5.11). Since the process Z is continuous and because of assumptions (i) and (ii), it is clear that Z non-decreasing.

Let $\rho, \alpha > 0, N \in \mathbb{N}^*$. We set $t_i^N = \dfrac{iT}{N}$, $0 \le i \le N$.

For almost all ω in Ω,

$$\sup_i |Z(t_{i+1}^N) - Z(t_i^N)|(\omega) \le \delta\Big(Z(\cdot,\omega)\,;\frac{T}{N}\Big),$$

where $\delta\Big(Z(\cdot,\omega)\,;\dfrac{T}{N}\Big)$ is the continuity modulus of $\big(Z(t,\omega), 0\le t\le T\big)$. The a.s. convergence of $\delta\Big(Z(.,\omega)\,;\dfrac{T}{N}\Big)$ to zero, when $N\to\infty$, implies the one in probability. Therefore, we choose N (fixed) so that

$$P\Big\{\delta\Big(Z(.)\,;\frac{T}{N}\Big) > \frac{\alpha}{4}\Big\} \le \rho.$$

We define

$$A = \Big\{\sup_{0\le t\le T} |Z_\ell(t)) - Z(t)| > \alpha\Big\}.$$

From now on, we will simply write $t_i = t_i^N, 0 \le i \le N$. Since

$$\sup_{0\le t\le T}|Z_\ell(t) - Z(t)| = \sup_{0\le i\le N-1}\Big(\sup_{t\in[t_i,t_{i+1}]}|Z_\ell(t) - Z(t)|\Big),$$

the event A is included in the union $\displaystyle\bigcup_{i=0}^{N-1} A_i$ where

$$A_i = \Big\{\sup_{t\in[t_i,t_{i+1}]}|Z_\ell(t) - Z(t)| > \alpha\Big\}\ ;\ 0\le i\le N-1.$$

Let $t\in[t_i, t_{i+1}]$. Since processes Z^ℓ and Z are increasing, we have

$$Z_\ell(t) - Z(t) \le Z_\ell(t_{i+1}) - Z(t_i).$$

We rearrange the right hand-side of the above inequality,

$$Z_\ell(t_{i+1}) - Z(t_i) = Z_\ell(t_{i+1}) - Z(t_{i+1}) + Z(t_{i+1}) - Z(t_i).$$

Then

$$Z_\ell(t) - Z(t) \leq Z_\ell(t_{i+1}) - Z(t_{i+1}) + \delta\Big(Z(.)\,;\frac{T}{N}\Big).$$

Similarly

$$Z_\ell(t) - Z(t) \geq Z_\ell(t_i) - Z(t_i) - \delta\Big(Z(.)\,;\frac{T}{N}\Big).$$

Therefore,

$$|Z_\ell(t)-Z(t)| \leq \delta\Big(Z(.)\,;\frac{T}{N}\Big)+|Z_\ell(t_{i+1})-Z(t_{i+1})|+|Z_\ell(t_i)-Z(t_i)| \ ;\ \forall t \in [t_i, t_{i+1}].$$

Consequently,

$$A_i \subset \Big\{\delta\Big(Z(.)\,;\frac{T}{N}\Big) > \frac{\alpha}{2}\Big\} \cup \widetilde{A}_i \cup \widetilde{A}_{i+1},$$

where

$$\widetilde{A}_i = \Big\{|Z_\ell(t_i) - Z(t_i)| > \frac{\alpha}{4}\Big\}.$$

Finally,

$$\begin{aligned} P(A) \quad &\leq \quad P\left(\delta\Big(Z(.)\,;\frac{1}{N}\Big) > \frac{\alpha}{4}\right) + P(\bigcup_{i=0}^{N} \widetilde{A}_i) \\ &\leq \quad P\left(\delta\Big(Z(.)\,;\frac{1}{N}\Big) > \frac{\alpha}{4}\right) + \sum_{i=0}^{N} P(\widetilde{A}_i). \end{aligned}$$

Since $Z_\ell(t_i)$ converges in probability to $Z(t_i)$, for any $0 \leq i \leq N-1$, ℓ going to ∞, yields $P(A) \leq \rho$. Since ρ is arbitrary then the result follows. ■

5.2 The quadratic variation of a (continuous) martingale and a semimartingale

Let $(M_t; t \geq 0)$ be a continuous local martingale. Then according to Proposition 4.32, $(M_t^2; t \geq 0)$, admits a Doob's type decomposition.

$$M_t^2 = N_t + A_t, \qquad t \geq 0, \tag{5.6}$$

where $(N_t; t \geq 0)$ is a continuous local martingale, $(A_t, t \geq 0)$ is a non-decreasing, adapted, continuous process and $A_0 = 0$.

The main result of this subsection is the following.

Theorem 5.13 *Let $(M_t; t \geq 0)$ be a continuous $(\mathcal{F}_t)$-local martingale. Then the process A coming from the Doob decomposition of M^2 coincides with $[M, M]$. In particular $(M_t^2 - [M, M]_t; t \geq 0)$ is a continuous $(\mathcal{F}_t)$-local martingale. If $(M_t; t \geq 0)$ is a continuous and square integrable martingale (see Definition 4.2) then $(M_t^2 - [M, M]_t; t \geq 0)$ is a continuous martingale and $E[[M, M]_t] < \infty$, for any $t \geq 0$.*

Corollary 5.14 *Let W be standard $(\mathcal{F}_t)$-Brownian motion. Then $[W]_t \equiv t$.*

Proof. By Proposition 4.6 $W_t^2 - t$ is a martingale. The result follows from Theorem 5.13. ■

The proof of Theorem 5.13 is adapted from Theorem 5.8 chap. 1 of [18]. The key fact employed here is that, when squaring sums of martingale increments and taking the expectation, one can neglect the cross-product terms. Before the sketch of the proof we propose an easy exercise.

Let A as the process appearing in the decomposition (5.6).

Exercise 5.15 *Let M be an $(\mathcal{F}_t)$-square integrable martingale. Let $0 \leq a_0 \leq a \leq b \leq u \leq v$. We have the following.*

(i) $E\left((M_v - M_u)(M_b - M_a)|\mathcal{F}_{a_0}\right) = 0.$

(ii)

$$E[(M_b - M_a)^2|\mathcal{F}_{a_0}] = E[M_b^2 - M_a^2|\mathcal{F}_{a_0}] = E(A_b - A_a|\mathcal{F}_{a_0}); \qquad (5.7)$$

Exercise 5.16 *Let $\Pi = \{0 = t_0 < \ldots < t_n = T\}$ be a subdivision of $[0, T]$.*

Let M be a square integrable martingale and let Doob decomposition of $M^2 = N + A$ where N is a martingale and A is an increasing process vanishing at zero. Then

$$E(C(\Pi, M, M)_t) = E(A_t).$$

Proof (of Theorem 5.13). Let $0 \leq t \leq T$. Let Π be a subdivision of $[0, T]$ as in (5.1). We denote Π_t the subdivision $\{0 = s_0 < s_1 \ldots < s_m\}$ where $\{s_i\}$ are the subset of $[0, T]$ of distinct elements obtained setting $s_i = t_i \wedge t$.

Lemma 5.17 *Let M be a square integrable $(\mathcal{F}_t)$-martingale and let K such that such that* $\sup_{t\leq T}|M_t| \leq K$. *Then, for every $t \in [0,T]$,*

$$E(C(\Pi, M, M)_t)^2) \leq 3K^4.$$

Proof. Using the linearity of conditional expectation and martingale property (see Exercise (5.15)), for $0 \leq k \leq m-1$ we have

$$\begin{aligned}
E\left(\sum_{j=k+1}^{m-1}(M_{s_{j+1}} - M_{s_j})^2|\mathcal{F}_{s_{k+1}}\right) &= \sum_{j=k+1}^{m-1} E(M_{s_{j+1}} - M_{s_j})^2|\mathcal{F}_{s_{k+1}}) \\
&= \sum_{j=k+1}^{m-1} E(M_{s_{j+1}}^2 - M_{s_j}^2|\mathcal{F}_{s_{k+1}}) \qquad (5.8) \\
&= E\left(\sum_{j=k+1}^{m-1}(M_{s_{j+1}}^2 - M_{s_j}^2)|\mathcal{F}_{s_{k+1}}\right) \\
&= E\left(M_{s_m}^2 - M_{s_{k+1}}^2)|\mathcal{F}_{s_{k+1}}\right) \leq E(M_{s_m}^2|\mathcal{F}_{s_{k+1}}) \leq K^2.
\end{aligned}$$

So

$$\begin{aligned}
& E\left(\sum_{k=0}^{m-1}\sum_{j=k+1}^{m-1}(M_{s_{j+1}} - M_{s_j})^2(M_{s_{k+1}} - M_{s_k})^2\right) \\
= \ & E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^2 E\left(\sum_{j=k+1}^{m-1}(M_{s_{j+1}} - M_{s_j})^2|\mathcal{F}_{s_{k+1}}\right)\right) \qquad (5.9) \\
\leq \ & K^2 E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^2\right) \leq K^4,
\end{aligned}$$

where the latter line can be justified taking the expectation in (5.8). We also have

$$E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^4\right) \leq K^2 E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^2\right) \leq K^4. \qquad (5.10)$$

Inequalities (5.9) and (5.10) imply

$$\begin{aligned}
E(C(\Pi, M, M)_t^2) &= E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^4\right) \\
&+ 2E\left(\sum_{k=0}^{m-1}\sum_{j=k+1}^{m-1}(M_{s_{j+1}} - M_{s_j})^2(M_{s_{k+1}} - M_{s_k})^2\right) \\
&\leq 3K^4.
\end{aligned}$$

■

Lemma 5.18 *Let M be a square integrable continuous martingale and let K such that $\sup_{t\leq T}|M_t| \leq K$ a.s. Then, given a sequence (Π) of subdivisions whose mesh converges to zero, for every $t \in [0,T]$, we have*

$$\lim_{|\Pi|\ \to 0} E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^4\right) = 0.$$

Proof. For any subdivision Π, we can write

$$\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^4 \leq C(\Pi, M, M)\delta^2(M, |\Pi|),$$

where $\delta(M,\varepsilon)$ is the modulus of continuity of M, i.e.

$$\delta(M,\varepsilon) := \sup\{|M_r - M_s|; 0 \leq r < s \leq T, s - r \leq \varepsilon\}. \tag{5.11}$$

That quantity is a measurable r.v. because the supremum can be restricted to rational s and r and the quantity of rational numbers is countable. Hölder inequality implies

$$E\left(\sum_{k=0}^{m-1}(M_{s_{k+1}} - M_{s_k})^4\right) \leq \left\{E(C(\Pi, M, M)^2)E(\delta^4(M; |\Pi|)\right\}^{\frac{1}{2}}.$$

As $|\Pi|$ approaches zero, the first factor on the right-hand side remains bounded by Lemma 5.17 and the second term tends to zero, by the uniform continuity of M and by Lebesgue dominated convergence theorem.

■

We go on with the proof of Theorem 5.13. Taking into account Dini's type Lemma 5.11 it is enough to show that for any fixed $t \in [0,T]$, $C(\Pi, M, M)_t - A_t$ converges to zero in probability. Let Π of the type (5.1) as in the beginning of the present theorem proof.

We start supposing that $\sup_{0\leq t\leq T}|M_t| \leq K$ and also that A intervening in the Doob decomposition $M^2 = N + A$ is also bounded by K, for some constant K. Then $N = M^2 - A$ is also bounded and it is of course even a square integrable martingale. Since for $0 \leq j \leq m-1$ we have

$$(M_{s_{j+1}} - M_{s_j})^2 = M_{s_{j+1}}^2 - M_{s_j}^2 - 2M_{s_j}(M_{s_{j+1}} - M_{s_j}),$$

and taking into account Exercise 5.15 it follows

$$E\{(C(\Pi,M,M)_t-A_t)^2\}=E\left(\sum_{j=0}^{m-1}\{(M_{s_{j+1}}-M_{s_j})^2-(A_{s_{j+1}}-A_{s_j})\}\right)^2$$
$$=\sum_{j=0}^{m-1}\sum_{k=0}^{m-1}E\left\{((M_{s_{j+1}}-M_{s_j})^2-(A_{s_{j+1}}-A_{s_j}))((M_{s_{k+1}}-M_{s_k})^2-(A_{s_{k+1}}-A_{s_k}))\right\}.$$

Now for fixed $j\neq k$ (say $k>j$) we get

$$\begin{aligned}&E\left\{((M_{s_{j+1}}-M_{s_j})^2-(A_{s_{j+1}}-A_{s_j}))((M_{s_{k+1}}-M_{s_k})^2-(A_{s_{k+1}}-A_{s_k}))\right\}\\=\;&E\left\{((M_{s_{j+1}}-M_{s_j})^2-(A_{s_{j+1}}-A_{s_j}))E((M_{s_{k+1}}-M_{s_k})^2-(A_{s_{k+1}}-A_{s_k})|\mathcal{F}_{s_k})\right\}.\end{aligned}$$

Morever by Exercise 5.15

$$E((M_{s_{k+1}}-M_{s_k})^2-(A_{s_{k+1}}-A_{s_k})|\mathcal{F}_{s_k})=0.$$

So

$$\begin{aligned}E\{(C(\Pi,M,M)_t-A_t)^2\}&=\sum_{j=0}^{m-1}E\left\{((M_{s_{j+1}}-M_{s_j})^2-(A_{s_{j+1}}-A_{s_j}))^2\right\}\\&\leq 2\sum_{j=0}^{m-1}E\left((M_{s_{j+1}}-M_{s_j})^4+(A_{s_{j+1}}-A_{s_j})^2\right)\\&\leq 2\sum_{j=0}^{m-1}E\left((M_{s_{j+1}}-M_{s_j})^4+E(A_t\delta(A;\Pi))\right).\end{aligned}$$

As the mesh of Π approaches zero, the first term on the right-hand side of this inequality converges to zero because of Lemma 5.18; the second term does as well, by the Lebesgue dominated convergence theorem and the sample path uniform continuity of A. Convergence in L^2 implies convergence in probability, so this proves the theorem for martingales which are uniformly bounded.

We treat now the general case. We proceed by localization. Let (τ^N) be the localizing sequence defined by

$$\tau^N:=\inf\{t||M_t|\;>N\}\wedge\inf\{t|A_t>N\}.$$

(τ^N) is a *suitable* sequence of stopping times in the sense that $\cup\Omega_N=\Omega$ a.s. where $\Omega_N:=\{\tau^N>T\}$. Since the stopped process

$$N^{\tau^N}=(M^2)^{\tau^N}-A^{\tau_N}=(M^{\tau^N})^2-A^{\tau_N},$$

is a martingale, by the first part of the proof we have

$$[M^{\tau_N}, M^{\tau_N}] = A^{\tau_N}.$$

Clearly, on Ω_N we have $M = M^{\tau_N}, \quad A = A^{\tau_N}$. Finally

$$1_{\Omega_N}[M, M] = 1_{\Omega_N}[M^{\tau^N}, M^{\tau^N}] = 1_{\Omega_N} A^{\tau_N} = 1_{\Omega_N} A.$$

■

Proposition 5.19 *Let $(\mathcal{F}_t)$ be a usual filtration and M is a càdlàg $(\mathcal{F}_t)$-local martingale vanishing at zero with $[M] = 0$. Then M is identically zero.*

Proof.

Let us consider the sequence (τ_k) intervening in the definition of local martingale. For every k, M^{τ_k} is a martingale so $(M^2)^{\tau_k}$ is a non-negative submartingale which is a martingale since $[M^{\tau_k}, M^{\tau_k}] = [M, M]^{\tau_k} \equiv 0$. See Doob decomposition and Theorem 5.13. Without restriction of generality, we can suppose that $\lim_{k \to +\infty} \tau_k(\omega) = \infty$, for every $\omega \in \Omega$. It will enough to show that, for every k,

$$1_{\{\tau_k > T\}} M \equiv 0,$$

or better that $M^{\tau_k} \equiv 0$. Finally, it will be sufficient to show the proposition when M is a martingale such that M^2 is a martingale.

Consequently, $E(M_t^2) = E(M_0^2) = 0$ and so $M_t = 0$ a.s. Since M is càdlàg then M will be indistinguishable from 0.

■

Corollary 5.20 *The decomposition of an $(\mathcal{F}_t)$-continuous semimartingale is unique.*

Proof. We write $0 = M + V$ where V is a finite variation process and M is an $(\mathcal{F}_t)$-local martingale. The bilinearity of the covariation implies

$$0 = [M, M] + 2[M, V] + [V, V].$$

This implies $[M, M] = 0$ because of Proposition 5.10. Proposition 5.19 alows to conclude. ■

Exercise 5.21 *Show that M, N are continuous $(\mathcal{F}_t)$-local martingales then (M, N) has all its mutual brackets.*

At this point we can easily identify the covariation of two semimartingales.

Proposition 5.22 *Let $S^i = M^i + V^i$ be two $(\mathcal{F}_t)$-semimartingales, $i = 1, 2$, where M^i are continuous local martingales and V^i bounded variation continuous processes. We have $[S^1, S^2] = [M^1, M^2]$.*

Proof. The result follows directly from Proposition 5.10 (v) and the bilinearity of the covariation. ■

Corollary 5.23 *Let M, N be two continuous $(\mathcal{F}_t)$-local martingales.*

(i) $MN - [M, N]$ is an $(\mathcal{F}_t)$-local martingale. In particular MN is an $(\mathcal{F}_t)$-semimartingale.

(ii) If M, N are square integrable martingale then $MN - [M, N]$ is an $(\mathcal{F}_t)$-martingale.

Proof. Items (i) and (ii) follow from the bilinearity of the covariation and Theorem 5.13. ■

6 Complements on martingale theory

6.1 A change of variable

Proposition 6.1 *Let $\varphi : \mathbb{R}_+ \to \mathbb{R}_+$, strictly increasing. Let $M = (M_t)$ be an $(\mathcal{F}_t)$-local martingale. We set $M^\varphi_t = M_{\varphi(t)}$. Then M^φ is an $(\mathcal{F}_{\varphi(t)})$-local martingale and $[M^\varphi]_t = [M]_{\varphi(t)}$.*

Proof. (of Proposition 6.1).
Through the usual localization procedure, one can reduce to the case that M is a martingale. This case can be treated since M^φ is obviously a martingale. Concerning the quadratic variation property, by Theorem 5.13 it is enough

to show that $(M^\varphi)^2 - [M]^\varphi$ is a martingale. This happens because $N = M^2 - [M]$ is a martingale and the definition of quadratic variation.

■

6.2 Burkholder-Davies-Gundy inequality

We start with an elementary result denominated again **Kunita-Watanabe**.

Proposition 6.2 *Let M, N be two processes such that (M, N) has all its mutal brackets. Let H, K be measurable processes such that $\int_0^T H_s^2 d[M,M]_s + \int_0^T K_s^2 d[N,N]_s < \infty$ a.s. Then for all $t \in [0,T]$*

$$|\int_0^t H_r K_r d[M,N]_r| \leq \sqrt{\int_0^t H_r^2 d[M,M]_r \int_0^t K_r^2 d[N,N]_r}.$$

In particular the left-hand side is well-defined.

Exercise 6.3 *Discuss the case $H = K = 1_{[a,b]}$, where $0 \leq a < b$.*

Among the most important inequality of martingale theory is Burkholder-Davies-Gundy (BDG) inequality.

Proposition 6.4 *Let $(M_t)_{t\geq 0}$ be a continuous $(\mathcal{F}_t)$-local martingale such that $M_0 = 0$. We set*

$$M_t^* = \max_{s\leq t}|M_s|.$$

For every $m > 0$, there are universal positive constants k_m, K_m such that

$$k_m E\left([M]_\tau^m\right) \leq E\left((M^*)_\tau^{2m}\right) \leq K_m E\left([M]_\tau^m\right), \tag{6.1}$$

takes place for any finite stopping time τ.

Proof. See [18], Theorem 3.28 chap. 3. ■

A first consequence of previous result is the following.

Proposition 6.5 *Let $(M_t)_{t\in[0,T]}$ be a continuous $(\mathcal{F}_t)$-local martingale such that $M_0 \in L^p, p \geq 1$. Suppose that one of the items below are valid.*

(i) $E([M]_T^{\frac{p}{2}}) < \infty$.

(ii) $\sup_{t\in[0,T]} |M_t| \in L^p$.

Then (M_t) is a martingale in L^p.

Corollary 6.6 *Let (M_t) be a continuous $(\mathcal{F}_t)$-local martingale such that $M_0 \in L^2$ and*

$$E([M]_T) < \infty, \forall T > 0.$$

Then (M_t) is a square integrable martingale.

Proof (of Proposition 6.5).

By subtraction, it is of course possible to suppose $M_0 = 0$.

- (i) and (ii) are equivalent by BDG.
- The result for $p = 1$ follows by Lemma 4.29.
- The case $p > 1$ follows since (ii) implies that $\sup_{t\in[0,T]} |M_t| \in L^1$.

■

7 Stochastic Itô integral

7.1 The case of a square integrable martingale

Let M be continuous square integrable martingale, i.e. such that $M_t \in L^2$, for all $t \geq 0$. By Theorem 5.13 we know that $[M,M]_T \in L^1$ and $M^2 - [M,M]$ is a martingale.

Definition 7.1 *We denote by $\mathcal{H}_T^2$ the set of square integrable continuous martingales M such that $M_0 = 0$, equipped with inner product*

$$\langle M, N\rangle_{\mathcal{H}_T^2} = E(M_T N_T). \tag{7.2}$$

We remark that, if $M, N \in \mathcal{H}^2_T$, then $M+N$ and $M-N$ belong to $\mathcal{H}^2_T$, and by Corollary 5.23 $MN - [M,N]$ is a martingale.

Proposition 7.2 *$\mathcal{H}^2_T$ is a Hilbert space.*

Proof. The positivity and the bilinearity properties for (7.2) are obvious. Suppose that $\langle M, M\rangle_{\mathcal{H}^2_T} = 0$. This implies that $E(M_T^2) = 0$. So M is a submartingale with

$$0 = E(M_0^2) \le E(M_t^2) \le E(M_T^2) = 0.$$

Finally $E(M_t^2) = 0$ and so $M_t \equiv 0$ a.s. which implies $M \equiv 0$ in the indistinguishable sense.

It remains to show $\mathcal{H}^2_T$ is complete. We consider a Cauchy sequence $(M^{(n)})$. Then $(M_T^{(n)})$ is a Cauchy sequence of $L^2(\Omega, \mathcal{F}, P)$ which is complete. There is then a r.v. that we will denote M_T such that

$$E((M_T^{(n)} - M_T)^2) \to 0,$$

when $n \to \infty$. By Doob's inequality we have

$$E(\sup_{s \le T} |M_s^{(n)} - M_s^{(m)}|^2) \le 4E(M_T^{(n)} - M_T^{(m)})^2,$$

which converges to zero when $n, m \to \infty$. Consequently the r.v. $\sup_{s\le T} |M_s^{(n)} - M_s^{(m)}|$ converges in probability to zero and so $M^{(n)}$ is a Cauchy sequence in $\mathcal{C}_{\mathcal{F}}([0,T])$ equipped with the usual metric d, related to the convergence ucp, so it converges to some (continuous) process M. It is square integrable since for every t $M_t^{(n)}$ converges in L^2 to M_t.
It remains to show that M is a martingale: indeed

$$M_t = \lim_{n\to\infty} M_t^{(n)} = \lim_{n\to\infty} E(M_T^{(n)} | \mathcal{F}_t) = E(M_T|\mathcal{F}_t),$$

by the L^2-continuity of the conditional expectation. Finally M belongs to $\mathcal{H}^2_T$ and

$$\|M^{(n)} - M\|^2_{\mathcal{H}^2_T} = E(M_T^{(n)} - M_T)^2 \to 0,$$

when $n \to \infty$. ■

Definition 7.3 *Given a continuous $(\mathcal{F}_t)$-local martingale M we denote $L^2(M)_T$ the set of progressively measurable processes H such that $E(\int_0^T H_s^2 d[M,M]_s) < \infty$ equipped with the inner product*

$$\langle H, K\rangle_{L^2(M)_T} = E\left(\int_0^T H_s K_s d[M,M]_s\right).$$

Definition 7.4 *We denote by $\mathcal{E}$ the set of elementary process, i.e. the set of processes H which are linear combination of processes of the type*

$$\sum_{i=0}^{n-1} H_i 1_{[t_i, t_{i+1}[}, \tag{7.3}$$

where Π is a subdivision of $[0,T]$ of the type (5.1) and every H_i is a bounded $\mathcal{F}_{t_i}$-measurable r.v.

Proposition 7.5 *$\mathcal{E}$ is dense in $L^2(M)_T$.*

Proof. Clearly $L^2(M)_T$ is a closed subspace of $L^2(\Omega\times[0,T], \mathcal{F}_T\otimes\mathcal{B}([0,T]), m)$ where m is the measure defined, by $A \in \mathcal{F}_T \otimes \mathcal{B}([0,T]), m)$

$$m(A) = E\left(\int_0^T 1_A(\omega, s) d[M,M]_s(\omega)\right).$$

Therefore it is a Hilbert space. It is enough to show that if $K \in L^2(M)_T$ is orthogonal to $\mathcal{E}$ then it is null. We set

$$X_t = \int_0^t K_s d[M,M]_s, t \in [0,T].$$

We show that X is a martingale. By Cauchy-Schwarz we get

$$\int_0^t |K_s| d[M,M]_s \leq \sqrt{E([M,M]_t)} E\left\{\left(\int_0^t K_s^2 d[M,M]_s\right)\right\}^{\frac{1}{2}} < \infty. \tag{7.4}$$

Consequently $X_t \in L^1$. Let $0 \leq s \leq t \leq T$. Let Z be a $\mathcal{F}_s$-measurable bounded r.v. The process $H_r = Z 1_{[s,t[}(r)$ is an element of $\mathcal{E}$; so, by assumption, it is orthogonal to K. Therefore

$$0 = E\left(\int_0^T H_r K_r d[M,M]_r\right) = E\left(Z \int_s^t K_r d[M,M]_r\right) = E(Z(X_t - X_s)).$$

This shows that X is a martingale. Since it has bounded variation it is null, see Proposition 5.10 item (v)(b) and Proposition 5.19. So $K = 0$ as an element of $L^2(M)_T$. ■

If H is of the type (7.3) and M is a continuous square integrable martingale then we denote

$$(H \cdot M)_t := \sum_{i=0}^{n-1} H_i (M_{t_{i+1} \wedge t} - M_{t_i \wedge t}).$$

Exercise 7.6 *Show that the process $H \cdot M$ is a square integrable martingale.*

Proposition 7.7 *If $H \in \mathcal{E}$ and M is a square integrable continuous martingale then*

$$E(H \cdot M)_T^2 = E\left(\int_0^T H_s^2 d[M,M]_s \right).$$

Proof. By the linearity of expectation and the orthogonality property stated in Exercise 5.15 applied to the martingale $H \cdot M$ we obtain

$$\begin{aligned}
E(H \cdot M)_T^2 &= \sum_{i=0}^{n-1} E\left((H \cdot M)_{t_{i+1}} - (H \cdot M)_{t_i}\right)^2 \\
&+ \sum_{i,j=0, i \neq j}^{n-1} E\left(((H \cdot M)_{t_{i+1}} - (H \cdot M)_{t_i})((H \cdot M)_{t_{j+1}} - (H \cdot M)_{t_j})\right) \\
&= \sum_{i=0}^{n-1} E\left((H \cdot M)_{t_{i+1}} - (H \cdot M)_{t_i}\right)^2 \\
&= \sum_{i=0}^{n-1} E\left(H_i^2 E((M_{t_{i+1}} - M_{t_i})^2 | \mathcal{F}_{t_i})\right) \\
&= \sum_{i=0}^{n-1} E\left(H_i^2 E([M,M]_{t_{i+1}} - [M,M]_{t_i} | \mathcal{F}_{t_i})\right),
\end{aligned}$$

where the latter passage can be explained by (5.7) and Theorem 5.13. Consequently this equals

$$\begin{aligned}
E(H \cdot M)_T^2 &= \sum_{i=0}^{n-1} E\left(E(H_i^2([M,M]_{t_{i+1}} - [M,M]_{t_i}) | \mathcal{F}_{t_i})\right) \\
&= \sum_{i=0}^{n-1} E\left(H_i^2([M,M]_{t_{i+1}} - [M,M]_{t_i})\right) \\
&= E\left(\int_0^T \sum_{i=0}^{n-1} H_i^2 1_{[t_i, t_{i+1}[}(s) d[M,M](s)) \right) \\
&= E\left(\int_0^T H_s^2 d[M,M]_s \right).
\end{aligned}$$

■

Theorem 7.8 *The application $\mathcal{E} \to \mathcal{H}^2_T$ defined by $H \mapsto H \cdot M$ extends to an isometry to $L^2(M)_T \to \mathcal{H}^2_T$.*

Remark 7.9 *The extended application will still be denoted by $H \mapsto H \cdot M$.*

Proof. By Proposition 7.7, the map $H \mapsto H \cdot M$ defined on $\mathcal{E}$ as a subspace of $L^2(M)_T$ is an isometry. Since $\mathcal{E}$ is dense and $\mathcal{H}^2_T$ is complete then we can extend it to the whole space $L^2(M)_T$. Indeed we have the general following result. ■

Proposition 7.10 *Let (F_1, d_1) and (F_2, d_2) be two metric spaces, D a dense subset of F_1. Let $T : D \to F_2$ for which there is $c > 0$ such that*

$$d_2(Tx, Ty) \leq cd_1(x, y) \text{ (resp. } d_2(Tx, Ty) = cd_1(x, y)), \tag{7.5}$$

for every $x, y \in D$.
If (F_2, d_2) is complete then the map T prolonges to the full space F_1 to a map for which (7.5) remains true.

Proof. Since D is dense in F_1, for every $x \in F_1$, there is a sequence (x_n) in D such that $d_1(x_n, x) \to 0$ when $n \to \infty$. The sequence (x_n) is Cauchy since it converges and (in)equality (7.5) implies that (Tx_n) is also Cauchy and it converges to some element $z \in F_2$, being F_2 complete.

(i) We set $Tx := z$. It is not difficult to see that choosing a different sequence (x'_n), the definition z is still the same, so Tx does not depend on the chosen subsequence.

(ii) Since d_2 and d_1 are continuous on respectively $F_1 \times F_1$ (resp. $F_2 \times F_2$), (7.5) is maintained for every $x, y \in F_1$.

■

Theorem 7.11 *Let $H \in L^2(M)_T$. $(H \cdot M)$ is the only continuous martingale Φ vanishing at zero such that, for every $N \in \mathcal{H}^2_T$*

$$[\Phi, N] = \int_0^\cdot H_s d[M, N]_s. \tag{7.6}$$

Proof. We set $\Phi := H \cdot M$ and we prove (7.6). Taking into account Corollary 5.23 and Corollary 5.20, we need to show that

$$(H \cdot M)N - \int_0^{\cdot} H_r d[M,N]_r \text{ is a martingale.} \tag{7.7}$$

For this let $0 \leq s < t \leq T$. We have to show that, for every ξ bounded $\mathcal{F}_s$-measurable we have

$$E\left(\left((H \cdot M)_t N_t - (H \cdot M)_s N_s - \int_s^t H_r d[M,N]_r\right)\xi\right) = 0. \tag{7.8}$$

The case $H = Z1_{[a,b[}$, $0 \leq a < b \leq T$, where Z is an $\mathcal{F}_a$-measurable random variable will be the object of Exercise 7.12 and so (7.7) for $H \in \mathcal{E}$ follows by linearity. If $H \in L^2(M)_T$, consider a sequence (H^n) in $\mathcal{E}$ converging to H in $L^2(M)_T$. By Theorem 7.8, Proposition 6.2 (Kunita-Watanabe) and Exercise 7.13 below, we have

$$[H \cdot M, N]_T = \lim_{n\to\infty} [H^n \cdot M, N]_T = \lim_{n\to\infty} \int_0^{\cdot} H_s^n d[M,N]_s = \int_0^{\cdot} H_s d[M,N]_s.$$

It remains to show the uniqueness. Let X be a martingale vanishing at zero, such that for each square integrable martingale N,

$$[X, N]_T = [H \cdot M, N]_T.$$

By localization, this extends to every martingale N. We take $N = X - H \cdot M$, we get $[X - H \cdot M, X - H \cdot M] = 0$ and therefore $X - H \cdot M = 0$.

■

Exercise 7.12 *Show* (7.8) *for* H *is an elementary process.* **Hint.** *Make use of* (5.7) *in Exercise 5.15.*

Exercise 7.13 *Let* $N \in \mathcal{H}_T^2$*. Show that the map* $M \mapsto [M,N]_T$ *is continuous from* $\mathcal{H}_T^2$ *to* $L^1(\Omega)$.

Corollary 7.14 *For every martingales* $M, N \in \mathcal{H}_T^2, H, K \in L^2(M)_T$*, we get*

$$[H \cdot M, K \cdot N] = \int_0^{\cdot} H_s K_s d[M,N]_s.$$

7.2 The case of local martingales

For this part we do not develop all the details, the interested reader can consult [18], chapter 3, section 3.2 D. Let M be a continuous $(\mathcal{F}_t)$-local martingale vanishing at zero and let $\mathcal{P}(M)$ be the collections of (equivalence classes) of all progressively measurable processes X satisfying

$$\int_0^T X_t^2 d[M]_t < \infty, \text{ a.s.} \tag{7.9}$$

The proposition below is the object of Proposition 2.24, Chapter 3 of [18].

Proposition 7.15 *Let $H \in \mathcal{P}(M)$. There is a unique continuous local martingale vanishing at zero Φ such that for every continuous martingale in $N \in \mathcal{H}_T^2$ we have*

$$[\Phi, N] = \int_0^\cdot H_s d[M,N]_s. \tag{7.10}$$

Corollary 7.16 *For every continuous local martingales M, N, $H \in \mathcal{P}(M), K \in \mathcal{P}(N)$ we get*

$$[H \cdot M, K \cdot N] = \int_0^\cdot H_s K_s d[M,N]_s.$$

Definition 7.17 *The process Φ characterized by Proposition 7.15 is called* **(Itô)-stochastic integral** *of H with respect to M and it is also denoted by $\int_0^\cdot H dM$.*

The next proposition is constitued by Proposition 2.26, chapter 3. of [18]

Proposition 7.18 *Let $\tau \leq T$ be a stopping time and let (H^n) be a sequence of elements in $\mathcal{P}(M)$ and $H \in \mathcal{P}(M)$ such that, in probability,*

$$\int_0^\tau |H_r^n - H_r|^2 d[M]_r \to 0.$$

Then

$$\lim_{n\to\infty} \sup_{t \leq \tau} |\int_0^t H_r^n dM_r - \int_0^t H_r dM_r| = 0,$$

in probability

Remark 7.19 *Taking $\tau = T$, previous convergence says that $\int_0^\cdot H^n dM$ converge to $\int_0^\cdot H dM$ ucp.*

Proposition 7.20 *Let M be a local martingale, $\alpha, \beta \in \mathbb{R}$ H^1, H^2 be elements of $\mathcal{P}(M)$ such that*

$$\int_0^T \left((H_s^1)^2 + (H_s^2)^2\right) d[M,M]_s < \infty, \text{ a.s.}$$

Then

$$\int_0^\cdot (\alpha H_s^1 + \beta H_s^2) dM_s = \alpha \int_0^\cdot H_s^1 dM_s + \beta \int_0^\cdot H_s^2 dM_s.$$

Proof. Exercise. ■

The stochastic integral with respect to a local martingale extends the one related to square integrable martingales. The proof of proposition below is an easy exercise.

Proposition 7.21 *Let M be a square integrable martingale, $H \in L^2(M)_T$. Then $\int_0^\cdot H dM = H \cdot M$.*

The following chain rule holds and it is an easy consequence of Theorem 7.11 and Proposition 7.15.

Proposition 7.22 *Let M be a continuous local martingale, let $H, K \in \mathcal{P}(M)$ such that $\int_0^T H_s^2 d[M,M]_s + \int_0^T H_s^2 K_s^2 d[M,M]_s < \infty$ a.s.*
Let $X_t := \int_0^t H dM, 0 \le t \le T$.
Then

$$\int_0^\cdot K dX = \int_0^\cdot HK dM.$$

8 Itô integral with respect to a semimartingale

Itô stochastic integral can be extended to the case where the integrator is a continuous $(\mathcal{F}_t)$-semimartingale (S_t). Let $S = M + V$ its decomposition where M is a $(\mathcal{F}_t)$-local martingale and V is an $(\mathcal{F}_t)$-progressively process finite variation process, $V_0 = 0$, for simplicity.

Example 8.1 Itô process

Let $(W_t)_{t\geq 0}$ be an $(\mathcal{F}_t)$-classical Brownian motion and X_0 be $\mathcal{F}_0$-measurable, (H_t), (K_t) $(\mathcal{F}_t)$-progressively measurable processes such that

$$\int_0^T H_s^2 ds + \int_0^T |K_s| ds < \infty \text{ a.s.}$$

Then

$$S_t = X_0 + \int_0^t H_s dW_s + \int_0^t K_s ds$$

is called $(W, \mathcal{F}_t)$-**Itô process** *or simply Itô process.*

Remark 8.2 *(i) An Itô process is of course an $(\mathcal{F}_t)$-semimartingale.*

(ii) We recall that $[W]_t \equiv t$.

It is possible to construct Itô stochastic integral related to S.

Let $(Y_t)_{t\geq 0}$ be an $(\mathcal{F}_t)$-progressively measurable process (V_t) an $(\mathcal{F}_t)$-progressively measurable finite variation continuous process and M a continuous $(\mathcal{F}_t)$-local martingale. If

$$\int_0^T |Y_s|^2 d[M]_s + \int_0^T |Y_s| d\|V\|_s < \infty \quad \text{a.s.} \tag{8.11}$$

and $S = M + V$, then

$$\int_0^t Y_s dS_s := \int_0^t Y_s dM_s + \int_0^t Y_s dV_s$$

is well-defined. Moreover it is a semimartingale, see Exercise 8.13 below.

Proposition 8.3 *Let S be a continuous semimartingale, let H, K progressively measurable processes such that $\int_0^T H_s^2 d[M,M]_s + \int_0^T H_s^2 K_s^2 d[M,M]_s + \int_0^T |H|_s d\|V\|_s + \int_0^T |H_s K_s|_s d\|V\|_s < \infty$ a.s. Let $X_t := \int_0^t H dS, 0 \leq t \leq T$. Then*

$$\int_0^\cdot K dX = \int_0^\cdot HK dS.$$

Proof. It follows by Proposition 7.22 and the definition of stochastic integral with respect to semimartingales. ∎

Application.

Let (W_t) an $(\mathcal{F}_t)$-standard Brownian motion. Let $S_t = X_0 + \int_0^t H_s dW_s + \int_0^t K_s ds$ an Itô stochastic process. Let (Y_t) be an $(\mathcal{F}_t)$-progressively measurable process such that $\int_0^T Y_s^2 H_s^2 ds + \int_0^T |Y_s K_s| ds < +\infty$ a.s. Then

$$\int_0^t Y dS = \int_0^t Y H dW + \int_0^t Y_s K_s ds.$$

Exercise 8.4 *Let $(Y_t^i), i = 1, 2$, $(\mathcal{F}_t)$-progressively measurable. Let $S^i, i = 1, 2$ be two continuous $(\mathcal{F}_t)$-semimartingales with decomposition $M^i + V^i, i = 1, 2$. We suppose*

$$\int_0^T |Y_s^i|^2 d[M^i]_s + \int_0^T |Y_s^i| d\|V^i\|_s, < \infty \quad \text{a.s.}, i = 1, 2. \tag{8.12}$$

Verify the following properties.

i) $(\int_0^t Y_s^1 dS_s^1)_{t \geq 0}$ *is an $(\mathcal{F}_t)$-semimartingale.*

ii)

$$\left[\int_0^\cdot Y^1 dS^1, \int_0^\cdot Y^2 dS^2\right]_t = \int_0^\cdot Y^1 Y^2 d[S^1, S^2]_s. \tag{8.13}$$

Solutions

i) We set $Y = Y^1, M = M^1, V = V^1$. The $(\mathcal{F}_t)$- local martingale part is $\left(\int_0^t Y dM\right)_{t \geq 0}$. The finite variation part is $\int_0^t Y_s dV_s$.

ii) We make use of Corollary 7.16, the definition of stochastic integral with respect to a semimartingale. We also take into account the fact that, whenever V is a bounded variation process, then $\int_0^\cdot Y^i dV$ is of finite variation, $i = 1, 2$ and so

$$\left[\int_0^\cdot Y^i dV\right] \equiv [V] \equiv 0.$$

On the other hand, the covariation of a finite variation process and a continuous process is zero.

An important property of localization for Itô integrals is given below.

Remark 8.5 *Let M, N be two $(\mathcal{F}_t)$-local martingales, $(Y_t), (Z_t)$ $(\mathcal{F}_t)$-progressively measurable processes such that $\int_0^T Y_s^2 d[M]_s + \int_0^T Z_s^2 d[N]_s < \infty$ a.s. Let $\Omega_0 \in \mathcal{F}$ such that*

$$M_t(\omega) = N_t(\omega), Y_t(\omega) = Z_t(\omega), \text{ for } t \leq T, \omega \in \Omega_0.$$

Then

$$1_{\Omega_0} \int_0^t Y_s dM_s = 1_{\Omega_0} \int_0^t Z_s dN_s, t \leq T.$$

(True for elementary processes, afterwards we go to the limit).

An important theorem which extends the classical theory of Lebesgue integration to the Itô integral is *stochastic Fubini's theorem.*

Proposition 8.6 *(Stochastic Fubini). Let $X = M+V$ be an $(\mathcal{F}_t)$-semimartingale, $H : \mathbb{R}^d \times \mathbb{R}_+ \times \Omega \to \mathbb{R}$, measurable, with respect to the product σ-field such that for every a, $(H(a, s)_{s\geq 0})$ is $(\mathcal{F}_t)$-progressively measurable. Let μ be a finite Borel measure on $\mathbb{R}^d$. We also suppose that $(H_t)_{0\leq t\leq T}$ is either bounded or satisfies*

$$\int_U \Big\{E\Big[\int_0^T H(u, s)^2 d[M, M]_s\Big] + \int_0^T |H(u, s)| d\|V\|_s\Big\}\mu(du) < \infty. \quad (8.14)$$

We set

$$Z_t^a = \int_0^t H(a, s) dX_s.$$

Then (Z_t^a) admits a measurable version from $\mathbb{R}_+ \times \mathbb{R}^d \times \Omega \to \mathbb{R}$ and

$$\int_{\mathbb{R}^d} Z_t^a \mu(da) = \int_0^t H(s) dX_s$$

where $H(s) = \int_{\mathbb{R}^d} H(a, s)\mu(da)$.

Proof. See [30], theorem 45, ch. 4. ■

At this stage, there is a very popular integral which is altenative to Itô integral.

Definition 8.7 *Let $(S_t)_{t\geq 0}, (Y_t)_{t\geq 0}$, be two $(\mathcal{F}_t)$-continuous semimartingales. We denote then*

$$\int_0^t Y \circ dS := \int_0^t Y dS + \frac{1}{2}[Y,S]_t. \tag{8.15}$$

This process will be called **Stratonovich** *integral of Y with respect to S.*

We remark that

- *Itô integral $\int_0^t Y dS$ exists,*
- *$[Y,S]$ exists.*

8.1 Connections between Itô, Stratonovich and integrals via discretization

In this section we want to explore the relation between *classical* integrals with respect to semimartingales and integrals via discretizations.

Theorem 8.8 *Let $(\mathcal{F}_t)$ be a usual filtration, X, Y be two continuous $(\mathcal{F}_t)$-semimartingales, Z be an $(\mathcal{F}_t)$-progressively measurable càglàd (or càdlàg) process. Then*

i) $\int_0^t Z d^- X = \int_0^t Z dX$,

ii) $\int_0^t Y d^\circ X = \int_0^t Y \circ dX$.

Moreover previous forward and symmetric integrals exist ucp.

Proof.

Suppose that Z is càglàd. If it were càdlàg we can replace Z with $Z_s^- = Z_{s-}, s \geq 0$ which is càglàd, see Exercise 8.9 below.

We use the fact that a càglàd function g is a pointwise limit of simple functions. We have $S^-(\Pi, Z, X)_t = \int_0^t Z^\Pi dX$ where

$$Z^\Pi(s) = \sum_{k=0}^{n-1} Z_{t_k} 1_{]t_k, t_{k+1}]}(s).$$

We need to show that $\int_0^t (Z^\Pi - Z) dX \to 0$ in the ucp sense. Now $Z^\Pi(s) \to Z_s$ pointwise a.s. We decompose $X = M + V$ where M is an $(\mathcal{F}_t)$-local martingale and V a bounded variation process. Taking in account Proposition 5.10 (v), it is enough to consider the case $X = M$.

We proceed via localization. For $k > 0$ we consider the stopping time

$$\tau_k = \inf\{t \geq 0 | \sup_{s \leq t} |Z_s| \geq k \quad \text{or} \quad [X]_t \geq k \quad \text{or} \quad |X_t| \geq k\}.$$

Indeed the càglàd process (Z_s) is locally bounded, see Remark 5.1. We set $\Omega_k = \{\tau_k > T\}$. If $\omega \in \Omega_k$, for $t \in [0, T]$, we have

$$\sup_{s \leq t} |Z_s|^2 \leq k^2 \quad [X]_t \leq k, |X_t| \leq k.$$

Now

$$1_{\Omega_k} \int_0^t (Z - Z^\Pi) dX = 1_{\Omega_k} \int_0^t (Z - Z^\Pi)^{\tau_k} dX^{\tau_k} \tag{8.16}$$

Moreover

$$\int_0^t \left((Z - Z^\Pi)_s^{\tau_k} \right)^2 1_{[0, \tau_k]} d[X^{\tau_k}] \leq k^3 \text{ a.s.} \tag{8.17}$$

We have, up to a negligible set

$$\Omega = \bigcup_k \Omega_k,$$

In fact it is enough to prove that for any $k > 0$

$$\sup_{t \leq T} 1_{\Omega_k} \left| \int_0^t (Z - Z^\Pi)_s dX_s \right|, \tag{8.18}$$

converges to zero in probability. This will hold if we show that the expectation of previous expression (8.18) goes to zero when $|\Pi| \to 0$. Because of (8.16), that expectation equals

$$E \left(\sup_{t \leq T} 1_{\Omega_k} \left| \int_0^t (Z - Z_s^\Pi)^{\tau_k} dX_s^{\tau_k} \right| \right)$$

$$\leq 4E \left| \int_0^T ((Z - Z^\Pi)^{\tau_k})_s^2 1_{[0, \tau_k]} d[X]_s^{\tau_k} \right|.$$

This converges to zero because of Lebesgue dominated convergence theorem taking into account (8.17).

ii) Consequence of previous point and of the definition of Stratonovich integral. ∎

Exercise 8.9 *Let $(\mathcal{F}_t), X, Y$ as in the statement of Theorem 8.8. Let Z be an $(\mathcal{F}_t)$-progressively measurable càdlàg process.*

(i) Show that the Itô integrals $\int_0^t Z dX$ and $\int_0^t Z^- dX$ are equal, where $Z_s^- = Z_{s-}, s \geq 0$.

(ii) Prove item i) of Theorem 8.8 by first proving that $\int_0^t Z d^- X = \int_0^t Z^- dX$ if Z is càdlàg.

(iii) Conclude to the validity of ii) in Theorem 8.8 when Z is càdlàg.

9 Stability of covariation and Itô formula

9.1 Recalls in real measure theory

A useful tool of probability theory is weak convergence of probability laws; that concept coincides with the convergence of the related distribution functions at each continuity point of the limiting distribution function. Here we will make use of such convergence for the paths realizations of a process. For reference on the subject we refer to [19, 32].

As mentioned earlier, we denote by BV the set of Borel real functions defined on $\mathbb{R}_+$ (resp. $[0, T]$) which have locally bouded variation.

Remark 9.1 *(i) A bounded variation function is always equal a.e. to a càdlàg bounded variation function. In fact it is difference of non-decreasing functions.*

(ii) For that reason, without restriction of generality, a bounded variation function can be considered càdlàg.

(iii) A càdlàg (càglàd) function is bounded and has at most countable jumps.

(iv) For basic properties (as those above), the reader can refer to [4], chapter 1.

We equip BV with the topology associated with the following convergence. A sequence (V_n) in BV converges to V if

- $V_n(0) \longrightarrow V(0)$,
- $dV_n \longrightarrow dV$ with respect to the vague topology, i.e. for every $\alpha \in C^0(\mathbb{R})$ with compact support
$$\int_0^\infty \alpha dV_n \longrightarrow \int_0^\infty \alpha dV.$$

The natural topology is the total variation on each compact but, it is too strong for our purposes.

Remark 9.2 *a) The sequence (dV_n) converges to dV if and only if for every $\alpha \in C^0(\mathbb{R})$*
$$\int_0^t \alpha dV_n \longrightarrow \int_0^t \alpha dV$$
holds (at every continuity point t of V).

b) If (V_n) is a sequence converging in BV, the total variations are uniformly bounded on each compact set K, i.e.
$$\sup_n \int_K d\|V_n\| < \infty. \tag{9.1}$$

c) $V_n \to V$ in BV if and only if $V_n(x) \to V(x)$ at every continuity point x of V and (9.1) holds.

d) We need also the following Dini's type lemma.
Let $(u_n(t), t \in [0,T])$ be a sequence of non-decreasing functions such that $\sup_{t\in[0,T]} |u_n(t)| < \infty$.
If $u_n(t) \longrightarrow u(t), t \in [0,T]$ (pointwise) then the convergence holds uniformly.

9.2 Stability of the covariation

One basic tool of calculus related to covariations is the following.

Proposition 9.3 *Let $\underline{X} = (X^1, \ldots, X^d)$ be a vector of processes having all its mutual covariations, $F, G \in C^1(\mathbb{R}^d)$. Then, $[F(\underline{X}), G(\underline{X})]$ exists and it is given by*
$$[F(\underline{X}), G(\underline{X})]_t = \sum_{i,j=1}^d \int_0^t \partial_i F(\underline{X}) \partial_j G(\underline{X}) d[X^i, X^j]. \tag{9.2}$$

We have the following particular cases.

Remark 9.4 *a) An interesting particular case, is given by* $d = 2$, $F(x_1, x_2) = f(x_1)$, $G(x_1, x_2) = g(x_2)$ *with* $f, g \in C^1(\mathbb{R})$.
Then (9.2) implies that $(f(X^1), g(X^2))$ *has all its mutual covariations and*

$$[f(X^1), g(X^2)]_t = \int_0^t f'(X^1)g'(X^2)d[X^1, X^2].$$

b) In particular, the family of finite quadratic variation processes is stable through C^1 *transformations. Let* $\phi_1, \phi_2 \in C^1(\mathbb{R})$.

$$[\phi_1(X), \phi_2(X)]_t = \int_0^t \phi_1'(X)\phi_2'(X)d[X].$$

We recall that the ucp convergence coïncides with the convergence in probability of random variables with the values in the metric space $E = C[0, T]$. We recall the following fact.

Remark 9.5 *A sequence* (Z_n) *of* E*-valued random variables converges in probability to an* E*-valued random variable* Z *if and only if for each subsequence* (n_k) *there is a subsequence* (n_{k_j}) *such that* $Z_{n_{k_j}} \longrightarrow Z$ *a.s. and in probability.*

Let $\Pi = \{0 = t_0 < ... < t_n = T\}$ be an element of a subdivisions sequence. We set

$$\mu^{\Pi}(t) = \sum_{i|t_i \le t} (X_{t_{i+1}} - X_{t_i})^2, t \in [0, T].$$

Exercise 9.6 *Let* (Π^ℓ) *be a sequence of subdivisions whose mesh converge to zero. Using the definition of* $[X, X]$ *and Remark 9.5, show that there is a subsequence* (ℓ_k) *such that a.s.* $\mu^{\Pi^{\ell_k}}$ *converges in* BV *to* $[X, X]$.

We will not prove Proposition 9.3 but only point a) of Remark 9.4. The proof of the Proposition and point b) can be proven analogously.

Proof *Sketch* of Remark 9.4 a).
Using polarization techniques (bilinearity arguments), it is enough to consider the case $X = X^1 = X^2, f = g$.

Let $\Pi = \{0 = t_0 < ... < t_n = T\}$ be an element of a subdivisions sequence, whose mesh converges to zero. We set $\Pi_t = \{0 = s_0 < ... < s_n = t\}$ obtained setting $s_i = t_i \wedge t$. We expand

$$f(X_{s_{i+1}}) - f(X_{s_i}) = f'(X_{s_i})(X_{s_{i+1}} - X_{s_i}) + R_\Pi(t)(X_{s_{i+1}} - X_{s_i}),$$

where

$$|R_\Pi(t)| = \left| \int_0^1 f'(X_{s_i} + a(X_{s_{i+1}} - X_{s_i}))da - f'(X_{s_i}) \right| \leq \delta(f', \delta(X, |\Pi|)),$$

and $\delta(g, \cdot)$ is the modulus of continuity of a function g.
We have $\sup_{t \leq T} |R_\Pi(t)| \to 0$ in probability since f' and X are uniformly continuous on each compact.

We recall that the notation $(R_\Pi(t))$ indicates a family of processes such that $\sup_{t \leq T} |R_\Pi(t)| \to 0$ in probability.

So,

$$\big(f(X_{s_{i+1}}) - f(X_{s_i})\big)^2 - f'(X_{s_i})^2 (X_{s_{i+1}} - X_{s_i})^2 = R_\Pi(t)(X_{s_{i+1}} - X_{s_i})^2.$$

Summing up from $i = 0$ to $i = n - 1$, we get

$$\sum_{i=0}^{n-1} \big(f(X_{s_{i+1}}) - f(X_{s_i})\big)^2 = I_1(t, \Pi) + I_2(t, \Pi),$$

with

$$\begin{aligned} I_1(t, \Pi) &= \int_0^t f'(X_s)^2 d\mu^\Pi(s) + I_{11}(t, \Pi), \\ I_2(t, \Pi) &\leq \sum_{i=0}^{n-1} (X_{t_{i+1} \wedge t} - X_{t_i \wedge t})^2 R_\Pi(t), \end{aligned}$$

where it is not difficult to show that

$$I_{11}(t, \Pi) \leq \sup_{t \in [0,T]} f'(X_t)^2 \delta(X, |\Pi|)^2.$$

Now

$$\sup_{t \leq T} |I_2(t, \Pi)| \leq \sup_{s \leq T} |R_\Pi(t)| \sum_{i=0}^{n-1} (X_{t_{i+1} \wedge t} - X_{t_i \wedge t})^2.$$

Since $[X,X]$ exists then $I_2(\cdot,\Pi) \xrightarrow{\text{ucp}} 0$, it remains to prove that

$$\int_0^t Y_s d\mu^{\Pi}(s) \to \int_0^t Y_s d[X,X]_s \text{ ucp} \tag{9.3}$$

and Y is a continuous non-negative process.

Taking in account Remark 9.5, it will be enough to show that there is a sequence (Π^n) such that

$$\int_0^t Y_s d\mu_{\Pi^n}(s) \longrightarrow \int_0^t Y_s d[X,X]_s \quad \text{ucp}. \tag{9.4}$$

By Dini's lemma 5.11, it will be enough to show that (9.4) holds in probability for every fixed t.

By Exercise 9.6, we can consider a null set N such that for $\omega \notin N$ the sequence of the real functions $(\mu_{\Pi^n}(\cdot)(\omega))$ in BV to $[X,X](\omega)$. Therefore

$$\int_0^t Y_s(\omega) d\mu_{\Pi^n}(s)(\omega) \longrightarrow \int_0^t Y(\omega) d[X,X](\omega),$$

because of Remark 9.2 a) and the fact that Y is continuous.

■

9.3 Dirichlet processes

The notion of Dirichlet process is the natural generalization of semimartingales. It was introduced by H. Föllmer, see [13] under the inspiration of the theory of Dirichlet forms.

Let $(\mathcal{F}_t)_{t\geq 0}$ be a fixed filtration fulfilling the usual hypotheses.

Definition 9.7 *A $(\mathcal{F}_t)$-adapted process X will be called (continuous) $(\mathcal{F}_t)$-* **Dirichlet process***, if*

$$X = M + A, \tag{9.5}$$

where

- $(M_t)_{t\geq 0}$ *is a continuous $(\mathcal{F}_t)$-local martingale,*

- *A is a zero (continuous) quadratic variation process such that $A_0 = 0$.*

Remark 9.8 *(A_t) is an $(\mathcal{F}_t)$-adapted process.*

Remark 9.9 *An $(\mathcal{F}_t)$-semimartingale is an $(\mathcal{F}_t)$-Dirichlet process.*

If $(\mathcal{F}_t)$ is the canonical filtration to X, we will simply say that X is a Dirichlet process, omitting the mention to the filtration.

Lemma 9.10 *Decomposition (9.5) is unique.*

Proof. We suppose that

$$X = M^i + A^i, i = 1, 2,$$

where M^i is an $(\mathcal{F}_t)$-local martingale, A^i an $(\mathcal{F}_t)$-adapted, (continuous) zero quadratic variation process, such that $A_0^i = 0$, By additivity, we have

$$M + A = 0, \quad M = M^1 - M^2, \quad A = A^1 - A^2.$$

Moreover $M_0 = 0$ since $M_0^1 = M_0^2 = X_0$.
The bilinearity of the covariation says that

$$\begin{aligned} 0 &= [M + A, M + A] \\ &= [M, M] + 2[M, A] + [A, A]. \end{aligned}$$

Since A has zero quadratic variation, Proposition 5.10 (iii) implies that $[M, A] = 0$ so that $[M] = 0$. Recalling that $M_0 = 0$, Proposition 5.19 yields that $M \equiv 0$. Finally A will also be zero. ■

We have already seen that the class of finite quadratic variation processes is stable under C^1 transformations. This property extends to the case of Dirichlet processes as the following result shows.

Proposition 9.11 *Let X be an $(\mathcal{F}_t)$-Dirichlet process, let $f \in C^1$. Then $f(X)$ is again an $(\mathcal{F}_t)$-Dirichlet process.*

Proof. Let $X = M + A$, as in (9.5).
We set
$$M_t^f = \int_0^t f'(X)dM + f(X_0).$$
To conclude, it is enough to show that
$$A_t^f = f(X_t) - M_t^f,$$
is a zero quadratic variation process.
For this, we again proceed using the bilinearity and symmetry of the covariation.

(i) Remark 9.4 2. implies that
$$\begin{aligned}[f(X), f(X)]_t &= \int_0^t f'^2(X_s)d[X]_s \\ &= \int_0^t f'^2(X_s)d[M]_s.\end{aligned}$$

(ii) Since M^f is a local martingale and taking into account Corollary 7.14 and Proposition 5.10 (v) it follows that
$$[M^f, M^f]_t = \int_0^t f'(X_s)^2 d[M]_s,$$

(iii) Morever, using similar arguments as before
$$\begin{aligned}[f(X), M^f]_t &= \int_0^t f'(X)d[X, M^f] \\ &= \int_0^t f'(X)d[M, M^f] \\ &= \int_0^t f'^2(X)d[M].\end{aligned}$$

This concludes the proof of Proposition 9.11. ∎

Natural examples of Dirichlet processes arise from Brownian motion and related processes.

Let W be a classical $(\mathcal{F}_t)$-Brownian motion.

Example 9.12 *Let f of class C^1. Then $X = f(W)$ is an $(\mathcal{F}_t)$-Dirichlet process.*

Remark 9.13 *Let $f \in C^0(\mathbb{R})$. We recall a celebrated result of [5]. $f(W)$ is a $(\mathcal{F}_t)$-semimartingale if and only if f is difference of two convex functions.*

Previous Example and Remark show easily that the class of $(\mathcal{F}_t)$-Dirichlet processes strictly include the class of $(\mathcal{F}_t)$-semimartingales.

Open question

Let $(\mathcal{F}_t)$ be a filtration such that X is a $(\mathcal{F}_t)$-Dirichlet process and let $(\mathcal{G}_t)$ be a subfiltration of $(\mathcal{F}_t)$ so that X is still $(\mathcal{G}_t)$-adapted.
Is X a $(\mathcal{G}_t)$-Dirichlet process ?
In general, at our knowledge the answer is unknown. However, at least in the particular case when X is $(\mathcal{F}_t)$-semimartingale, then X is a $(\mathcal{G}_t)$-Dirichlet process; in fact Stricker Theorem 4.28 asserts that if X is an $(\mathcal{F}_t)$-semimartingale then it is also a $(\mathcal{G}_t)$-semimartingale.

9.4 Itô formulae for finite quadratic variation process

We start with the one-dimensional case. Let $X = (X_t)_{t\geq 0}$ be a continuous process.

Proposition 9.14 *Suppose that $[X, X]$ exists and let $f \in C^2(\mathbb{R})$. Then*

$$\int_0^{\cdot} f'(X)d^-X \quad \text{and} \quad \int_0^{\cdot} f'(X)d^\circ X \quad \text{exist.} \tag{9.6}$$

Moreover

a) $f(X_t) = f(X_0) + \int_0^t f'(X)d^-X + \frac{1}{2}\int_0^t f''(X_s)d[X,X]_s.$

b) $f(X_t) = f(X_0) + \int_0^t f'(X)d^-X + \frac{1}{2}[f'(X), X]_t.$

c) $f(X_t) = f(X_0) + \int_0^t f'(X)d^\circ X.$

Proof. We use the same notations as before. Let $t \geq 0$. Π is a subdivision $\Pi = \{0 = t_0 < ... < t_m = T\}$, $s_i = t_i \wedge t$.

b) follows from a) because Remark 9.4 a) implies that

$$[f'(X), X]_t = \int_0^t f''(X) d[X, X].$$

c) follows from b) and Proposition 5.10 (i). It remains to justify a) and (9.6). We write Taylor expansion until the second order to obtain, for $t \geq 0$

$$\begin{aligned} f(X_{s_{i+1}}) &= f(X_{s_i}) + f'(X_{s_i})(X_{s_{i+1}} - X_{s_i}) \\ &+ \frac{f''(X_{s_i})}{2}(X_{s_{i+1}} - X_{s_i})^2 + R_\Pi(t)(X_{s_{i+1}} - X_{s_i})^2, \end{aligned} \tag{9.7}$$

where again $\sup_{t \leq T} |R_\Pi(t)| \longrightarrow 0$ a.s. because f'' is uniformly continuous on each compact. Summing (9.7) from $i = 0$ to $n - 1$, we obtain

$$\sum_{i=0}^{n-1} \big(f(X_{s_{i+1}}) - f(X_{s_i})\big) = I_1(t, \Pi) + I_2(t, \Pi) + I_3(t, \Pi), \tag{9.8}$$

where

$$\begin{aligned} I_1(t, \Pi) &= \sum_{i=0}^{n-1} f'(X_{s_i})(X_{s_{i+1}} - X_{s_i}) \\ I_2(t, \Pi) &= \frac{1}{2} \sum_{i=0}^{n-1} f''(X_{s_i})(X_{s_{i+1}} - X_{s_i})^2, \\ I_3(t, \Pi) &\leq C(\Pi, X, X)(t)|R_\Pi(t)|, \end{aligned}$$

with the same notations as in previous subsection. $I_2(\cdot, \Pi)$ converges ucp to $\frac{1}{2}\int_0^\cdot f''(X_s) d[X, X]$ and $I_3(\cdot, \Pi)$ goes ucp to zero by the same arguments as the proof of the C^1 stability of the covariation.

The left member of (9.8) equals (for each Π) $f(X_t) - f(X_0)$. Therefore the ucp limit of $I_1(\cdot, \Pi)$ is forced to exist and it will be of course $\int_0^\cdot f'(X) d^- X$. This establishes a) at the same time. The existence of $\int_0^\cdot f'(X) d^\circ X$ follows again by Proposition 5.10 (i) and Remark 9.4. ■

Before writing the generalization to dimension d, we observe the following.

Lemma 9.15 *Let X be a continuous process.*
The following properties are equivalent.

a) $[X,X]$ *exists.*

b) $\int_0^\cdot X d^- X$ *exist.*

Proof. a) $\Leftrightarrow$ b) follows by the following remark:

$$(X_{s_{i+1}} - X_{s_i})^2 = X_{s_{i+1}}^2 - X_{s_i}^2 - 2X_{s_i}(X_{s_{i+1}} - X_{s_i}). \tag{9.9}$$

■

Corollary 9.16 *Let X be a continuous process.*
The following properties are equivalent.

a) $[X,X]$ *exists.*

b) $\int_0^\cdot g(X) d^- X$ *exists* $\forall g \in C^1$.

Proof. a) $\Rightarrow$ b) follows by Itô formula stated in Proposition 9.14 a).
b) $\Rightarrow$ a) follows setting $g(x) = x$ and using Lemma 9.15. ■

Itô formula admits an extension to the multidimensional case.

Theorem 9.17 *Let $\underline{X} = (X^1, \cdots, X^d)$ be a vector of processes having all its mutual brackets.*
Let $f \in C^2(\mathbb{R}^d)$. Suppose that

$$\int_0^\cdot \partial_i f(\underline{X}) d^- X^i \tag{9.10}$$

exist $2 \le i \le d$. *Then*

$$\begin{aligned} f(\underline{X}_t) &= f(\underline{X}_0) + \sum_{i=1}^d \int_0^t \partial_i f(\underline{X}_s) d^- X_s^i \\ &+ \frac{1}{2} \sum_{i,j=1}^d \int_0^t \partial_{i,j}^2 f(\underline{X}_s) d[X^i, X^j]_s \end{aligned} \tag{9.11}$$

In particular

$$\int_0^\cdot \partial_1 f(\underline{X}_s) d^- X^1 \quad \text{exists.}$$

Remark 9.18 *a) One could define a vector stochastic integral of an $\mathbb{R}^d$-valued process $\underline{Y}$ with respect to an $\mathbb{R}^d$-valued process $\underline{X}$ denoted $\int_0^t \underline{Y} \cdot d^-\underline{X}$. It would be defined as the limit in probability of*

$$\sum_{i=0}^{n-1} \sum_{\ell=1}^{d} Y_{s^i}^{\ell} (X_{s_{i+1}}^{\ell} - X_{s_i}^{\ell}), \tag{9.12}$$

where $s_i = t_i \wedge t$ and (t_i) is a element of a sequence of subdivisions whose mesh goes to zero. With this convention, we would not need to suppose (9.10) in Theorem 9.17 and $\sum_{i=1}^{d} \int_0^{\cdot} \partial_i f(\underline{X}) d^- X^i$ would be replaced with $\int_0^{\cdot} \nabla f(\underline{X}) \cdot d^- X$.
Similar considerations as for (9.12) *can be done for the symmetric integral $\int_0^{\cdot} Y \cdot d^{\circ} X$.*

b) Let $f \in C^{2,1}(\mathbb{R} \times \mathbb{R}^{d-1})$.
Suppose that $X^2, \cdots, X^d$ are bounded variation processes. In this case, $\int_0^{\cdot} \partial_i f(\underline{X}) d^- X^i$ coincides with the classical Lebesgue-Stieltjes integral $\int_0^{\cdot} \partial_i f(\underline{X}) dX^i$. We can show that Itô's formula conclusion (see Theorem 9.17) is verified; in particular

$$\int_0^{\cdot} \partial_1 f(\underline{X}) d^- X^1 \quad \text{exists}$$

and

$$\begin{aligned} f(\underline{X}_t) &= f(\underline{X}_0) + \int_0^t \partial_1 f(\underline{X}) d^- X^1 + \sum_{i=2}^{n} \int_0^{\cdot} \partial_i f(\underline{X}_s) d\underline{X}_s^i \\ &+ \frac{1}{2} \int_0^t \partial_1^2 f(X_s) d[X^1, X^1]_s. \end{aligned}$$

c) Let V be a real bounded variation process with values in $\mathbb{R}_+$ and a process (X_t) with values in $\mathbb{R}^{d-1}$ having all its mutual covariations. Then a Itô formula also holds under the only assumption $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^{d-1})$. Indeed we have

$$\begin{aligned} f(V_t, X_t) &= f(V_0, X_0) + \int_0^t \partial_v (V_s, X_s) dV_s + \int_0^t \nabla_x f(V_s, X_s) \cdot d^- X_s \\ &+ \frac{1}{2} \sum_{i,j=2}^{d} \int_0^t \partial_{ij}^2 f(V_s, X_s) d[X^i, X^j]_s. \end{aligned}$$

d) In reality, if one knows a priori that the process (X_t) will remain in an open set O and setting $V_t = t$, we only need to suppose $f \in C^{1,2}(\mathbb{R}_+ \times O)$ in order to expand $f(t, X_t)$. For instance if X is a strictly positive finite quadratic variation process and $f(t,x) := \log x$, we can write

$$\log X_t = \log X_0 + \int_0^t \frac{1}{X_s} d^- X_s - \frac{1}{2} \int_0^t \frac{1}{X_s^2} d[X]_s.$$

Taking in account the relation between symmetric and forward integrals, see Proposition 5.10 (i) and Proposition 9.3, we can easily prove the following.

Proposition 9.19 *Let $\underline{X} = (X^1, \cdots, X^n)$ be a vector of processes having all its mutual brackets.*
Let $f \in C^2(\mathbb{R}^n)$. Then

$$f(\underline{X}_t) = f(\underline{X}_0) + \int_0^t \nabla f(\underline{X}_s) \cdot d^\circ \underline{X}_s.$$

9.5 Applications to semimartingales and Itô processes

Let $(S_t)_{t \geq 0}$ be an $(\mathcal{F}_t)$-semimartingale and $(W_t)_{t \geq 0}$ be an $(\mathcal{F}_t)$-Brownian motion. Let $(X_t)_{t \in [0,T]}$ an Itô process of the form

$$X_t = X_0 + \int_0^t K_s ds + \int_0^t H_s dW_s, \quad t \in [0,T]. \tag{9.13}$$

We recall that (X_t) is an $(\mathcal{F}_t)$-semimartingale with bounded variation part $V_t = \int_0^t K_s ds$.

Proposition 9.20 *Let $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$, $(s,x) \mapsto f(s,x)$. We have the following.*

i)

$$\begin{aligned} f(t, S_t) &= f(0, S_0) + \int_0^t \frac{\partial f}{\partial s}(s, S_s) ds \\ &+ \int_0^t \frac{\partial f}{\partial x}(s, S_s) dS_s + \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x^2}(s, S_s) d[S,S]_s. \end{aligned}$$

ii)

$$
\begin{aligned}
f(t, X_t) &= f(0, X_0) + \int_0^t \frac{\partial f}{\partial s}(s, X_s) ds \\
&+ \int_0^t \frac{\partial f}{\partial x}(s, X_s)(H_s dW_s + K_s ds) + \frac{1}{2}\int_0^t \frac{\partial^2 f}{\partial x^2}(s, X_s) H_s^2 ds.
\end{aligned}
$$

iii) Let S^0 be another $(\mathcal{F}_t)$-semimartingale. The following integration by parts holds.

$$
S_t S_t^0 = S_0 S_0^0 + \int_0^t S_s dS_s^0 + \int_0^t S_s^0 dS_s + [S, S^0]_t.
$$

Proof.

i) We recall that Itô and forward integral coincide, see Theorem 8.8. The result follows then from Remark 9.18 c) which follows Theorem 9.17 with $d = 2$, $V_t = t, X_t = S_t$.

ii) We write the Chain rule Proposition 8.3, the fact that $[W, W]_t = t$, and Corollary 7.16 which implies that $[X, X]_t = \int_0^t H_s^2 d[W]_s$.

iii) It is a consequence of integration by parts formula Proposition 5.10 (iii) on forward integral, and the fact that again in this case Itô and forward integrals coincide.

■

Proposition 9.21 *a) The product of semimartingales is still a semimartingale.*

b) The product of Itô processes is again a Itô process.

c) In particular, the family of $(\mathcal{F}_t)$-semimartingale (resp. $((\mathcal{F}_t), W_t)$-Itô processes) is a vector algebra, which is a subalgebra of the algebra $\mathcal{C}_{\mathcal{F}}$ of $(\mathcal{F}_t)$-adapted continuous processes.

Proof. It is a consequence of integration by parts. In fact, Y^1 and Y^2 are two $(\mathcal{F}_t)$-semimartingales, resp. Itô processes; Proposition 9.20 iii) gives

$$
Y_t^1 Y_t^2 = Y_0^1 Y_0^2 + \int_0^t Y^1 dY^2 + \int_0^t Y^2 dY^1 + [Y^1, Y^2]_t.
$$

"Chain rule" Proposition 8.3 and Exercise 8.4 give the result. ■

Exercise 9.22 *a) Evaluate the quadratic variation of $[sin(W), sin(W)]$. Let A be a zero quadratic variation process and set $D = W + A$. Deduce $[sin(D), sin(D)]$.*

b) If X is a Itô process in the form given above, express $\int_0^t X \circ dW$.

c) Suppose that F is of class $C^2(\mathbb{R})$. Verify that

$$M_t = F(W_t) - \frac{1}{2}\int_0^t F''(W_s)ds, t \in [0, T],$$

is an $(\mathcal{F}_t)$-martingale if $\int_{\mathbb{R}} |F'|^2(y)\frac{p(\frac{y}{\sqrt{T}})}{1+y^2}dy < \infty$, where p is the law density of $N(0, 1)$.

9.6 A first glance to stochastic differential equations

Let W be an $(\mathcal{F}_t)$-Brownian motion, ξ be an $\mathcal{F}_0$-measurable random variable. Let us consider some Borel functions $a, b : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$.

Definition 9.23 *We will say that a process $(X_t)_{t\in[0,T]}$ is a solution to*

$$\begin{cases} dX_t &= b(t, X_t)dt + a(t, X_t)dW_t \\ X_0 &= \xi \end{cases} \tag{9.14}$$

in the Itô sense, *if*

- *X is $(\mathcal{F}_t)$-adapted;*
- $\int_0^T a^2(s, X_s)ds + \int_0^T |b|(s, X_s)ds < \infty$ a.s.
- $X_t = \xi + \int_0^t a(s, X_s)dW_s + \int_0^t b(s, X_s)ds, \quad$ a.s., $\forall t \in [0, T]$.

In particular, a solution has a continuous modification which is an Itô process. Very often ξ, which is in general only an $\mathcal{F}_0$-random variable, will be a real deterministic x_0.

We would like to discuss here a first method for the research of a solution of equation (9.14).

We first start to find out a solution in the linear case: let $\mu, \sigma \in \mathbb{R}, s_0 > 0$.

Let $(S_t)_{t\geq 0}$ be a strictly positive solution (if it exists) of

$$\begin{cases} dS_t &= \mu S_t dt + \sigma S_t dW_t \\ S_0 &= s_0. \end{cases} \tag{9.15}$$

We proceed now using the Itô formula in an open set, i.e. Remark 9.18 d), since $f(x) = \log(x)$ is not of class $C^2(\mathbb{R})$. Such a solution is a Itô process. We have

$$\log(S_t) = \log(s_0) + \int_0^t \frac{dS_s}{S_s} + \frac{1}{2}\int_0^t \Big(\frac{-1}{S_s^2}\Big)\sigma^2 S_s^2 ds. \tag{9.16}$$

Using (9.15), for $Y_t = \log S_t$ we have

$$Y_t = Y_0 + \int_0^t \Big(\mu - \frac{\sigma^2}{2}\Big)ds + \int_0^t \sigma dW_s. \tag{9.17}$$

We deduce

$$Y_t = \log S_t = \log S_0 + \Big(\mu - \frac{\sigma^2}{2}\Big)t + \sigma W_t. \tag{9.18}$$

It seems that

$$S_t = s_0 \exp\Big\{\Big(\mu - \frac{\sigma^2}{2}\Big)t + \sigma W_t\Big\} \tag{9.19}$$

is a candidate solution of (9.15). We will verify rigorously that this is even the unique solution to (9.15).

We start with the existence property which will be stated in a more general framework in the next Proposition 9.26.

Definition 9.24 *Suppose a and b as in the previous definition being continuous. Let ξ be an $\mathcal{F}_0$-measurable random variable. Let $A = (A_t)_{t\in[0,T]}$, $(\mathcal{F}_t)$-adapted measurable process. We will say that a process $(X_t)_{t\in[0,T]}$ is a solution to*

$$\begin{cases} dX_t &= b(t, X_t)dt + a(t, X_t)d^- A_t \\ X_0 &= \xi, \end{cases} \tag{9.20}$$

in the forward sense *if*

- $\int_0^T |b|(s, X_s)ds < \infty$ a.s.
- $X_t = X_0 + \int_0^t a(s, X_s)d^- A_s + \int_0^t b(s, X_s)ds$, a.s., $\forall t \geq 0$.

Remark 9.25 *Suppose a, b continuous functions as before.*

If A is a $(\mathcal{F}_t)$- semimartingale, an $(\mathcal{F}_t)$-adapted process X is a solution to the equation in the Itô sense if and only if it is a solution in the forward sense.

Proposition 9.26 *Let A be an $(\mathcal{F}_t)$-adapted finite quadratic variation process. Then*

$$S_t = s_0 \exp\left\{\mu t - \frac{\sigma^2}{2}[A, A]_t + \sigma A_t\right\} \tag{9.21}$$

is a solution of (9.20) with $a(t, x) = \mu x$ and $b(t, x) = \sigma x$.

Proof. We have $S = f(V, A)$, where $f(v, x) = s_0 \exp(v + \sigma x)$ and $V_t = \mu t - \frac{\sigma^2}{2}[A]_t$. We have

$$\begin{aligned}
\partial_x f(v, x) &= \sigma f(v, x) \\
\partial^2_{xx} f(v, x) &= \sigma^2 f(v, x) \\
\partial_v f(v, x) &= f(v, x).
\end{aligned}$$

Consequently, using Itô formula (Remark 9.18 c)), we obtain

$$\begin{aligned}
S_t = f(V_t, A_t) &= \underbrace{f(0,0)}_{s_0} + \int_0^t \partial_v f(V_s, A_s) dV_s + \int_0^t \partial_x f(V_s, A_s) d^- A_s \\
&+ \frac{1}{2}\int_0^t \partial^2_{xx} f(V_s, A_s) d[A, A]_s \\
&= s_0 + \int_0^t f(V_s, A_s)(\mu ds - \frac{\sigma^2}{2} d[A]_s) + \int_0^t \sigma f(V_s, A_s) d^- A_s \\
&+ \frac{1}{2}\int_0^t \sigma^2 f(V_s, A_s) d[A, A]_s \\
&= s_0 + \mu \int_0^t S_s ds + \sigma \int_0^t S_s d^- A_s.
\end{aligned}$$

■

Exercise 9.27 *Let $b, a : \mathbb{R} \to \mathbb{R}$ respectively continuous and of class C^1. Let also $x_0 \in \mathbb{R}$, $a > 0$. Let $(X_t)_{t\in[0,T]}$ be a strictly positive solution to*

$$\begin{cases} dX_t = b(X_t)dt + a(X_t)dW_t \\ X_0 = \xi. \end{cases} \tag{9.22}$$

Set $H(x) = \int_0^x \frac{1}{a}(y)dy, x \in \mathbb{R}$ and set $Y = H(X)$. Show explicitly that Y solves one equation directed by a bounded variation process (so not involving stochastic integrals).

Remark 9.28 *In particular, Proposition 9.26 implies that* (9.19) *is a solution of* (9.15). *Next argument will show that it is the unique solution.*

So

$$S_t = s_0 \exp\left\{ \Big(\mu - \frac{\sigma^2}{2}\Big)t + \sigma W_t \right\}$$

is a solution of (9.15); we will suppose that $(X_t)_{t\geq 0}$ is another one. We will try to express the "Itô stochastic differential" of $X_t S_t^{-1}$. We set

$$Z_t = s_0 S_t^{-1} = \exp\left\{ \Big(-\mu + \frac{\sigma^2}{2}\Big)t - \sigma W_t \right\};$$

if $\mu' = -\mu + \sigma^2$ et $\sigma' = -\sigma$ then $Z_t = \exp\left\{ \Big(\mu' - \frac{\sigma'^2}{2}\Big)t + \sigma' W_t \right\}$. Proposition 9.26 shows again that

$$Z_t = 1 + \int_0^t Z_s \Big(\mu' ds + \sigma' dW_s\Big) = 1 + \int_0^t Z_s \Big((\sigma^2 - \mu)ds - \sigma dW_s\Big). \quad (9.23)$$

We can express now $X_t Z_t$ through integration by parts for semimartingales, see Proposition 9.20. We formally obtain

$$d(X_t Z_t) = X_t dZ_t + Z_t dX_t + d[X, Z]_t.$$

Since (X_t) is a solution to equation (9.15) and having in mind (9.23), we have

$$[X, Z]_t = \left[\int_0^t X_s \sigma dW_s, -\int_0^t Z_s \sigma dW_s\right] = -\int_0^t \sigma^2 X_s Z_s ds.$$

We deduce that

$$\begin{aligned}
X_t Z_t &= s_0 + \int_0^t X_s \underbrace{dZ_s}_{Z_s((\sigma^2-\mu)ds - \sigma dW_s)} + \int_0^t Z_s \underbrace{dX_s}_{X_s(\mu ds + \sigma dW_s)} - \int_0^t \sigma^2 X_s Z_s ds \\
&= s_0 + \int_0^t (\sigma^2 - \mu) X_s Z_s ds - \sigma \int_0^t Z_s X_s dW_s + \int_0^t Z_s X_s (\mu - \sigma^2) ds \\
&+ \int_0^t \sigma Z_s X_s dW_s \\
&= s_0.
\end{aligned}$$

So $X_t Z_t \equiv s_0, \quad \forall t \geq 0 \quad, P$-a.s.
This yields

$$X_t = s_0 Z_t^{-1} = S_t, \quad \forall t \geq 0, \quad P-\text{a.s.}$$

We have finally been able to prove the following result.

Theorem 9.29 *Let $\sigma \in \mathbb{R}, \mu \in \mathbb{R}, s_0 \in \mathbb{R}, (W_t)$ is an $(\mathcal{F}_t)$ standard Brownian motion and $T > 0$. There is a unique Itô process $(S_t)_{t\in[0,T]}$ (up to indistinguishability) which is solution to*

$$\left\{ \begin{array}{rcl} dX_t & = & \mu X_t dt + \sigma X_t dW_t \\ X_0 & = & s_0. \end{array} \right. \tag{9.24}$$

This process is given by

$$S_t = s_0 \exp\left\{ \left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t \right\}.$$

Remark 9.30 *(i) Process S_t that we have provided constitutes the classical model of the price of a stock in the financial model of Black-Scholes.*

(ii) When $\mu = 0, S_t$ is a martingale, see Proposition 4.6. This kind of processus is called **exponential martingale**.

We construct now a classical process of stochastic analysis.

Definition 9.31 *(***Generalized Ornstein-Uhlenbeck process***). Let $(A_t)_{t\geq 0}$ be a general (continuous) process such that $A_0 = \xi$, ξ being any random variable, $c \in \mathbb{R}$. A solution X to*

$$X_t = A_t + c \int_0^t X_s ds, \tag{9.25}$$

is called **Ornstein-Uhlenbeck process**.

Proposition 9.33 will show that it is legitimate to denote the previous integral equation with

$$\left\{ \begin{array}{rcl} dX_t & = & cX_t dt + d^- A_t \\ X_0 & = & \xi \end{array} \right. \tag{9.26}$$

Remark 9.32 *Previous equation admits a unique solution in the class of continuous processes. This is an elementary consequence of a fixed point theorem on each path.*

Proposition 9.33 *The unique solution (X_t) of previous equation is given by*

$$X_t = e^{ct}\xi + \int_0^t e^{c(t-s)} d^- A_s.$$

Remark 9.34 *The most classical example of Ornstein-Uhlenbeck comes out from the case $A = \xi + \sigma W$ where W is a classical Brownian motion with canonical filtration $(\mathcal{F}_t)$. Let ξ be an $\mathcal{F}_0$-measurable r.v.*

In this case previous expression gives

$$X_t = e^{ct}\xi + \sigma \int_0^t e^{c(t-s)} dW_s.$$

Proof (of Proposition 9.33).

Uniqueness and existence have already been discussed. Let X be a solution. We set $Y_t = X_t e^{-ct}$. Since $t \mapsto e^{-t}$ has bounded variation, we have $[e^{-c\cdot}, X] \equiv 0$. Integration by parts in Proposition 5.10, and the fact that X is a solution of (9.26) gives

$$\begin{aligned} Y_t &= \xi + \int_0^t e^{-cs} d^- X_s - c \int_0^t X_s e^{-cs} ds \\ &= \xi + \int_0^t e^{-cs} d^- A_s + c \int_0^t e^{-cs} X_s ds - c \int_0^t e^{-cs} X_s ds \\ &= \xi + c \int_0^t e^{-cs} d^- A_s. \end{aligned}$$

Since $X_t = e^{ct} Y_t$, we obtain the expression stated in Proposition 9.33.

■

9.7 Multidimensional semimartingales and Itô processes

Before evaluating the covariation of two independent stochastic processes, we profit to give a foundational theorem of probability, see for instance Section 4.3 in [1].

Definition 9.35 *(i) A non-empty class $\mathcal{D}$ of subsets of Ω is called* Π-*system if*

$$A, B \in \mathcal{D} \Rightarrow A \bigcap B \in \mathcal{D}.$$

(ii) A non empty class Λ is called λ-system if

(a) $\Omega \in \Lambda$;

(b) $A, B \in \Lambda, B \subset A \Rightarrow A - B \in \Lambda$;

(c) If (A_n) is an increasing sequence of elements of Λ then $\bigcup_n A_n \in \Lambda$.

The following theorem is called **Monotone class theorem**.

Theorem 9.36 *Let $\mathcal{D}$ be a Π-system, Λ a λ-system such that $\mathcal{D} \subset \Lambda$. Then $\sigma(\mathcal{D}) \subset \Lambda$.*

An easy observation is the following.

Proposition 9.37 *Let $\mathcal{A}, \mathcal{B}$ be two σ -fields, $\Lambda = \sigma(\mathcal{A} \cup \mathcal{B})$. We set $\mathcal{M} = \sigma(A \bigcap B, A \in \mathcal{A}, B \in \mathcal{B})$. Then $\Lambda = \mathcal{M}$.*

Proof. It is enough to show $\Lambda \subset \mathcal{M}$, the other sense being obvious. This follows since every element A of $\mathcal{A}$ (resp. B of $\mathcal{B}$) can be written as $A = A \cap \Omega$ (resp. $B = \Omega \cap B$). ■

Proposition 9.38 *Let $(M_t)_{t \in [0,T]}$ be an continuous $(\mathcal{F}_t)$-local martingale, $(Y_t)_{t \in [0,T]}$ be a càdlàg or càglàd $(\mathcal{F}_t)$-adapted process. If M and Y are independent then $[M, Y] = 0$.*

Proof. Let $\mathcal{Y}_T$ be the σ-field generated by $(Y_t)_{t \in [0,T]}$. We denote $\mathcal{M}_t = \sigma(\mathcal{Y}_T \bigcup \mathcal{F}_t^M)$, where $(\mathcal{F}_t^M)$ is the canonical filtration associated with M. Via localization, it is possible to suppose M to be an $\mathcal{F}_t^M$-martingale.

We prove that (M_t) is an $(\mathcal{M}_t)$-martingale, that is to say

$$E(M_t | \mathcal{M}_s) = M_s, t \geq s.$$

In other words

$$E(M_t 1_C) = E(M_s 1_C), \quad \forall C \in \mathcal{M}_s. \tag{9.27}$$

Now $\mathcal{M}_s = \sigma(\mathcal{D})$ where

$$\mathcal{D} = \{A \bigcap B, A \in \mathcal{F}_s^M, B \in \mathcal{Y}_T\},$$

see Proposition 9.37. Let Λ be the family of C for which (9.27) is verified.

- $\Omega \in \Lambda$;
- by additivity, if A, B verifying (9.27) with $B \subset A$ then it is true for $C = A - B$;
- if (C_n) is an increasing sequence of elements fulfilling (9.27) then it will be the same for $C = \bigcup_n C_n$.

We observe that $\mathcal{D} \subset \Lambda$: let $C = A \cap B, A \in \mathcal{F}_s^M, B \in \mathcal{Y}_T$. We have

$$\begin{aligned} E(1_A 1_B M_t) &= E(1_A M_t E(1_B | \mathcal{F}_t^M)) = E(1_A M_t P(B)) \\ &= P(B)E(1_A M_t) = P(B)E(1_A M_s) = E(1_A P(B) M_s) = E(1_A M_s E(1_B | \mathcal{F}_s^M)) \\ &= E(1_A 1_B M_s). \end{aligned}$$

So by monotone class theorem, the set Λ contains $\sigma(\mathcal{D}) = \mathcal{M}_s$, so (9.27) is proved.

Finally (M_t) is also an $(\mathcal{M}_t)$-martingale. By definition we know that $C_t(\Pi; Y, M) = S_t^+(\Pi, Y, M) - S_t^-(\Pi, Y, M), t \in [0, T]$. Theorem 8.8 implies that the limit when $|\Pi|$ goes to zero of $S_t^-(\Pi, Y, M)$, which is the forward integral $\int_0^t Y d^- M$, equals the $(\mathcal{M}_t)$-Itô integral $\int_0^t Y dM$.

Without restriction of generality, we can suppose that and $M_0 = 0$ and Y is bounded and càglàd.

For fixed t we have

$$S_t^+(\Pi, Y, M) = \int_0^t Y_s^{\Pi,+} dM_s + R(|\Pi|, t),$$

where

$$Y_s^{\Pi,+} = \sum_k Y_{s_{k+1}} 1_{]s_k, s_{k+1}]}(s),$$

where $s_k = t_k \wedge t, 1 \leq k \leq n$.

Since $(Y_s^{\Pi,+})$ is $\mathcal{M}_s$-adapted this equals the Itô integral

$$\int_0^t Y_s^{\Pi,+} dM_s.$$

By a similar argument as for the proof of Theorem 8.8, this can be shown to converge ucp again to Itô integral $\int_0^t Y dM$.

This concludes the proof. ■

Corollary 9.39 *Let S^1, S^2 be two $(\mathcal{F}_t)$-continuous semimartingales whose martingale parts are independent. Then $[S^1, S^2] = 0$.*

Theorem 8.8, Theorem 9.17 and Remark 9.18 c), give the following multidimensional Itô formula.

Proposition 9.40 *Let $X^1, \dots, X^m$ be $(\mathcal{F}_t)$-semimartingales. Let $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^m)$. Then*

$$\begin{aligned} f(t, X_t^1, \dots, X_t^m) &= f(0, X_0^1, \dots, X_0^m) + \int_0^t \frac{\partial f}{\partial s}(s, X_s^1, \dots, X_s^m) ds \\ &+ \sum_{i=1}^m \int_0^t \frac{\partial f}{\partial x_i}(s, X_s^1, \dots, X_s^m) dX_s^i \\ &+ \frac{1}{2} \sum_{i,j=1}^m \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(s, X_s^1, \dots, X_s^m) d[X^i, X^j]_s. \end{aligned}$$

Definition 9.41 *A vector $(W^1, \dots, W^p)$ of independent classical Brownian motions is called* ***p*-dimensional classical Brownian motion**. *We can of course define similarly an $(\mathcal{F}_t)$- p*-**dimensional classical Brownian motion**.

Remark 9.42 *What is the value of $[W^i, W^j]$ when $i \neq j$ and W^i, W^j are two independent classical Brownian motions? Corollary 9.39 affirms that its value is zero.*

We continue with an extension of the notion of Itô process.

Definition 9.43 *Let $W = (W^1, \ldots, W^p)$ be an $(\mathcal{F}_t)$-p-dimensional classical Brownian motion. We say that a process $(X_t)_{t \geq 0}$ is a $(W, \mathcal{F})$-Itô process (or simply general Itô process) if*

$$X_t = X_0 + \int_0^t K_s ds + \sum_{i=1}^p \int_0^t H_s^i dW_s^i,$$

where

- X_0 *est* $\mathcal{F}_0$*-measurable;*
- $(K_t), (H_t^i)$ *are* $(\mathcal{F}_t)$*-progressively measurable processes,*
- $\int_0^T |K_s| ds < \infty$ *a.s.*
- $\int_0^T \left(H_s^i\right)^2 ds < \infty$ *a.s.,* $1 \leq i \leq p$.

Remark 9.42 and Corollary 7.16 (which allows to calculate brackets of Itô integrals related to local martingales) provide the following result.

Proposition 9.44 *Let $(H_t^i)_{t \in [0,T]}, i = 1, 2$ be two $(\mathcal{F}_t)$-progressively measurable processes, such that $\int_0^T (H_s^i)^2 ds < \infty$ a.s. $i = 1, 2$. Let W^1, W^2 be two independent $(\mathcal{F}_t)$- Brownian motions. Then*

$$\left[\int_0^\cdot H_s^1 dW_s^1, \int_0^\cdot H_s^2 dW_s^2\right] \equiv 0.$$

Exercise 9.45 *Let W be a classical p-dimensional Brownian motion, $F \in C^2(\mathbb{R}^p)$. Show that*

$$F(W_t) - F(0) - \frac{1}{2} \int_0^t \Delta F(W_s) ds$$

defines an $(\mathcal{F}_t)$-martingale if ∇F is bounded.

Proposition 9.44 also allows to evaluate the covariation of two general Itô processes. Consider in fact

$$X_t^i = X_0^i + \int_0^t K_s^i ds + \sum_{j=1}^p \int_0^t H_s^{i,j} dW_s^j, \quad i = 1, 2.$$

Then

$$[X^1, X^2]_t = \sum_{j=1}^{p} \int_0^t H_s^{1,j} H_s^{2,j} ds.$$

10 Stochastic differential equations

At Section 9, we have studied in detail the solutions of equations

$$X_t = x_0 + \int_0^t X_s(\mu ds + \sigma dW_s)$$

and the Ornstein-Uhlenbeck processes; we recall that examples of such processes are provided by

$$X_t = x_0 + c\int_0^t X_s ds + \sigma W_t,$$

where W is a classical Brownian motion and c, σ, x_0 are real constants.

At this point we will study first equations with Lipschitz coefficients, i.e.

$$X_t = Z + \int_0^t b(s, X_s)ds + \int_0^t a(s, X_s)dW_s. \qquad (10.1)$$

We recall that those equations are called **Itô stochastic differential equations**. A solution of previous equation is named **diffusion**.

10.1 Existence and uniqueness for Itô stochastic differential equations

Definition 10.1 *A function $\gamma : [0,T] \times \mathbb{R}^m \longrightarrow \mathbb{R}^d$ is called* **Lipschitz** *(with respect to the second variable x uniformly with respect to the first t) if there is a constant $k > 0$ such that*

$$\sup_{t\in[0,T]} |\gamma(t,x) - \gamma(t,y)| \leq k|x-y|, \quad \forall x, y \in \mathbb{R}^m. \qquad (10.2)$$

A function $\gamma : \mathbb{R}_+ \times \mathbb{R}^m \longrightarrow \mathbb{R}^d$ is called **Lipschitz** *(with respect to the second variable x uniformly with respect to the first t on every compact interval) if for every $T > 0$, γ restricted to $[0,T] \times \mathbb{R}^m$ is Lipschitz in previous sense.*

A function $\gamma : [0,T] \times \mathbb{R}^m \longrightarrow \mathbb{R}^d$ is said to have linear growth (with respect to x uniformly with respect to t), if there is a constant $C > 0$ with

$$\sup_{t\in[0,T]} |\gamma(t,x)| \leq C(1+|x|). \tag{10.3}$$

A function $\gamma : \mathbb{R}_+ \times \mathbb{R}^m \longrightarrow \mathbb{R}^d$ is said to have linear growth (with respect to x uniformly for t on each compact interval), if for every $T > 0$, the restriction of γ with respect to $[0,T] \times \mathbb{R}^m$ has linear growth in previous sense.

Remark 10.2 *If $\gamma : [0,T] \times \mathbb{R}^m$ is continuous and Lipschitz then it has linear growth.*

In fact, if $x, y \in \mathbb{R}^m, t \geq 0$

$$\begin{array}{rcl} |\gamma(t,x)| & \leq & |\gamma(t,x) - \gamma(t,0)| + |\gamma(t,0)| \\ & \leq & k|x| + \sup\limits_{t\in[0,T]} |\gamma(t,0)| \\ & \leq & C(|x|+1), \quad \forall x \in \mathbb{R}^m \\ \text{avec} \quad C & = & \max\left(\sup\limits_{t\in[0,T]} |\gamma(t,0)|; k\right). \end{array}$$

As usual we will consider a probability space $(\Omega, \mathcal{F}, P)$ equipped with a usual filtration $(\mathcal{F}_t)_{t\geq 0}$.

Let $a, b : \mathbb{R}_+ \times \mathbb{R} \longrightarrow \mathbb{R}$ be Borel functions, Z a random variable $\mathcal{F}_0$-measurable and $(W_t)_{t\geq 0}$ a classical $(\mathcal{F}_t)$-Brownian motion.

Remark 10.3 *a) We recall that (10.1) can be formally expressed by*

$$\left\{ \begin{array}{rcl} dX_t & = & b(t,X_t)dt + a(t,X_t)dW_t \\ X_0 & = & Z. \end{array} \right.$$

b) If a continuous process fulfills (10.1) P a.s. for every $t \geq 0$ then equality (10.1) holds $\forall t \geq 0$ P a.s. since both members are continuous and so indistinguishable.

Theorem 10.4 *We suppose $a, b : \mathbb{R}_+ \times \mathbb{R} \longrightarrow \mathbb{R}$ Lipschitz with linear growth. Let Z be a square integrable r.v. that is $\mathcal{F}_0$-measurable. Then (10.4) has* *a (unique up to indistinguishability) continuous solution X. Moreover, for every $T > 0$*

$$E\left(\sup_{s\leq T}|X_s|^2\right) < \infty.$$

Proof It will be enough to discuss existence and uniqueness in $[0,T]$, where $T > 0$. We denote

$$\mathcal{A} = \Big\{(X_t)_{t\in[0,T]}; (\mathcal{F}_t)\text{-adapted continuous such that}\\ \mathbb{E}\left(\sup_{t\leq T}|X_t|^2\right) < \infty\Big\}.$$

$\mathcal{A}$ is a Banach space equipped with the norm

$$||X|| = \sqrt{E\left(\sup_{s\leq T}|X_s|^2\right)}.$$

If A is a positive constant, $\mathcal{A}$ can itself be equipped with the norm $||\cdot||_{\sim,A}$ (equivalent to $||\cdot||$):

$$||X||^2_{\sim,A} = \sup_{t\in[0,T]} e^{-\mathrm{A}t}E\left(\sup_{s\leq t}|X_s|^2\right).$$

In fact

$$||X||^2 = \sup_{t\in[0,T]} E\left(\sup_{s\leq t}|X_s|^2\right).$$

Let Φ be the application that to a process $(X_s)_{s\in[0,T]}$ in $\mathcal{A}$ associates a process $(\Phi(X)_t)_{t\in[0,T]}$ defined by

$$\Phi(X)_t = Z + \int_0^t b(s,X_s)ds + \int_0^t a(s,X_s)dW_s.$$

Remark 10.5 Φ *sends an element of $\mathcal{A}$ into $\mathcal{A}$, i.e., if $(X_t)_{t\in[0,T]} \in \mathcal{A}$ then*

$$E\left(\sup_{t\leq T}|\Phi(X)_t|^2\right) < \infty. \tag{10.4}$$

In fact, Doob inequality says

$$
\begin{aligned}
E\left(\sup_{t\le T}|\Phi(X)_t|^2\right) &\le 4E(Z^2)+4E\left(\sup_{t\le T}\left|\int_0^t a(s,X_s)dW_s\right|^2\right)\\
&+ 4E\left(\int_0^T |b(s,X_s)|ds\right)^2\\
&\le 4E(|Z|^2)+16E\left(\int_0^T a^2(s,X_s)ds\right)+4TE\left(\int_0^T b^2(s,X_s)ds\right)
\end{aligned}
$$

(provided the expectation in the middle is finite).
Since a and b have linear growth, previous expression is inferior to

$$
4\mathbb{E}(|Z|^2)+C^2E\left(\int_0^T(1+|X_s|^2)ds(32+8T)\right),
$$

where C is a linear growth constant.
Since $X\in\mathcal{A}$, previous expression is finite. □

On the other hand, if $X,Y\in\mathcal{A}, 0\le t\le u$, we have

$$
\begin{aligned}
|\Phi(X)_t-\Phi(Y)_t|^2 &\le 2\left(\int_0^u|b(s,X_s)-b(s,Y_s)|ds\right)^2\\
&+ 2\sup_{t\in[0,u]}\left|\int_0^t(a(s,X_s)-a(s,Y_s))dW_s\right|^2
\end{aligned}
$$

and consequently for $u\in[0,T]$

$$
\begin{aligned}
E\left(\sup_{t\le u}|\Phi(X)_t-\Phi(Y)_t|^2\right) &\le 2TE\left(\int_0^u|b(s,X_s)-b(s,Y_s)|^2ds\right)\\
&+ 8E\left(\int_0^u(a(s,X_s)-a(s,Y_s))^2ds\right)\\
&\le \tilde{K}E\left(\int_0^u|X_s-Y_s|^2ds\right),
\end{aligned}
$$

where $\tilde{K}$ is a constant depending on Lipschitz constants and T.
Previous expression equals

$$
\tilde{K}\int_0^u ds\ \underbrace{E(|X_s-Y_s|^2)e^{-\mathrm{A}s}}_{\le E\left(\sup_{r\le s}|X_r-Y_r|^2\right)e^{-\mathrm{A}s}}\ e^{\mathrm{A}s}\le\tilde{K}||X-Y||^2_{\sim,A}\frac{e^{\mathrm{A}u}-1}{\mathrm{A}}.
$$

Therefore

$$\begin{aligned}
||\Phi(X)-\Phi(Y)||^2_{\sim,A} &= \sup_{u\in[0,T]} E\left(\sup_{s\le u}|\Phi(X)_s-\Phi(Y)_s|^{\ 2}\right)e^{-\mathrm{A}u}\\
&\le \sup_{u\in[0,T]}\frac{\tilde K}{\mathrm{A}}\left(1-e^{-\mathrm{A}u}\right)||X-Y||^2_{\sim,A}\\
&\le \frac{\tilde K}{\mathrm{A}}||X-Y||^2_{\sim,A}.
\end{aligned}$$

Choosing $A>\tilde K$, the application Φ is a contraction on $\mathcal{A}$ equipped with the norm $||\cdot||_{\sim,A}$.

By Banach fixed point theorem there is a "unique process in $\mathcal{A}$" (X_t) such that

$$X_t = Z+\int_0^t a(s,X_s)dW_s+\int_0^t b(s,X_s)ds, \quad \forall t\in[0,T] \quad \text{Pa.s.} \tag{10.5}$$

The proof of uniqueness is omitted. ■

10.2 Vector valued Itô stochastic differential equations

It is possible to define and study Itô stochastic differential equation where the process solution propagates in $\mathbb{R}^m$. We consider the following objects.

- $W=(W^1,\dots,W^p)$ a p-dimensional $(\mathcal{F}_t)$-Brownian motion;
- a Borel function $b:\mathbb{R}_+\times\mathbb{R}^m\longrightarrow\mathbb{R}^m$, $\quad b(s,x)=(b^1(s,x),\dots,b^m(s,x))$;
- $a:\mathbb{R}_+\times\mathbb{R}^m\longrightarrow\mathbb{R}^{m\times p}$ (the set of $m\times p$ real matrices) $a(s,x)=(a_{ij}(s,x))_{1\le i\le m,1\le j\le p}$ being Borel functions;
- an $\mathcal{F}_0$-measurable $Z=(Z^1,\dots,Z^m)$ with values in $\mathbb{R}^m$.

We consider now the stochastic differential equation

$$X^i_t = Z^i+\int_0^t b^i(s,X_s)ds+\sum_{j=1}^p\int_0^t a_{ij}(s,X_s)dW^j_s,\ 1\le i\le m, \tag{10.6}$$

where by solution we intend again a continuous process $(X_t)_{t\ge0}$ with values in $\mathbb{R}^m$, adapted to $(\mathcal{F}_t)_{t\ge0}$ and such that (10.6) is verified P a.s. $\forall t\in[0,T], i\in\{1,\dots,m\}$.

(10.6) can be compactified as

$$X_t = Z + \int_0^t b(s, X_s)ds + \int_0^t a(s, X_s)dW_s. \tag{10.7}$$

Existence and uniqueness theorem can be generalized in the following way. If $x \in \mathbb{R}^m, a \in \mathbb{R}^{m\times p}$, we denote by $|x|$ the Euclidean norm of x and $|a|^2 = \sum_{1\le i\le m, 1\le j\le p} a_{ij}^2$.

Theorem 10.6 *We suppose a, b Lipschitz with linear growth and $E(|Z|^2) < \infty$. Then, there is a solution to (10.7), unique up to indistinguishability. Moreover, that solution verifies*

$$E\left(\sup_{t\le T} |X_t|^2\right) < \infty, \quad \forall T > 0.$$

Remark 10.7 *The proof is fairly similar to the real case.*

Remark 10.8 *The assumptions of Theorem 10.6 (and of course of Theorem 10.4) can be weakened assuming that coefficients a and b are Borel, locally Lipschitz and with linear growth. The idea is that the locally Lipschitz property provides local existence and uniqueness of possibly exploding solutions and the linear growth condition guarantees the non-explosion of the solution. Concerning uniqueness (under locally Lipschitz conditions) we refer for instance to [30], Theorem 38, Chapter 7, which also discusses existence of possibly exploding solutions.*

10.3 Dirichlet processes and SDEs

Next exercise motivates the introduction of Dirichlet processes for studying generalized stochastic differential equations.

Exercise 10.9 *(***Zvonkin transformation***).*

Let (W_t) be an $(\mathcal{F}_t)$-classical Brownian motion. Let $a, b : \mathbb{R} \to \mathbb{R}$ be respectively be of class C^1 and C^0, such that $a > 0$. We fix $x \in \mathbb{R}$. Let $(X_t)_{t\ge 0}$ be a solution of

$$X_t = x + \int_0^t b(X_s)ds + \int_0^t a(X_s)dW_s. \tag{10.8}$$

(i) Determine one particular strictly increasing solution h of equation

$$\frac{1}{2}a^2h'' + bh' = 0.$$

We set $\tilde{a}(x) = (ah')(h^{-1}(x))$ where h^{-1} is the inverse of h.

(ii) Verify that $Y_t = h(X_t)$ defines a solution of

$$\begin{cases} dY_t &= \tilde{a}(Y_t)dW_t \\ Y_0 &= h(x). \end{cases} \tag{10.9}$$

We suppose now that

- *a has linear growth;*
- *$y \mapsto \int_0^y \frac{b(s)}{a^2(s)}ds$ is a bounded function.*

(iii) Verify that problem (10.9) has a unique solution.

(iv) Explain why previous arguments provide uniqueness for problem (10.8).

(v) Establish existence for (10.8).

One motivation for studying Dirichlet processes comes from the irregular medium, see i.e. [34], [24, 25].
Those authors consider formally one equation of the type

$$dX_t = dW_t + \gamma'(X_t)dt, \tag{10.10}$$

where γ is only a continuous function; γ could be for instance the realization of an independent Brownian motion W. When $a = 1, b = \gamma'$, formulating stochastic differential equation (10.9), $h'(y) = e^{-2\gamma}(y)$ still makes sense. If Y is a solution of (10.9) then $X = h^{-1}(Y)$ provides in *a sense to be precised* a solution to (10.8).

We will see that problem (10.9) admits *existence and uniqueness in the sense of distribution laws*. [10, 11] make rigorous the previous statement, in particular they investigated generalized SDEs as (10.10). More recent papers are [33, 9].

Remark that X, being only a C^1 function of the semimartingale Y, will only be a Dirichlet process.

11 Change of probability et martingale representation theorems

11.1 Equivalent probabilities

Let $(\Omega, \mathcal{F})$ be a set equipped with a σ-field, let P, Q be two probabilities. We recall that Q is **absolutely continuous** with respect to P if

$$P(N) = 0 \quad \Rightarrow \quad Q(N) = 0 \quad \forall N \in \mathcal{F}.$$

We symbolize this by $Q << P$. We will say that P and Q are **equivalent** if $P << Q$ and $Q << P$; this will be denoted by $P \sim Q$.

Remark 11.1 *(Radon-Nikodym Theorem).*
$Q << P$ if and only if it exists $Z \geq 0$ on $(\Omega, \mathcal{F})$ such that

$$Q(A) = \int_A Z(\omega) dP(\omega), \forall A \in \mathcal{F}.$$

Z is called the **(Radon-Nikodym) density** *of Q with respect to P, denoted by $\frac{dQ}{dP}$.*

Lemma 11.2 *Suppose $Q << P$ with Radon-Nikodym density Z. Then $P \sim Q \Leftrightarrow P\{Z > 0\} = 1$.*

Proof. $\Leftarrow$. Soit $A \in \mathcal{F}$. It is enough to see that $Q(A) = 0 \Rightarrow P(A) = 0$. Suppose $Q(A) = \int \mathbf{1}_A Z dP = 0$. This implies $\mathbf{1}_A Z = 0$, P a.s. But $Z > 0$ a.s. yields $\mathbf{1}_A = 0$ a.s. and so $P(A) = 0$.
$\Rightarrow$ We set $C = \{Z = 0\}$. We have to show that $P(C) = 0$.
Since $P << Q$, it is enough to show that $Q(C) = 0$.
Now $Q(C) = \int Z \mathbf{1}_{\{Z=0\}} dP = 0$. ■

Exercise 11.3 *Verify that a discrete probability on the Borel sets of $\mathbb{R}$ is not absolutely continuous with respect to Lebesgue measure.*

11.2 Girsanov Theorem

Let $(\Omega, \mathcal{F}, P)$ be a probability space with $\mathcal{F} = \mathcal{F}_T$ and let $(\mathcal{F}_t)_{t\in[0,T]}$ be a filtration fulfilling the usual conditions and consider a classical $(\mathcal{F}_t)_{t\in[0,T]}$-Brownian motion $(W_t)_{t\in[0,T]}$. Next theorem is known under the name of Girsanov theorem.

Theorem 11.4 *Let $(\theta_t)_{t\in[0,T]}$ be a progressively measurable process such that $\int_0^T \theta_s^2 ds < \infty$ a.s. We set*

$$L_t = \exp\left(\int_0^t \theta_s dW_s - \frac{1}{2}\int_0^t |\theta_s|^2 ds\right).$$

We suppose that $(L_t)_{t\in[0,T]}$ is a martingale.
Then, under probability $P^\star$ of density L_T with respect to P, the process $(B_t)_{t\in[0,T]}$ defined by $B_t = W_t - \int_0^t \theta_s ds$ is an $(\mathcal{F}_t)_{0\le t\le T}$-standard Brownian motion.

Remark 11.5 *i) $P^\star(\Omega) = 1$ since (L_t) is a martingale.*

ii) Girsanov theorem admits an obvious generalization to the case of a multi-dimensional Brownian motion.

iii) For the proof, the reader can consult for instance [18], section 3.5.

Remark 11.6 *i) (***Novikov condition***).*

A sufficient condition such that $(L_t)_{0\le t\le T}$ is a martingale is the following:

$$E\left\{\exp\left(\frac{1}{2}\int_0^T \theta_t^2 dt\right)\right\} < \infty.$$

ii) It is easily possible to show that

$$dL_t = \theta_t L_t dW_t.$$

This only shows that $(L_t)_{t\in[0,T]}$ is a local martingale. **Exercise**.

The proposition below constitutes a slight generalization of Novikov condition, which also admits a multi-dimensional generalization.

Theorem 11.7 *Let W be a standard Brownian motion and θ be a progressively measurable process such that there are instants $0 = t_0 < t_1 < \ldots < t_n = T$ such that for every $i \in \{1, \ldots, n\}$*

$$E\left(\exp\left(\frac{1}{2}\int_{t_{i-1}}^{t_i} |\theta_s|^2 ds\right)\right) < \infty.$$

Then the process

$$L_t = \exp\left(\int_0^t \theta_s dW_s - \frac{1}{2}\int_0^t |\theta_s|^2 ds\right), 0 \le t \le T,$$

is a martingale. In particular $E(L_T) = 1$.

Proof. For every $i \in \{1, \ldots, n\}$, we set $\theta(i)_t = \theta_t 1_{[t_{i-1}, t_i[}$. Novikov condition is fulfilled for $\theta(i)$ and so, the process

$$L_t(\theta(i)) := \exp\left(\int_0^t \theta(i)_s dW_s - \frac{1}{2}\int_0^t |\theta(i)_s|^2 ds\right)$$

is a martingale, so

$$E(L_{t_i}(\theta(i))|\mathcal{F}_{t_{i-1}}) = L_{t_{i-1}}(\theta(i)) = 1.$$

By recurrence on $i \le n$ one can show that $E(L_{t_i}) = 1$. Indeed, for every $i \in \{1, \ldots, n\}$

$$E(L_{t_i}) = E(L_{t_{i-1}} E(L_{t_i}(\theta(i))|\mathcal{F}_{t_{i-1}})) = E(L_{t_{i-1}}).$$

Since $t_n = T$, we deduce that $E(L_T) = 1$. Since L is a non-negative local martingale, it is a supermartingale. **Exercise.** Then L is a martingale.

■

The following application is due to to Benes.

Proposition 11.8 *Let W be a standard Brownian motion and $b : [0, T] \times \mathbb{R} \to \mathbb{R}$ a Borel function such that there is a constant K such that*

$$\sup_{t \in [0,T]} |b(t, x)| \le K(1 + |x|).$$

We set

$$L_t := \exp\left(\int_0^t (b(s, W_s) dW_s - \frac{1}{2} b^2(s, W_s)) ds\right), t \in [0, T].$$

Then L is a martingale. In particular $E(L_T) = 1$.

Remark 11.9 *Let N be a standard Gaussian r.v. If $c_0 < \frac{1}{2}$, then*

$$E\left(\exp(c_0 N^2)\right) < \infty.$$

Exercise.

Proof (of Proposition 11.8). Let n be a positive integer and $t_i = \frac{Ti}{n}, i \in \{1, \ldots, n\}$. For every $i \in \{1, \ldots, n\}$,

$$\int_{t_{i-1}}^{t_i} |b(s, W_s)|^2 ds \leq 2K^2 \frac{T}{n} \left(1 + \sup_{0 \leq s \leq T} |W_s|^2\right).$$

We set $c = \frac{K^2 T}{n}$. We choose n large enough so that $cT < \frac{1}{2}$. By Remark 11.9, for $t \geq 0$

$$E(\exp(c(1 + |W_t|^2))) = \exp(c)E(\exp(ctN^2)) \leq \exp(c)E(\exp(cTN^2)) < \infty,$$

where N is a standard Gaussian random variable. Since $x \mapsto \exp(c(1 + x^2))$ is convexe then $Y_t = \exp(c(1+|W_t|^2)$ is a submartingale. By Doob inequality Proposition 4.20 we get

$$E\left(e^{c(1+\sup_{0\leq s\leq T}|W_s|^2)}\right) = E\left(\sup_{0\leq t\leq T} e^{c(1+|W_t|^2)}\right) \leq 4E\left(e^{c(1+|W_T|^2)}\right) < \infty.$$

The result follows by Theorem 11.7 with $\theta_s = b(s, W_s)$.

■

An important remark concerns the change of probability which can be established by the approximation arguments of the construction of Itô integral.

Remark 11.10 *Let $(X_t)_{t\in[0,T]}$ be an $((\mathcal{F}_t)_{t\in[0,T]}, P)$-semimartingale. Suppose that $(X_t)_{t\in[0,T]}$ is still a semimartingale with respect to an equivalent probability measure Q. Let $(H_t)_{t\geq 0}$ be such that*

$$\int_0^t H_s dX_s$$

exists with respect to P. Then

$$\int_0^t H_s dX_s$$

exists also with respect to Q and the two integrals coincide.

Example 11.11 *Let* $(X_t)_{t\in[0,T]}$ *be a solution to*

$$dX_t = a(X_t)dW_t + b(X_t)dt, \quad X_0 = x_0, \tag{11.1}$$

where a, b *are Lipschitz real functions,* $a > 0$, $\frac{b}{a}$ *is bounded and* x_0 *is a real number.*

In this case

$$dX_t = a(X_t)dB_t, \tag{11.2}$$

where

$$B_t = W_t + \int_0^t \frac{b}{a}(X_s)ds.$$

We set

$$L_t = \exp\left(-\int_0^t \theta_s dW_s - \frac{1}{2}\int_0^t |\theta_s|^2 ds\right),$$

with $\theta := \frac{b}{a}$*. Since* $\frac{b}{a}$ *is bounded, Novikov condition is verified and* $(L_t)_{t\in[0,T]}$ *is an* $(\mathcal{F}_t)_{t\geq 0}$*- martingale. Therefore under* Q *defined so that* $dQ = L_T dP$, $(B_t)_{t\in[0,T]}$ *is an* $(\mathcal{F}_t)_{t\geq 0}$*- Brownian motion.*

The local martingale $(X_t)_{t\in[0,T]}$ *is (under* Q*) the unique solution of the SDE* (11.2)*. By Theorem 10.4, it is an* $(\mathcal{F}_t)_{t\geq 0}$*- square integrable martingale (still with respect to* Q*).*

11.3 Representation theorem for Brownian martingales

We suppose here that $(\mathcal{F}_t)$ is the canonical filtration of a classical Brownian motion (W_t). We will say in this case that $(\mathcal{F}_t)$ is a Brownian filtration.

We recall that, if $(H_t)_{t\geq 0}$ is a progressively measurable process such that $E\left(\int_0^T H_t^2 dt\right) < \infty$, the process $\left(\int_0^t H_s dW_s\right)_{t\in[0,T]}$ is a (square integrable) martingale vanishing at 0.
Next theorem shows that all martingales with respect to a Brownian filtration can be represented with the help of a stochastic integral.

Theorem 11.12 *Let* $(M_t)_{0\leq t\leq T}$ *be an* $(\mathcal{F}_t)_{t\in[0,T]}$*-martingale such that* $E(M_t^2) < \infty$ *(i.e. square integrable).*
There is a progressively measurable process $(H_t)_{t\in[0,T]}$ *such that*

$$E\left(\int_0^T H_s^2 ds\right) < \infty \quad \text{and}$$

$$M_t = M_0 + \int_0^t H_s dW_s \quad \text{a.s.} \quad \forall t \in [0,T]. \tag{11.3}$$

In particular the martingale admits a continuous modification.

Remark 11.13 *This representation is only possible for martingales related to a Brownian filtration (natural filtration of a Brownian motion). Such martingales are called* **Brownian martingales**.

From the theorem, it comes out that whenever U is a $\mathcal{F}_T$-measurable square integrable random variable, then we can express it as

$$U = E(U) + \int_0^T H_s dW_s \quad \text{a.s.} \tag{11.4}$$

where (H_t) is a progressively measurable process such that $E\left(\int_0^T H_s^2 ds\right) < \infty$.

In fact, the point consists in verifying the hypotheses of Theorem 11.12 for $M_t = E(U|\mathcal{F}_t)$.

- $(M_t)_{t\geq 0}$ is square integrable because

$$\begin{aligned} E(M_t^2) &= E\Big(E(U|\mathcal{F}_t)^2\Big) \leq E\Big(E(U^2|\mathcal{F}_t)\Big) \\ &= E(U^2). \end{aligned}$$

- $(M_t)_{t\geq 0}$ is an $(\mathcal{F}_t)$-martingale since

$$E(M_t|\mathcal{F}_s) = E\Big(E(U|\mathcal{F}_t)\Big|\mathcal{F}_s\Big) = E(U|\mathcal{F}_s) = M_s, t \geq s.$$

Exercise 11.14 *Let U be an $\mathcal{F}_T$-measurable r.v. Show that there exists at most a unique couple (h_0, H) where $h_0 \in \mathbb{R}$ and H is progressively measurable with $E(\int_0^T H_s^2 ds) < \infty$ such that*

$$U = h_0 + \int_0^T H_s dW_s.$$

The uniqueness holds $dtdP$-a.e.

12 Markov property of solutions of stochastic differential equations (SDEs)

12.1 Definition

A process $(X_t)_{t\geq 0}$ with values in $\mathbb{R}^m$ is said to fulfill the **Markov property** and it is called **Markov process** when its future behaviour after an instant t only depends on X_t and not from the past before t.

Mathematically speaking, a process $(X_t)_{t\geq 0}$ verifies the Markov property with respect to a filtration $(\mathcal{F}_t)_{t\geq 0}$ to which it is adapted, if for any Borel bounded function $f : \mathbb{R}^m \longrightarrow \mathbb{R}$ and for any $s \leq t$ we have

$$E\Big(f(X_t)\Big|\mathcal{F}_s\Big) = E\Big(f(X_t)\Big|X_s\Big).$$

We will see that solutions of SDEs are Markov. We consider the solution (X_t) of an SDE of the type

$$X_t = x_0 + \int_0^t a(u, X_u)dW_u + \int_0^t b(u, X_u)du, \tag{12.1}$$

where a, b are Lipschitz with linear growth, $x_0 \in \mathbb{R}, (W_t)$ is a classical Brownian motion with respect to a filtration $(\mathcal{F}_t)$.

Remark 12.1 *(i)* $(X_t)_{t\geq 0}$ *can be considered as a r.v.* $\omega \mapsto X_t(\omega)$ *from* Ω *with values in* $C(\mathbb{R}_+)$ *(equipped with the topology of uniform convergence on each compact and of the related Borel* σ*-field). Indeed we can write*

$$X = \Phi(x, W_\cdot),$$

where is $\Phi : \mathbb{R} \times C(\mathbb{R}_+) \longrightarrow C(\mathbb{R}_+)$ *is measurable (proof omitted).*

(ii) Let $0 \leq s \leq t$. *Similar considerations as in Section 10.1 are valid for equation*

$$Y_t = Z + \int_s^t a(u, Y_u)dW_u + \int_s^t b(u, Y_u)du, \quad t \geq s, \tag{12.2}$$

Z *being* $\mathcal{F}_s$*-measurable and square integrable.*

Since Itô integral coincides with forward integral, the stochastic integral *with respect to W coincides with the integral with respect to $(\bar{W}_u)_{u\geq 0}$, where $\bar{W}_u = W_{s+u} - W_s$. We remark that $\bar{W}$ is a classical Brownian motion independent of $\mathcal{F}_s$.*

Consequently, (12.2) with $Z = y$ can also be written as

$$Y_t = y + \int_s^t a(u, Y_u) d\bar{W}_u + \int_s^t b(u, Y_u) du, \quad t \geq s.$$

Via point (i), there is $\bar{\Phi} : \mathbb{R} \times C([0, \infty[) \longrightarrow C([s, \infty[)$ such that $Y = \bar{\Phi}(y, \bar{W})$. In other words, Y is measurable with respect to y and to the process of Brownian motion increments.

(iii) *The considerations of this section extend to the case of multi-dimensional SDEs.*

For $x \in \mathbb{R}$, $(X_t^{s,x})_{t\geq s}$ will denote the unique solution $(Y_t)_{t\geq s}$ of

$$Y_t = x + \int_s^t a(u, Y_u) dW_u + \int_s^t b(u, Y_u) du, \quad t \geq s.$$

The solution $X^{0,x}$ of (12.1) will simply be denoted by X^x.

Remark 12.2 *It is possible to show the existence of a continuous modification in (s, t, x) of $(X_t^{s,x})$ still denoted by the same symbol. It is a delicate result to establish, see [22]. Similarly, $(s, t, x) \mapsto \int_s^t a(u, X_u^{s,x}) dW_u$ admits a continuous modification.*

12.2 Flow of a stochastic differential equation

This notion generalizes the flow property fulfilled in the case of ordinary differential equations.

Lemma 12.3 *Under the conditions of existence and uniqueness of Theorem 10.4, we have*

$$X_t^{0,x} = X_t^{s, X_s^x}, \quad t \geq s \geq 0, \quad x \in \mathbb{R} \quad P\text{-a.s.}$$

Proof. For every $y \in \mathbb{R}$, $\quad t \geq s \geq 0$,

$$X_t^{s,y} = y + \int_s^t b(u, X_u^{s,y}) du + \int_s^t a(u, X_u^{s,y}) dW_u \quad \text{a.s.} \tag{12.3}$$

Since left and right members are continuous in (s, t, y), we have (12.3) for every $t \geq s \geq 0, \quad y \in \mathbb{R}$ P-a.s.
In fact, two continuous processes (random fields) which are modification one of the other are indeed indistinguishable.
Finally, for $y = X_s^x$

$$X_t^{s,X_s^x} = X_s^x + \int_s^t b\left(u, X_u^{s,X_s^x}\right) du + \int_s^t a\left(u, X_u^{s,X_s^x}\right) dW_u. \qquad (12.4)$$

Now X_t^x is also solution of previous equation because, if $t \geq s$,

$$\begin{aligned} X_t^x &= x + \int_0^t b(u, X_u^x)du + \int_0^t a(u, X_u^x)dW_u \\ &= \underbrace{x + \int_0^s b(u, X_u^x)du + \int_0^s a(u, X_u^x)dW_u}_{X_s^x} + \int_s^t b(u, X_u^x)du \\ &+ \int_s^t a(u, X_u^x)dW_u. \end{aligned}$$

We recall that equation (12.4), which is (12.2) with initial condition $Z = X_s^x$, has a unique solution. This implies the result. ■

The Markov property assumes consequently the following form.

Theorem 12.4 *Let $(X_t)_{t \geq 0}$ be a solution of (12.1). Then it has the Markov property with respect to the filtration $(\mathcal{F}_t)$. More precisely, for every bounded Borel function f, we have*

$$E\Big(f(X_t)\Big|\mathcal{F}_s\Big) = \phi(X_s), \qquad (12.5)$$

where

$$\phi(y) = E\left(f\left(X_t^{s,y}\right)\right).$$

Remark 12.5 *We often express previous equality as*

$$E\left(f(X_t)\Big|\mathcal{F}_s\right) = E\left(f\left(X_t^{s,y}\right)\right)\Big|_{y=X_s}.$$

Proof. The flow property says that

$$X_t^x = X_t^{s,X_s^x}. \qquad (12.6)$$

By Remark 12.1, there is $\Phi : \mathbb{R} \times C(\mathbb{R}_+) \longrightarrow C([s, +\infty[)$ such that

$$X^{s,y} = \Phi(y, W_{s+u} - W_s; u \geq 0).$$

Therefore, by (12.5), for fixed s, t

$$X_t^x = \Phi(X_s^x, W_{s+u} - W_s; u \geq 0),$$

with X_s^x being $(\mathcal{F}_s)$-measurable and $(W_{s+u} - W_s)$ independant of $\mathcal{F}_s$.
We apply the well-known *freezing* Theorem 2.13 for the conditional expectation. So the left-hand side of (12.5) equals

$$\begin{aligned} E\Big\{f\Big(\Phi(X_s^x, W_{s+u} - W_s \quad ; \quad u \geq 0)\Big)\Big|\mathcal{F}_s\Big\} & \\ &= E\left\{f\Big(\Phi(y, W_{s+u} - W_s; u \geq 0)\Big)\right\}\Big|_{y=X_s^x} \\ &= E\left(f\left(X_t^{s,y}\right)\right)\Big|_{y=X_s^x}. \end{aligned}$$

■

The previous result can be generalized to the case when Φ depends on the whole path of the diffusion after time s.

Theorem 12.6 *Let $(X_t)_{t\geq 0}$ be a solution to (12.1) and $r(s,x)$ be a non-negative measurable fonction. If $t > s$ we have*

$$E\left(e^{-\int_s^t r(u,X_u)du} f(X_t)\Big|\mathcal{F}_s\right) = \phi(X_s) \quad \text{P a.s.}$$

with

$$\phi(y) = E\left(e^{-\int_s^t r(u,X_u^{s,y})du} f\left(X_t^{s,y}\right)\right).$$

We also express that equality in the form

$$E\left(e^{-\int_s^t r(u,X_u)du} f(X_t)\Big|\mathcal{F}_s\right) = E\left(e^{-\int_s^t r(u,X_u^{s,y})du} f\left(X_t^{s,y}\right)\right)\Big|_{y=X_s}$$

Remark 12.7 *We can in fact establish a more general result than the one previously stated. Omitting technical details, we can affirm that, given a function Φ of the whole path of X_t after s, we have*

$$E\left(\Phi\Big(X_t^x; t \geq s\Big)\Big|\mathcal{F}_s\right) = E\left(\Phi\Big(X_t^{s,y}, t \geq s\Big)\right)\Big|_{y=X_s} \quad \text{P a.s.}$$

12.3 Infinitesimal generator of a diffusion

Suppose $a, b : \mathbb{R} \to \mathbb{R}$ Lipschitz.
We denote by $(X_t)_{t\geq 0}$ a solution of

$$dX_t = b(X_t)dt + a(X_t)dW_t. \tag{12.7}$$

Proposition 12.8 *We denote by $\mathcal{A}$ the differential operator that to a function g of class C^2 associates*

$$(\mathcal{A}g)(x) = \frac{a^2(x)}{2}g''(x) + b(x)g'(x).$$

Then, if $f \in C^2(\mathbb{R})$ with bounded first order derivative then $M_t = f(X_t) - \int_0^t \mathcal{A}f(X_s)ds$ is an $(\mathcal{F}_t)$-martingale.

Proof. Exercise. ∎

Remark 12.9 *We denote by $(X_t^x)_{t\geq 0}$ the solution of the SDE (12.7) such that $X_0^x = x$.*
Let $f \in C^2(\mathbb{R})$ with bounded derivatives. Let K_f be a constant bounding the first and second derivatives and K such that $|b(x)| + |a(x)| \leq K(1 + |x|)$. Theorem 10.4 of existence and uniqueness for SDEs says that

$$E\left(\sup_{s\leq T}\left|\mathcal{A}f\left(X_s^x\right)\right|\right) \leq K'_f\left(1 + E\left(\sup_{s\leq T}\left|X_s^x\right|^2\right)\right) < \infty,$$

where K'_f is another constant depending on K and K_f.

From Proposition 12.8, we deduce that

$$E\left(f\left(X_t^x\right)\right) = f(x) + E\left(\int_0^t \mathcal{A}f\left(X_s^x\right)ds\right).$$

Consequently, we can apply Lebesgue dominated convergence theorem since $x \mapsto \mathcal{A}f(x)$ and $s \mapsto X_s^x$ are continuous functions. We deduce that

$$\frac{d}{dt}E\left(f\left(X_t^x\right)\right)_{t=0} = \lim_{t\longrightarrow 0} E\left(\frac{1}{t}\int_0^t \mathcal{A}f\left(X_s^x\right)ds\right) = \mathcal{A}f(x).$$

The differential operator $\mathcal{A}$ is called **infinitesimal generator** of the diffusion.

Proposition 12.8 extends to the case where the coefficients of the SDE (12.7) depend on time.

Proposition 12.10 *Let $(t,x) \mapsto u(t,x)$ of class $C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$. We suppose moreover that first order derivative with respect to x is bounded. Then the process*

$$M_t = u(t, X_t) - \int_0^t \left(\frac{\partial u}{\partial t} + \mathcal{A}_s u \right)(s, X_s)ds$$

is a martingale, where $\mathcal{A}_s$ is the operator acting on variable x defined by

$$(\mathcal{A}_s f)(x) = \frac{a^2(s,x)}{2} f''(x) + b(s,x)f'(x).$$

Proof. It is analogous to the one of Proposition 12.8.
We use this time the time-dependent Itô formula Proposition 9.20. ■

In reality, we need a more general result.

Proposition 12.11 *Under assumptions of Proposition 12.10, if $r : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}, \quad (t,x) \mapsto r(t,x)$ is bounded and continuous, the process*

$$M_t = e^{-\int_0^t r(s,X_s)ds} u(t, X_t) - \int_0^t e^{-\int_0^s r(v,X_v)dv} \left(\frac{\partial u}{\partial t} + \mathcal{A}_s u - ru \right)(s, X_s)ds,$$

is a martingale.

Proof. That proposition can be established differentiating the product

$$e^{-\int_0^t r(s,X_s)ds} u(t, X_t), \tag{12.8}$$

through integration by parts formula Proposition 9.20 iii); so expression (12.8) becomes

$$\begin{aligned} u(0, \underbrace{X_0}_{x_0}) \quad &- \quad \int_0^t u(s,X_s) e^{-\int_0^s r(v,X_v)dv} r(s,X_s)ds \\ &+ \quad \int_0^t e^{-\int_0^s r(v,X_v)dv} du(s,X_s) + \underbrace{[\quad]}_{0}. \end{aligned}$$

Subsequently, we apply Itô formula to process $u(t, X_t)$. ■

Previous result generalizes to the case of vector valued diffusions. Consider the stochastic differential equation

$$dX_t^i = b^i(t, X_t)dt + \sum_{j=1}^p a_{ij}(t, X_t)dW_t^j, \quad i = 1, \ldots, m. \tag{12.9}$$

. We make the assumptions of existence and uniqueness of Section 10.2. For every t, we introduce the operator $\mathcal{A}_t$ which to a function $f \in C^2(\mathbb{R}^m;\mathbb{R})$ associates the function

$$\mathcal{A}_t f(x) = \frac{1}{2}\sum_{i,j=1}^{n} \ell_{ij}(t,x)\frac{\partial^2 f}{\partial x_i \partial x_j}(x) + \sum_{j=1}^{n} b_j(t,x)\frac{\partial f}{\partial x_j}(x),$$

where $(\sigma_{ij}(t,x))$ is the matrix defined by

$$\sigma_{ij}(t,x) = \sum_{k=1}^{p} a_{ik}(t,x)a_{jk}(t,x).$$

With matrix notations, we will have $\Sigma(t,x) = (\sigma_{ij}(t,x))$ with $\Sigma(t,x) = A(t,x)A^\top(t,x)$, where $A^\top(t,x)$ is the transposed matrix of $A(t,x) = (a_{ij}(t,x))$.

Proposition 12.12 *Let (X_t) be a solution of system (12.9), $u : \mathbb{R}_+ \times \mathbb{R}^m \longrightarrow \mathbb{R}, \quad (t,x) \mapsto u(t,x)$ of class $C^{1,2}$. We suppose moreover that the first order derivatives with respect to x are bounded.*
Let $r : \mathbb{R}_+ \times \mathbb{R}^m \longrightarrow \mathbb{R}, \quad (t,x) \mapsto r(t,x)$ be a bounded continuous function. Then the process

$$M_t = e^{-\int_0^t r(v,X_v)dv}u(t,X_t) - \int_0^t e^{-\int_0^s r(v,X_v)dv}\left(\frac{\partial u}{\partial t} + \mathcal{A}_s u - ru\right)(s,X_s)ds$$

is a martingale.

Proof. It follows applying integration by parts and multi-dimensional Itô formula. ■

Remark 12.13 *The differential operator $\dfrac{\partial}{\partial t} + \mathcal{A}_t$ is called the Dynkin operator of the diffusion.*

Example 12.14 *We consider the p-dimensional Brownian motion issued from x, obtained in (12.9) setting $p = m, \quad b^i = 0, \quad a_{ij} = \delta_{ij}$. In this case, the generator is $\mathcal{A}_t \equiv \frac{1}{2}\Delta$.*

12.4 Parabolic partial differential equations and expectation evaluations

We consider a diffusion $(X_t)_{t\geq 0}$ with values in $\mathbb{R}^m$, $f(x)$ from $\mathbb{R}^m$ to $\mathbb{R}$ non-negative Borel, $r(s,x)$ a real continuous and bounded function. One of the classical problems of the theory of financial mathematics consists in evaluating

$$V_s = E\left(e^{-\int_s^T r(t,X_t)dt} f(X_T) \Big| \mathcal{F}_s\right).$$

As we have seen at the beginning of this chapter for the case $m = 1$, we have equally

$$V_s = F(s, X_s),$$

where

$$F(s,x) = E\left(e^{-\int_s^T r(t,X_t^{s,x})dt} f\left(X_T^{s,x}\right)\right), \tag{12.10}$$

$X^{s,x}$ being the unique solution to system (12.9) issued from x at time s.

We can extend this result and show that, if r is a function only depending on x (time-homogeneity), we have

$$E\left\{e^{-\int_s^{s+t} r\left(X_u^{s,y}\right)du} f\left(X_{s+t}^{s,y}\right)\right\} = E\left\{e^{-\int_0^t r\left(X_u^{0,y}\right)du} f\left(X_t^{0,y}\right)\right\}.$$

In this case the equality in Theorem 12.6 can be rewritten as follows:

$$E\left\{e^{-\int_s^t r(X_u)du} f(X_t) \Big| \mathcal{F}_s\right\} = E\left\{e^{-\int_0^{t-s} r\left(X_u^{0,y}\right)du} f\left(X_{t-s}^{0,y}\right)\right\}\Big|_{y=X_s}.$$

Next result allows to relate F defined in (12.10) to the solution of a parabolic partial differential equation.

Theorem 12.15 *(***Feynmann-Kac***) Let $u : \mathbb{R}_+ \times \mathbb{R}^m \longrightarrow \mathbb{R}$ in (s,x) of class $C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^m)$.*
We suppose moreover that the first order derivatives with respect to x are bounded.
If

$$\left\{\begin{array}{rcll} u(T,x) & = & f(x), & \forall x \in \mathbb{R}^m \\ \left(\dfrac{\partial u}{\partial s} + \mathcal{A}_s u - ru\right)(s,x) & = & 0, & \forall (s,x) \in]0,T[\times \mathbb{R}^m, \end{array}\right.$$

then

$$u(s,x) = F(s,x) := E\left(e^{-\int_s^T r(t,X_t^{s,x})dt} f\left(X_T^{s,x}\right)\right).$$

Proof. We prove equality $u(s,x) = F(s,x)$ for $s = 0$.
By Proposition 12.12, we know that the process

$$M_t = e^{-\int_0^t r(v,X_v^{0,x})dv} u(t, X_t^{0,x})$$

is a martingale. Writing $E(M_0) = E(M_T)$ we obtain

$$\begin{aligned} u(0,x) &= E\left(e^{-\int_0^T r(v,X_v^{0,x})dv} u(T, X_T^{0,x})\right) \\ &= E\left(e^{-\int_0^T r(v,X_v^{0,x})dv} f(X_T^{0,x})\right), \end{aligned}$$

since $u(T,x) = f(x)$. The proof is similar when $s > 0$. ■

Remark 12.16 *In order to calculate $F(s,x)$ given in (12.10), previous theorem suggests to find out the solution of the following problem.*

$$\begin{cases} \dfrac{\partial u}{\partial s} + \mathcal{A}_s u - ru = 0 & \text{in} \quad]0,T] \times \mathbb{R}^m \\ u(T,x) = f(x), & \forall x \in \mathbb{R}^m. \end{cases} \tag{12.11}$$

Remark 12.17 *When $r = 0$ and $\mathcal{A}_s$ is the laplacian, the following property holds. If*

$$\begin{cases} \dfrac{\partial u}{\partial s} + \dfrac{1}{2}\Delta u = 0, & \text{in} \quad]0,T] \times \mathbb{R}^m, \\ u(T,x) = f(x), & \forall x \in \mathbb{R}^m, \end{cases} \tag{12.12}$$

we have

$$u(s,x) = E\left(f(x + W_{T-s})\right),$$

$(W_s) = (W_s^1, \ldots, W_s^p)$ being a classical Brownian motion.

Problem (12.11) consists in one equation of parabolic type with terminal condition (whenever the function $u(T,\cdot)$ is given).

(12.11) will be well-posed, if we formulate a suitable functional framework; there we can provide existence (and uniqueness) theorems.
Moreover, we will be able to affirm that the solution of (12.11) equals F if we can prove that this solution is regular enough in order to apply Proposition 12.12. This kind of result can be generally obtained under an ellipticity

condition for operator $\mathcal{A}_t$, which has the form

$$\begin{aligned} &\exists c > 0, \quad \forall (t,x) \in [0,T] \times \mathbb{R}^m \\ &\forall (\xi_1, \ldots, \xi_n) \in \mathbb{R}^m : \quad \sum_{i,j} a_{ij}(t,x)\xi_i\xi_j \geq c \sum_{i=1}^{n} |\xi_i|^2 \end{aligned} \tag{12.13}$$

and some regularity assumptions on b and a.

12.5 Partial differential equations with boundary conditions and expectation calculation

In the sequel of this chapter, we will have a process $(X_t)_{t\geq 0}$ which is solution of a stochastic differential equation with coefficients a, b which do not depend on time. Let also consider $r : \mathbb{R} \to \mathbb{R}$ bounded and continuous.
So we have

$$dX_t = a(X_t)dW_t + b(X_t)dt. \tag{12.14}$$

We consider the differential operator

$$\mathcal{A}g(x) = \frac{1}{2}a^2(x)g''(x) + b(x)g'(x). \tag{12.15}$$

We denote $\tilde{\mathcal{A}}g(x) = \mathcal{A}g(x) - r(x)g(x)$. Equation (12.11) can be written as

$$\left\{ \begin{array}{rcll} \dfrac{\partial u}{\partial t}(t,x) + \tilde{\mathcal{A}}u(t,x) & = & 0, & \text{in }]0,T] \times \mathbb{R} \\ u(T,x) & = & f(x), & \forall x \in \mathbb{R}. \end{array} \right. \tag{12.16}$$

If we state Problem (12.16) not any more on the whole real line, but on $\mathcal{O} =]x_0, x_1[$, we need to impose boundary conditions on x_0 and x_1.
We will be more particularly interested to the case when we impose zero boundary conditions (of Dirichlet type). Let $f : \mathcal{O} \to \mathbb{R}$. We try in this case to solve

$$\left\{ \begin{array}{rclll} \dfrac{\partial u}{\partial t}(t,x) + \tilde{\mathcal{A}}u(t,x) & = & 0, & & \text{in }]0,T] \times \mathcal{O} \\ u(t,x_0) & = & u(t,x_1) & = 0, & \forall t \leq T \\ u(T,x) & = & f(x) & & \forall x \in \mathcal{O}. \end{array} \right. \tag{12.17}$$

A regular solution of (12.17) can so be interpreted with the help of the diffusion equation (12.14). We denote by $X^{s,x}$ the solution of (12.14) issued from x at time s.

Theorem 12.18 *Let* $u : \mathbb{R}_+ \times \bar{\mathcal{O}} \longrightarrow \mathbb{R}$ *be a function in* (t,x) *of class* $C^{1,2}([0,T] \times \bar{\mathcal{O}})$.
We suppose moreover u *to be solution of (12.17).*
Then

$$u(s,x) = E\left\{ \mathbf{1}_{\{\forall t \in [s,T], X_t^{s,x} \in \mathcal{O}\}} e^{-\int_s^T r(X_t^{s,x})dt} f(X_T^{s,x}) \right\}.$$

Proof. We check the result when $s = 0$, the proof being similar in the other cases.
We can extend u from $[0,T] \times \bar{\mathcal{O}}$ to $[0,T] \times \mathbb{R}$ conserving the $C^{1,2}$ character of u on $[0,T] \times \mathbb{R}$. We continue symbolizing u such a prolongation. Through Proposition 12.11, we know that the process

$$\begin{aligned} M_t &= e^{-\int_0^t r(X_v^{0,x})dv} u(t, X_t^{0,x}) \\ &- \int_0^t \exp\left\{ -\int_0^v r(X_\alpha^{0,x}) d\alpha \right\} \left(\frac{\partial u}{\partial t} + \mathcal{A}u - ru \right)(v, X_v^{0,x}) dv \end{aligned}$$

is a martingale. Moreover, we set

$$T^x = \begin{cases} \inf\{0 \le t \mid X_t^{0,x} \notin \mathcal{O}\} & : \text{ if } \{0 \le t \mid X_t^{0,x} \notin \mathcal{O}\} \neq \emptyset \\ \infty & : \text{ otherwise.} \end{cases}$$

T^x is a stopping time because $T^x = T^x_{x_0} \wedge T^x_{x_1}$, where

$$T^x_\ell = \inf\{t \ge 0 \;:\; X_t^{0,x} = \ell\}$$

and T^x_ℓ are stopping times, see Proposition 4.11.
We set $\tau^x = T^x \wedge T$; it is a bounded stopping time.

Applying the Doob stopping time Theorem 4.9 between 0 and τ^x, we obtain $E(M_0) = E(M_{\tau^x})$; taking into account that, if $t \in]0, \tau^x[$,

$$\left(\frac{\partial u}{\partial t} + \mathcal{A}u - ru \right)(t, X_t^{0,x}) = 0,$$

we have

$$\begin{aligned} u(0,x) &= E\left\{ e^{-\int_0^{\tau^x} r(X_v^{0,x})dv} u(\tau^x, X_{\tau^x}^{0,x}) \right\} \\ &= E\left\{ \mathbf{1}_{\{\forall t \in [0,T], X_t^{0,x} \in \mathcal{O}\}} e^{-\int_0^T r(X_v^{0,x})dv} u(T, X_T^{0,x}) \right\} \\ &+ E\left\{ \mathbf{1}_{\{\exists t \in [0,T], X_t^{0,x} \notin \mathcal{O}\}} e^{-\int_0^{\tau^x} r(X_v^{0,x})dv} u(\tau^x, X_{\tau^x}^{0,x}) \right\}. \end{aligned}$$

Now, on the event $\{\exists t \in [0,T], X_t^{0,x} \notin \mathcal{O}\}, u(\tau^x, X_{\tau^x}^{0,x}) = u(T^x, X_{T^x}^{0,x}) = 0$. On the other hand $f(x) = u(T,x)$, so that

$$u(0,x) = E\left\{\mathbf{1}_{\{\forall t \in [0,T], X_t^{0,x} \in \mathcal{O}\}} e^{-\int_0^T r(X_v^{0,x})dv} f(X_T^{0,x})\right\}.$$

This proves the result for $s = 0$. ■

13 Stochastic differential equations with non-Lipschitz coefficients

Let $a : [0,T] \times \mathbb{R}^m \to \mathbb{R}^{m \times p}, b : [0,T] \times \mathbb{R}^m \to \mathbb{R}^m$ Borel functions.

13.1 Generalities

The study of stochastic differential equations is much richer than the study of ordinary differential equations which are well-posed essentially when the coefficients are Lipschitz with very few exceptions.

We consider first the following well-known problem with $m = p = 1$.

$$\begin{cases} \dot{x}(t) & = & \sqrt{x(t)} \\ x(0) & = & 0. \end{cases}$$

This problem is not well-posed since the family of solutions is provided by $x \equiv 0$ and

$$x(t) = \begin{cases} \frac{(t-C)^2}{4} & : & t \geq C \\ 0 & : & t \in [0,C], \end{cases}$$

where C is a general constant. It is possible to show that previous system perturbed by a white noise (even with small intensity) provides uniqueness as we will discuss later. Some authors have examined the asymptotic behavior of the unique solution when one switches off the noise. Among the first authors who have examined this phenomenon when the noise goes to zero are those of [2].

We come back now to the stochastic case.

Let $(\Omega, \mathcal{F}, P)$ be a probability space, a usual filtration $(\mathcal{F}_t)_{t\geq 0}$, an $(\mathcal{F}_t)_{t\geq 0}$-classical p-dimensional Brownian motion $W = (W_t^1, \ldots, W_t^p)_{t\geq 0}$. Let $T > 0$.

Generally speaking, we will be interested in a SDE of Itô type as follows:

$$dX_t = a(t, X_t)dW_t + b(t, X_t)dt \qquad (E(a,b)). \tag{13.1}$$

Definition 13.1 *A stochastic process $(X_t)_{t\in[0,T]}$ is said to be* **solution** *of (13.1) if the following holds.*

i) (X_t) *is* $(\mathcal{F}_t)_{t\in[0,T]}$*-adapted and measurable.*

ii) $\int_0^T (|b(t,X_t)| + |a|^2(t,X_t))dt < \infty$ *a.s.*

iii) $X_t = X_0 + \int_0^t a(s,X_s)dW_s + \int_0^t b(s,X_s)ds, \forall t \in [0,T]$, *P-a.s.,*

where the stochastic integral above is defined componentwise as in (10.7)*. In an obvious way, we can define the notion of solution $(X_t)_{t\geq 0}$ on the whole real line.*

Remark 13.2 *Item iii) means that the left and right-hand side are indistinguishable. If X is a continuous process then the equality holds if for any t $X_t = X_0 + \int_0^t a(s,X_s)dW_s + \int_0^t b(s,X_s)ds$ a.s.*

Definition 13.3 *(***Strong existence***). If for any probablity space $(\Omega, \mathcal{F}, P)$, an $(\mathcal{F}_t)_{t\geq 0}$-classical Brownian motion $(W_t)_{t\geq 0}$-classical Brownian motion, with respect to a filtration $(\mathcal{F}_t)_{t\geq 0}$ (fulfilling the usual conditions), γ_0, $\mathcal{F}_0$-measurable square integrable random variable, there is a process $(X_t)_{t\geq 0}$ solution to E(a,b) with $X_0 = \gamma_0$ a.s., we will say that equation E(a,b) admits* **strong existence**.

Definition 13.4 *(***Pathwise uniqueness***). We will say that equation $E(a,b)$ admits pathwise uniqueness if the following property is fulfilled.*

Let $(\Omega, \mathcal{F}, P)$ be a probability space, a usual filtration $(\mathcal{F}_t)_{t\geq 0}$, an $(\mathcal{F}_t)_{t\geq 0}$-classical Brownian motion $(W_t)_{t\geq 0}$. Let γ_0 be a (square integrable) $\mathcal{F}_0$-r.v. If two processes $X, \tilde{X}$ are two solutions of E(a,b), such that $X_0 = \tilde{X}_0 = \gamma_0$ a.s. then X and $\tilde{X}$ are indistinguishable.

Definition 13.5 *(***Existence in law** *or* **Weak existence***). Let ν be a Borel probability law (on $\mathbb{R}^m$). We will say that $E(a,b;\nu)$ admits weak existence if there is a probability space $(\Omega,\mathcal{F},P)$, a usual filtration $(\mathcal{F}_t)_{t\geq 0}$, an $(\mathcal{F}_t)_{t\geq 0}$-p-dimensional classical Brownian motion $(W_t)_{t\geq 0}$ and a process $(X_t)_{t\geq 0}$ solution of $E(a,b)$ where ν is the law of X_0.*

We say that $E(a,b)$ admits weak existence if $E(a,b;\nu)$ admits weak existence for every ν.

Definition 13.6 *(***Uniqueness in law***). Let ν be a probability law (on $\mathbb{R}^m$). We say that $E(a,b;\nu)$ has a* **unique solution in law** *if the following holds. Let $(\Omega,\mathcal{F},P)$ (resp. $(\tilde{\Omega},\tilde{\mathcal{F}},\tilde{P})$,) be a probability space; consider an usual filtration $(\mathcal{F}_t)_{t\geq 0}$ (resp. $(\tilde{\mathcal{F}}_t)_{t\geq 0}$), an $(\mathcal{F}_t)_{t\geq 0}$-classical Brownian motion $(W_t)_{t\geq 0}$ (resp. an $(\tilde{\mathcal{F}}_t)_{t\geq 0}$- classical Brownian motion $(\tilde{W}_t)_{t\geq 0}$), and a process $(X_t)_{t\geq 0}$ (resp. $(\tilde{X}_t)_{t\geq 0}$) solution of $E(a,b)$, such that both the law of X_0 and $\tilde{X}_0$ are identical to ν. Then X and $\tilde{X}$ have the same law as r.v. with values in $E = C(\mathbb{R}_+)$ (or $C[0,T]$).*

We say that $E(a,b)$ has a unique solution in law if $E(a,b;\nu)$ has a unique solution in law for every ν. Sometimes it is reasonnable to suppose ν is a probability measure having a second moment.

Proposition 13.7 *(Yamada-Watanabe). The two properties are valid.*

i) Pathwise uniqueness imply uniqueness in law.

ii) Weak existence and pathwise uniqueness imply strong existence.

A version can be stated for $E(a,b;\nu)$ where ν is a fixed probability law.

Proof. See [18] Proposition 5.3.20 and Corollary of this result. ■

In Section 7., we considered in some details the case where a and b are Lipschitz.

Definition 13.8 *A function $\gamma : [0,T]\times\mathbb{R}^m \to \mathbb{R}^\ell$ (resp. $\gamma : \mathbb{R}_+\times\mathbb{R}^m \to \mathbb{R}^\ell$) is said to be* **locally Lipschitz** *(with respect to x uniformly with respect to* *t), if for every $K > 0$ (resp. for every $T > 0, K > 0$) $\gamma_{|[0,T]\times[-K,K]^m}$ is Lipschitz (with respect to x uniformly with respect to t).*

We come back to notations introduced at the beginning of the chapter.

Remark 13.9 *i) If a and b are Lipschitz with linear growth, E(a,b) admits a unique solution, of course $(\mathcal{F}_t)_{t\geq 0}$- adapted. This means that E(a,b) admits strong existence and pathwise uniqueness. This result can be deduced from Theorem 10.4.*

ii) If a and b are only locally Lipschitz, it is possible to show existence until some suitable (explosion) stopping time. The local Lipschitz property guarantees however pathwise uniqueness.

iii) Condition i) on a and b may be replaced by the following condition:

a, b locally Lipschitz with linear growth.

A simple but very efficient inequality in the theory of ordinary differential equations, also useful in stochastic differential equations is Gronwall's lemma (inequality) whose first version was established by Haken Gronwall in 1919.

Lemma 13.10 *(Gronwall). Let $g : [0;T] \to \mathbb{R}$ be an integrable Borel function such that there exist constants $a, b \geq 0$,*

$$g(t) \leq a + b\int_0^t g(s)ds, 0 \leq t \leq T. \tag{13.2}$$

Then $g(t) \leq ae^{bt}, t \in [0,T]$.

Proof.

We set $G(t) = a + b\int_0^t g(s)ds$. Then by assumption $g(t) \leq G(t)$.

(i) If g is continuous then G is differentiable and

$$(e^{-bt}G(t))' = -be^{-bt}G(t) + e^{-bt}G'(t) = -be^{-bt}G(t) + be^{-bt}g(t) \leq 0. \tag{13.3}$$

So $t \mapsto e^{-bt}G(t)$ is non-increasing and

$$e^{-bt}g(t) \leq e^{-bt}G(t) \leq G(0) = a,$$

which proves the result in this case.

(ii) Suppose now that g is only Lebesgue integrable. Then G is continuous and verifies by assumption

$$G(t) = a + b\int_0^t g(s)ds \leq a + b\int_0^t G(s)ds.$$

But now the continuous function G fulfills (13.2) so we can apply the first part of the proof. Finally

$$g(t) \leq G(t) \leq ae^{bt}, t \in [0,T].$$

■

13.2 Existence and uniqueness in law: an example

We start with an example where $E(a,b)$ *does not admit* pathwise uniqueness, even though it admits uniqueness in law.

Before this, we need to state an important result due to Paul Lévy, called **Lévy characterization of Brownian motion**.

Theorem 13.11 *Let $(M_t)_{t\geq 0}$ be an $(\mathcal{F}_t)_{t\geq 0}$- continuous local martingale such that $M_0 = 0$. Then $(M_t)_{t\geq 0}$ is an $(\mathcal{F}_t)_{t\geq 0}$- classical Brownian motion if and only if $[M,M]_t \equiv t$.*

Proof. See [31] Theorem 3.6. ■

We fix $m = p = 1$. We set $b(x) = 0$ and

$$a(x) = sign^+(x) = \begin{cases} 1 & : \quad x \geq 0 \\ -1 & : \quad x < 0. \end{cases}$$

Equation $E(a,b;\delta_0)$ admits uniqueness in law. In fact, every solution X is a local martingale, vanishing at zero with quadratic variation $[X]_t = t$. So by Lévy characterization theorem X is a classical Brownian motion.

Remark 13.12 *A solution X of equation E(a,b;δ_0) is also a solution to*

$$X_t = \int_0^t sign^-(X_s)dW_s,$$

and

$$X_t = \int_0^t sign(X_s)dW_s,$$

where

$$sign^-(x) = \begin{cases} 1 & : \ x > 0 \\ -1 & : \ x \leq 0 \end{cases}$$

and

$$sign(x) = \begin{cases} 1 & : \ x > 0 \\ -1 & : \ x < 0 \\ 0 & : \ x = 0. \end{cases}$$

In fact, since X is a Brownian motion then $\int_0^t 1_{\{X_s=0\}} dW_s = 0$ for any $t \geq 0$. In fact, using Fubini's theorem and the isometry of stochastic integral,

$$E\left(\int_0^t 1_{\{X_s=0\}} ds\right) = \int_0^t P\{X_s = 0\} ds = 0,$$

since the law of X_s admits a density and therefore it is non-atomic.

Later we will show that $E(a, 0; \delta_0)$ admits weak existence. Let now $(\Omega, \mathcal{F}, P)$ be a probability space, an $(\mathcal{F}_t)_{t\geq 0}$- classical Brownian motion (W_t) with respect to a usual filtration and $(X_t)_{t\geq 0}$ be a solution of $E(a, b; \delta_0)$. Since X is a solution to $X_t = \int_0^t sign^-(X_s)dW_s, t \geq 0$, then $\tilde{X}_t = -X_t$ verifies of $\tilde{X}_t = \int_0^t sign^+(\tilde{X}_s)dW_s, t \geq 0$. Since X and $\tilde{X}$ are both solutions of $E(a, b; \delta_0)$, then $E(a, b; \delta_0)$ does not admit pathwise uniqueness.

13.3 Existence and uniqueness in law in the multi-dimensional case.

We recall here the known results about existence and uniqueness in law in the multi-dimensional case. Let $a : [0, T] \times \mathbb{R}^m \to \mathbb{R}^{m\times p}, b : [0, T] \times \mathbb{R}^m \to \mathbb{R}^m$ be two Borel functions.

We start introducing the so called *Non-degeneracy condition* for a given matrix function σ.

Assumption 13.13 *A function* $\sigma : [0,T] \times \mathbb{R}^m \to \mathbb{R}^{m\times m}$ *is said to fulfill the* **Non-degeneracy condition** *if*

$$\exists C > 0, \forall (t,x) \in [0,T]\times\mathbb{R}^m, \forall(\xi_1, \cdots, \xi_m) \in \mathbb{R}^m : \sum_{i,j=1}^{m} \sigma_{ij}(t,x)\xi_i\xi_j \geq C \sum_{i=1}^{m} |\xi_i|^2. \tag{13.4}$$

The first result of existence in law, that we state, is taken from the Theorem 1 in the Section 2.6 in [20].

Theorem 13.14 *We suppose* $m = p$*,* a*,* b *bounded and Borel measurable with respect to* (t,x) *such that* a *fulfills Assumption 13.13. Then* $E(a,b)$ *admits existence in law.*

In Exercise 7.3.3 in [36], we can find a theorem for the uniqueness in law only for $m = p = 1$.

Theorem 13.15 *We suppose* $m = p = 1$ *and* a*,* b *bounded and Borel measurable with respect to* (t,x) *such that for each compact* $K \subset \mathbb{R}$*, there is a constant* $C = C(K)$ *such that* $a^2(t,x) \geq C, \forall(t,x) \in [0,T]\times K$*. Then* $E(a,b)$ *admits existence and uniqueness in law.*

Under the non-degeneracy condition (13.13) on $aa^\top$ and a minimal regularity conditions we have the following well-statement result.

Theorem 13.16 *We suppose* a*,* b *bounded and Borel measurable such that* $\sigma = aa^\top$ *fulfills Assumption 13.13 and*

$$\lim_{y\to x} \sup_{0\leq s\leq T} |\sigma(s,y) - \sigma(s,x)| = 0, \; x \in \mathbb{R}^m. \tag{13.5}$$

Then the problem $E(a,b)$ *admits weak existence and uniqueness in law.*

Remark 13.17 *If* $aa^\top$ *is continuous,* (13.5) *is verified.*

A last interesting result is Theorem 12.2.3 in [36] about existence in law.

Theorem 13.18 *We suppose* a*,* b *continuous bounded. Then* $E(a,b)$ *admits existence in law.*

We remark that in Theorem 13.18, a could be also degenerate. A consequence of this theorem is the following.

Theorem 13.19 *We suppose a, b continuous with linear growth. Let ν be a Borel probability measure on $\mathbb{R}^m$ such that $\int_{\mathbb{R}^d} |x|^\gamma \nu(dx) < \infty$ for some $\gamma > 2m$. Then $E(a, b)$ admits existence in law.*

Proof (Sketch). We proceed according to the following steps.

(i) We truncate the coefficients a and b defining, for each $N > 0$, $a^N(t, x) = (a_{ij}^N(t, x))$ where $a_{ij}^N(t, x) = (a_{ij}(t, x) \wedge N) \vee (-N)$ and $b = (b_1, \ldots, b_m)^\top$ $b_i^N(t, x) = (b_i(t, x) \wedge N) \vee (-N)$, $1 \le i, j \le m$.

(ii) According to Theorem 13.18 we are allowed to consider (weak) solutions X^N of $E(a^N, b^N; \nu)$. A priori, by BDG and Jensen's inequalities, the moment $E(\sup_{t \le T} |X_s^N|^\gamma)$ is finite but it depends on N.

(iii) On the other hand previous item, the same BDG and Jensen's inequalities, together with Gronwall's lemma allow to show the existence of a constant $C(\gamma, T, c_a, c_b), E(|X_0|^\gamma))$, where c_a, c_b are the linear growth coefficients related to a and b such that

$$\sup_{t \le T} E(|X_t^N|^\gamma) \le C(\gamma, T, c_a, c_b), E(|X_0|^\gamma)). \tag{13.6}$$

(iv) At this point we can show that the laws of X^N as r.v. with values in $C([0, T])$ are tight. For this we make use of the Prohorov and Kolmogorov theorems whose consequences for us are summarized in Problem 4.11 in Section 2.4 in [18]. It is enough to show

$$E(|X_t^N - X_s^N|^\alpha) \le \text{const}|t - s|^{m+\beta}, \ s, t \in [0, T], |t - s| \le 1, \tag{13.7}$$

for some $\alpha, \beta > 0$. By BDG and Jensen inequalities and (13.6),

$$E(|X_t^N - X_s^N|^\gamma) \le \text{const}|t - s|^{\frac{\gamma}{2}}, \ s, t \in [0, T], |t - s| \le 1, \tag{13.8}$$

where the constant const does not depend on N. So (13.7) holds with $\alpha = \gamma, \beta = \frac{\gamma}{2} - m$.

(v) Since the laws of X^N are tight, there exists a subsequence (still denoted by X^N) which converges in law. By Skorohod theorem there is a sequence of copies $\tilde{X}^N$ of copies of X^N with the same distributions as X^N such that $\tilde{X}^N$ converge a.s., in particular ucp to some process X. Without restriction of generality we still write $X^N = \tilde{X}^N$.

(vi) We set $\sigma = aa^\top$. Let $f \in C^2(\mathbb{R}^m)$. By Itô formula

$$f(X_t^N) - f(X_0^N) - \int_0^t \nabla f(X_s^N) b(s, X_s^N) ds - \frac{1}{2} \sum_{i,j}^m \int_0^t \partial_{ij}^2 f(s, X_s^N) \sigma_{ij}(s, X_s^N) ds,$$

is a martingale. By previous point, taking $N \to \infty$ previous processes converge ucp to

$$f(X_t) - f(X_0) - \frac{1}{2} \sum_{i,j=1}^m \int_0^t \partial_{ij}^2 f(X_s) \sigma_{ij}(s, X_s) ds - \frac{1}{2} \int_0^t \nabla f(X_s) b(X_s) ds.$$

This can be shown to be a local martingale, using the definition of martingale. In other words (X, P) solves **the martingale problem** related to σ, b.

(vii) Since the concept of solution in law is equivalent to the one of martingale problem, see Proposition 4.16 of [18], it is possible to show that that X solves $E(a, b; \nu)$. See also Exercise 13.20 below.

■

Exercise 13.20 *Let $(\Omega, \mathcal{F}, P)$ and a filtration $(\mathcal{F}_t)$ fulfilling the usual conditions. Let $a : [0,T] \times \mathbb{R}^m \to \mathbb{R}^{m \times p}$, $b : [0,T] \times \mathbb{R}^m \to \mathbb{R}^m$ be bounded Borel functions.*

(i) Let X be a solution of $E(a, b)$. Show that for every $f \in C^2(\mathbb{R}^m)$ then

$$f(X_t) - f(X_0) - \int_0^t \nabla f(X_s) \cdot b(s, X_s) ds - \frac{1}{2} \sum_{i,j}^m \int_0^t \partial_{x_i, x_j}^2 f(s, X_s) (aa^\top)_{ij}(s, X_s) ds \tag{13.9}$$

is a local martingale, where $a^\top$ is the transposed of matrix a.
If (13.9) is verified for every $f \in C^2(\mathbb{R}^m)$, we say that (X, P) solves the **martingale problem** *related to b and $aa^\top$.*

(ii) Suppose now (X,Q) solves previous martingale problem, where X a continuous process and Q is a probability on $(\Omega,\mathcal{F})$. Q is the reference probability for the items below.

(a) Show that the processes

$$M_t^i := X_t^i - \int_0^t b_i(s,X_s)ds, t\geq 0, 1\leq i\leq m,$$

are $(\mathcal{F}_t^X)$-local martingales under Q and $(\mathcal{F}_t^X)$ is the canonical filtration associated with X.

(b) Show that for every $1\leq i,j\leq m$, $[M^i,M^j]=[X^i,X^j]$.

(c) Let $1\leq i,j\leq m$. Express (13.9) for $f(x)=x_ix_j$ to show that

$$[M^i,M^j]_t = \int_0^t (aa^\top)_{ij}(s,X_s)ds.$$

Consider now for simplicity the case $m=p=1$ and $a>0$. Let $M=M^1$. Show that the process

$$W_t := \int_0^t \frac{1}{a(s,X_s)}dM_s, t\geq 0,$$

is an $(\mathcal{F}_t^X)$-Brownian motion.

(d) Deduce finally that X is a solution of $E(a,b)$ and $E(a,b)$ admits weak existence.

Exercise 13.21 *Show* (13.6) *and* (13.8) *in the case $m=p=1$ and $T=1$ (without restriction of generality).*

13.4 The Engelbert-Schmidt criteria

We proceed with results which are more specifically unidimensional stating some results from K.J. Engelbert and W. Schmidt, see [7]. These two authors furnished necessary and sufficient conditions for weak existence and uniqueness in law of SDEs with zero drift.

For a Borel function $\gamma:\mathbb{R}\to\mathbb{R}$, we first define

$$Z(\gamma)=\{x\in\mathbb{R}|\gamma(x)=0\};$$

then we define the set $I(\gamma)$ as the set of real numbers x such that

$$\int_{x-\varepsilon}^{x+\varepsilon} \frac{dy}{\gamma^2(y)} = \infty, \forall \varepsilon > 0.$$

For a clear exposition of the next result, see Theorems 5.4 and 5.5 of [18].

Proposition 13.22 *(Engelbert-Schmidt). Suppose that $a : \mathbb{R} \to \mathbb{R}$, i.e. not depending ont time and consider equation $E(a, 0; \nu)$ for some law ν on $\mathbb{R}$.*

i) $E(a, 0; \nu)$ admits weak existence (without explosion) if and only if

$$I(a) \subset Z(a). \tag{13.10}$$

ii) E(a,0; ν) admits weak existence and uniqueness in law if and only if

$$I(a) = Z(a). \tag{13.11}$$

Remark 13.23 *i) If a is continuous then (13.10) is always verified. In fact, if $a(x) \neq 0$, there is $\varepsilon > 0$ such that*

$$|a(y)| > 0, \forall y \in [x - \varepsilon, x + \varepsilon].$$

Therefore x cannot belong to $I(a)$.

ii) (13.10) is verified also for some discontinuous functions as for instance $sign^+$. This confirms what was affirmed previously, i.e. the weak existence for $E(a, 0)$, see Theorem 13.15.

iii) If $a(x) = 1_{\{0\}}(x)$, (13.10) is not verified.

iv) If $a(x) = |x|^\alpha, \alpha \geq \frac{1}{2}$ then

$$Z(a) = I(a) = \{0\}.$$

So there is at most one solution in law for $E(a, 0; \nu)$ independently from initial condition.

v) The proof is technical and makes use of Lévy characterisation theorem 13.11 of Brownian motion.

13.5 Feller test for explosion

Let us consider Borel functions $a, b : \mathbb{R} \to \mathbb{R}$ (not depending on time). Suppose moreover $a > 0$ and that $a, b, \frac{b}{a^2}$ are locally integrable. We consider the following infinitesimal generator:

$$\mathcal{A}g = \frac{a^2}{2} g'' + b g'.$$

We denote

$$\Sigma(x) := 2 \int_0^x \frac{b}{a^2}(y)\, dy. \tag{13.12}$$

Let $\ell \in C^0(\mathbb{R})$ and $u_0, u_1 \in \mathbb{R}$. Then there is a unique solution to

$$\begin{cases} \mathcal{A}u &= & \ell \\ u(0) &= & u_0, \quad u'(0) = u_1, \end{cases} \tag{13.13}$$

which is given by

$$\begin{cases} u(0) &= & u_0 \\ u'(x) &= & e^{-\Sigma(x)} \left(\int_0^x \frac{2\ell}{a^2} e^{\Sigma}(y)\, dy + u_1 \right). \end{cases}$$

Indeed

$$\mathcal{A}f = \left(\frac{e^{\Sigma} f'}{2} \right)' e^{-\Sigma} a^2.$$

Remark 13.24 *This procedure can be extended to the case when b is the derivative of a continuous function β, therefore a distribution, see [10].*

Proposition 13.25 *(***Feller test for explosion***). Suppose $a > 0$ such that $a, b, \frac{b}{a^2}$ are locally integrable. Let v be the unique solution to*

$$\begin{cases} \mathcal{A}v &= & 1 \\ & & \\ v(0) &= & 0, \quad v'(0) = 0. \end{cases}$$

Then weak existence and uniqueness in law for $E(a, b)$ takes place (without explosion, for every $t \geq 0$) if and only if

$$v(-\infty) = v(+\infty) = +\infty. \tag{13.14}$$

Proof. Theorem 5.5.29 in [18] and the remark below which follows from (13.13) and (13.14) give the result. ■

Remark 13.26 *v given above is explicitely given by*

$$\begin{cases} v(0) &= 0 \\ v'(x) &= e^{-\Sigma(x)} \left(\int_0^x \frac{2e^{\Sigma}}{a^2}(y)\, dy \right). \end{cases}$$

Exercise 13.27 *i) Verify that whenever*

$$\int_0^{\infty} e^{-\Sigma(y)}\, dy = \int_{-\infty}^{0} e^{-\Sigma(y)}\, dy = \infty, \tag{13.15}$$

then (13.14) is verified.

ii) Deduce in particular that a unique (non-exploding) solution exists in the three following cases.

a) $b = 0$*;*

b)Σ *is upper bounded;*

c) $\frac{b}{a^2}$ *is integrable.*

iii) Suppose a *of class* C^1*. Find necessary and sufficient conditions so that the Stratonovich type equation*

$$X_t = X_0 + \int_0^t a(X_s) \circ dW_s$$

has a (unique Itô process) non-exploding solution.

Solutions

i) It will be enough to check that $v(+\infty) = +\infty$, since the $-\infty$ case can be justified similarly. For $z \geq 1$ we have

$$\begin{aligned} v(z) &= 2\int_0^z dx e^{-\Sigma(x)} \left(\int_0^x \frac{e^{\Sigma}}{a^2}(y)\, dy\right) \\ &= 2\int_0^z dy \frac{e^{\Sigma}}{a^2}(y) \int_y^z e^{-\Sigma(x)} dx \\ &\geq 2\int_1^z dy \frac{e^{\Sigma}}{a^2}(y) \int_1^z e^{-\Sigma(x)} dx. \end{aligned}$$

Therefore

$$\text{liminf}_{z\to+\infty} v(z) \geq 2\int_1^{\infty} dy \frac{e^{\Sigma}}{a^2}(y) \int_1^{\infty} e^{-\Sigma(x)} dx.$$

So, if condition (13.15) is verified then $v(+\infty) = +\infty$.

ii) This includes the following cases:

a) $\Sigma = 0$ since $b = 0$;

b) Σ is bounded from above since

$$e^{-\Sigma} \geq e^{-\sup\Sigma};$$

c) $\frac{b}{a^2} \in L^1(\mathbb{R})$ because in that case Σ is bounded.

iii) We have

$$\begin{aligned}\int_0^t a(X_s) \circ dW_s &= \int_0^t a(X_s) dW_s + \frac{1}{2}[a(X), W]_t \\ &= \int_0^t a(X_s) dW_s + \int_0^t \left(\frac{a^2}{4}\right)' (X_s) ds.\end{aligned}$$

So, X can be considered as solution to $E(a, b)$ with $b = \left(\frac{a^2}{4}\right)'$. We have

$$\begin{aligned}\Sigma(x) &= 2\int_0^x \frac{(\frac{a^2}{4})'}{a^2}(y) dy = \frac{1}{2}\int_0^x \frac{(a^2)'}{a^2}(y) dy \\ &= \frac{1}{2}(\log a^2)(y)|_0^x = \log\frac{a(x)}{a(0)},\end{aligned}$$

so that

$$e^{\Sigma(x)} = \frac{a(x)}{a(0)}$$

and we have

$$v'(x) = \frac{1}{a(x)}\int_0^x \frac{1}{a(y)} dy.$$

(13.14) becomes

$$\begin{aligned}\int_0^\infty dx \frac{1}{a(x)}\int_0^x \frac{1}{a(y)} dy &= \frac{1}{2}\left(\int_0^\infty dx \frac{1}{a(x)}\right)^2 = +\infty; \\ \int_{-\infty}^0 dx \frac{1}{a(x)}\int_x^0 \frac{1}{a(y)} dy &= \frac{1}{2}\left(\int_{-\infty}^0 dx \frac{1}{a(x)}\right)^2 = +\infty.\end{aligned}$$

Conditions (13.14) become

$$\int_0^\infty \frac{1}{a(x)} dx = \int_{-\infty}^0 \frac{1}{a(x)} dx = +\infty.$$

Proof (of Proposition 13.25).

(Sketch). Without restriction of generality we suppose a to be continuous. We only discuss uniqueness in the particular case when (13.15) is verified, i.e.

$$\int_0^{\infty} e^{-\Sigma(y)}\, dy = \int_{-\infty}^0 e^{-\Sigma(y)}\, dy.$$

Let $(X_t)_{t\geq 0}$ be a solution to $E(a,b)$ related to some filtered probability space and an $(\mathcal{F}_t)_{t\geq 0}$- classical Brownian motion $(W_t)_{t\geq 0}$.

We denote $h : \mathbb{R} \to \mathbb{R}$ such that $h' = e^{-\Sigma}, h(0) = 0$; we remark that $h \in C^2(\mathbb{R})$ and $\mathcal{A}h = 0$. We set $Y_t = h(X_t)$; using Itô formula we obtain

$$\begin{aligned} Y_t &= h(X_0) + \int_0^t h'(X_s) dX_s + \frac{1}{2}\int_0^t h''(X_s) d[X]_s \\ &= h(X_0) + \int_0^t (ah')(X_s) dW_s + \frac{1}{2}\int_0^t \underbrace{\mathcal{A}h}_{=0}(X_s) ds \\ &= h(X_0) + \int_0^t a_0(Y_s) dW_s, \end{aligned}$$

where $a_0 = (ah') \circ h^{-1}$. Since Assumption (13.15) is verified, then h is surjective hence bijective. So we have

$$Y_t = Y_0 + \int_0^t a_0(Y_s) dW_s. \tag{13.16}$$

Uniqueness is verified. Indeed a_0 is continuous and strictly positive. We can trivially apply Engelbert-Schmidt criterion. ∎

Remark 13.28 *If (13.15) is not verified and $h(-\infty) = a, h(+\infty) = b$, the solution X explodes if and only if Y attains a or b with a positive probability.*

13.6 Results on pathwise uniqueness

We remain in the one-dimensional case.

Proposition 13.29 *(Yamada-Watanabe, [38]).*

Let $a, b : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$ and consider again E(a,b). Suppose b globally Lipschitz and $h : \mathbb{R}_+ \to \mathbb{R}_+$ strictly increasing continuous such that

- $h(0) = 0$;

- $$\int_0^{\varepsilon} \frac{1}{h^2}(y)dy = \infty, \forall \varepsilon > 0; \tag{13.17}$$
- $|a(t,x) - a(t,y)| \leq h(|x-y|)$.

Then $E(a,b)$ admits pathwise uniqueness.

Remark 13.30 *In the obvious case that a is Lipschitz one can choose $h(x) = x$.*

Proposition 13.31 *Under the assumption of Proposition 13.29 we suppose moreover that a, b are continuous and a has linear growth. Then $E(a,b)$ admits strong existence.*

Proof. It follows from Theorem 13.19, Proposition 13.29 and Proposition 13.7 ii). ■

Remark 13.32 *In Proposition 13.29, one typical choice is*

$$h(u) = u^{\alpha}, \alpha \geq \frac{1}{2}.$$

Proof (of Proposition 13.29). Taking into account the conditions imposed on h, there is a strictly decreasing sequence (a_n) in $[0,1]$, with

$$a_0 = 1, \lim_{n\to\infty} a_n = 0, \int_{a_n}^{a_{n-1}} \frac{1}{h^2}(x)dx = n,$$

for every $n \geq 1$; in fact for a fixed a_n, we have

$$\lim_{t\to 0+} \int_t^{a_n} \frac{1}{h^2}(x)dx = \infty$$

and the function $t \mapsto \int_t^{a_n} \frac{1}{h^2}(x)dx$ is continuous. For every $n \geq 1$, we define a continuous function $\rho_n : \mathbb{R}_+ \to \mathbb{R}_+$ such that

- its support is in $[a_n, a_{n-1}]$;
- $0 \leq \rho_n(x) \leq \frac{2}{nh^2(x)}, \forall x > 0$;
- $\int_{a_n}^{a_{n-1}} \rho_n(x)dx = 1$.

We can show how such a construction is possible. We fix n and, for $u \in [0,1]$, we set

$$\begin{aligned}\phi(u) &= a_n\frac{1+u}{2} + a_{n-1}\frac{1-u}{2};\\ \psi(u) &= a_n\frac{1-u}{2} + a_{n-1}\frac{1+u}{2}.\end{aligned}$$

We define the family of functions $\rho_n(\cdot,u)$ such that

$$\rho_n(x,u) = \begin{cases} 0 & : & x \notin [a_n, a_{n-1}] \\ \frac{2u}{nh^2(x)} & : & x \in [\phi(u), \psi(u)] \\ \text{linear affine} & : & x \in [a_n, \phi(u)[\cup]\psi(u), a_{n-1}]. \end{cases}$$

In fact we have:

$$\begin{aligned}\text{if} u = 0, \rho_n(x,u) &\equiv 0;\\ \text{if} u = 1, \rho_n(x,u) &= \frac{2}{nh^2(x)}, x \in [a_n, a_{n-1}].\end{aligned}$$

For $u \in [0,1[$, $\rho_n(\cdot,u)$ is continuous. We set

$$\Phi(u) = \int_{a_n}^{a_{n-1}} \rho_n(x,u)dx.$$

Now, $\Phi : [0,1] \to \mathbb{R}_+$ is continuous and $\Phi(0) = 0, \Phi(1) = 2$.

By the mean value theorem, there is $u_n \in]0,1[$ such that $\Phi(u_n) = 1$.

We choose $\rho_n(x) = \rho(x, u_n)$. We define $\psi_n : \mathbb{R} \to \mathbb{R}$ by

$$\psi_n(x) = \int_0^{|x|} dy \int_0^y \rho_n(z)dz, \quad x \in \mathbb{R}; \tag{13.18}$$

ψ_n is a even function with $|\psi_n'(x)| \leq 1$. On the other hand, if $y > 0$, there is $N(y)$ with

$$n > N(y) \Rightarrow \int_0^y \rho_n(z)dz = \int_{a_n}^{a_{n-1}} \rho_n(z)dz = 1.$$

Consequently we have

$$\text{lim}_{n\to\infty}\psi_n(x) = |x|, \forall x \in \mathbb{R}, \psi_n(x) \leq |x|.$$

Suppose now to have a probability space $(\Omega, \mathcal{F}, P)$ and an $(\mathcal{F}_t)$- Brownian motion $(W_t)_{t\geq 0}$ and two solutions $X^{(1)}$ and $X^{(2)}$ of $E(a,b)$ with $X_0^{(1)} = X_0^{(2)}$ a.s. It is enough to show that $X^{(1)}$ and $X^{(2)}$ are indistinguishable under the assumption

$$\begin{aligned} E\left(\int_0^t a^2(s, X_s^{(i)})ds\right) &< \infty \\ E\left(\int_0^t |b|(s, X_s^{(i)})ds\right) &< \infty, \end{aligned} \tag{13.19}$$

for every $0 \leq t < \infty, i = 1, 2$. Indeed, using a localization argument, it is possible to reduce everything to (13.19); we explain in Remark 13.33 the details.

We set $\Delta_t = X_t^{(2)} - X_t^{(1)}$ so that we have

$$\Delta_t = \int_0^t (a(s, X_s^{(1)}) - a(s, X_s^{(2)}))dW_s + \int_0^t (b(s, X_s^{(1)} - b(s, X_s^{(2)}))ds.$$

In particular, Δ_t is integrable. Using Itô formula for semimartigales and the fact that ψ_n is of class $C^2(\mathbb{R})$, we get

$$\begin{aligned} \psi_n(\Delta_t) &= \int_0^t \psi_n'(\Delta_s)\left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] ds \\ &+ \frac{1}{2}\int_0^t \psi_n''(\Delta_s)\left[a(s, X_s^{(1)}) - a(s, X_s^{(2)})\right]^2 ds \\ &+ \int_0^t \psi_n'(\Delta_s)\left[a(s, X_s^{(1)}) - a(s, X_s^{(2)})\right] dW_s. \end{aligned} \tag{13.20}$$

The expectation of stochastic integral in (13.20) is zero because of (13.19). On the other hand, the expectation of the second integral is bounded by

$$E\left(\int_0^t \underbrace{\psi_n''(\Delta_s)}_{\rho_n(|\Delta_s|) \leq \frac{2}{nh^2(|\Delta_s|)}} h^2(|\Delta_s|)ds\right) \leq \frac{2t}{n}.$$

We conclude that

$$E(\psi_n(\Delta_t)) \leq E\left(\int_0^t \psi_n'(\Delta_s)\left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})ds\right]\right) + \frac{t}{n}. \tag{13.21}$$

If k is the Lipschitz constant of b we get

$$E(\psi_n(\Delta_t)) \le \frac{t}{n} + k\int_0^t E(|\Delta_s|)ds.$$

We let n go to ∞ which by Lebesgue dominated convergence theorem gives

$$E(|\Delta_t|) \le k\int_0^t E(|\Delta_s|)ds.$$

By Gronwall lemma, we easily obtain $E(|\Delta_t|) \equiv 0$ and finally $\Delta_t = 0$ a.s. Since (Δ_t) is continuous, it is indistinguishable from zero. ∎

Remark 13.33 *We explain here how we can reduce to the hypothesis* (13.19) *the proof of Proposition 13.29.*

(i) First we reduce to the case when the initial condition X_0 belongs to a compact interval $[-M, M]$. For this we replace the initial probability P with the conditional expectation $P^M := P(\cdot|X_0 \in [-M, M])$. Under P^M the Brownian motion remains a Brownian motion and the solutions of the SDE remain solutions of the SDE.

(ii) Suppose now that X_0 lives in a compact interval. For each positive integer N we define the stopping times

$$\begin{aligned}\tau_i^N &:= \inf\{t \in [0,T] | X_t^i| \ge N\}, i = 1,2\\ \tau^N &:= \tau_1^N \vee \tau_2^N,\end{aligned}$$

with the convention the infimum of the empty set is $+\infty$. We remark that (up to a null set) $\Omega = \cup\Omega_N$, $\Omega_N := \{\tau^N > T\}$. Consequently, it is enough to show that $X^1 \equiv X^2$ on Ω_N. On Ω_N, we have (for $i = 1, 2$)

$$X_t^i = X_0 + \int_0^t a_N(s, X_s^i)dW_s + \int_0^t b_N(s, X_s^i)ds,$$

where

$$\begin{aligned}a_N(t,x) &= a(t,(x \wedge N) \vee (-N)),\\ b_N(t,x) &= b(t,(x \wedge N) \vee (-N)).\end{aligned}$$

Now b_N is Lipschitz and a_N fulfill the assumption of a. Consequently (13.19) *holds since a_N and b_N are bounded.*

A direct consequence of Propositions 13.29 and Corollary 13.31 we get the following.

Corollary 13.34 *Suppose the assumption of Proposition 13.29 verified and a, b are continuous with linear growth. Then $E(a,b)$ admits strong existence and pathwise uniqueness.*

Example 13.35

$$X_t = \int_0^t |X_s|^\alpha dW_s, t \geq 0. \tag{13.22}$$

We set $a(x) = |x|^\alpha, 0 < \alpha < 1, b = 0$.

According to Engelbert-Schmidt notations, we have $Z(a) = \{0\}$. Moreover we have the following.

- If $\alpha \geq \frac{1}{2}$ then $I(a) = \{0\}$.
- If $\alpha < \frac{1}{2}$ then $I(a) = \emptyset$.

On the other hand, if $\alpha \geq \frac{1}{2}$,

$$||x|^\alpha - |y|^\alpha| \leq h(|x-y|), \tag{13.23}$$

where $h(z) = z^\alpha$.

This follows from the fact that, if $\beta \geq 1, a, t \geq 0$,

$$(a+t)^\beta - a^\beta - t^\beta \geq 0. \tag{13.24}$$

In fact for a fixed, deriving, we easily see that $\Phi(t) = (a+t)^\beta - a^\beta - t^\beta$ is increasing; since $\Phi(0) = 0$, (13.24) follows. Consequently, setting $\beta = \frac{1}{\alpha}$, $a = |x|^\alpha, t = |y-x|^\alpha$, we get

$$(|y-x|^\alpha + |x|^\alpha)^{\frac{1}{\alpha}} \geq |x| + |y-x| \geq |y|.$$

Therefore

$$|x|^\alpha + |y-x|^\alpha \geq |y|^\alpha.$$

Replacing x with y we also obtain

$$|y|^\alpha + |y-x|^\alpha \geq |x|^\alpha.$$

The two inequalities above imply (13.23).

By Corollary 13.34, (13.22) admits strong existence and pathwise uniqueness.

The unique solution is $X \equiv 0$.

If $\alpha < \frac{1}{2}$, $X \equiv 0$ is always a solution. This is not the only one; even uniqueness in law is not true, by Engelbert-Schmidt criterion.

Exercise 13.36 *(Itô-Watanabe, 1978, [16]). We consider the equation*

$$X_t = 3\int_0^t X_s^{\frac{1}{3}}\, ds + 3\int_0^t X_s^{\frac{2}{3}}\, dW_s. \tag{13.25}$$

i) Verify if the conditions of Proposition 13.29 are verified.

ii) Show that (13.25) admits an infinity of solutions of the type

$$X_t^{(\theta)} = \begin{cases} 0 & : \quad 0 \le t < \beta_\theta; \\ W_t^3 & : \quad \beta_\theta \le t < \infty, \end{cases}$$

where $0 \le \theta < \infty$ *and*

$$\beta_\theta = \inf\{s \ge \theta | W_s = 0\}.$$

Under assumptions of Proposition 13.29, we also have a comparison theorem, see [31] (Theorem 3.7, Chap. IX) or [18] (Proposition 2.18, Chap. 5) for the case of time dependent coefficients.

Theorem 13.37 *Let $b^1, b^2 : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$ be Borel real functions. Let a fulfilling the same assumption of Proposition 13.29.*
Let (X_t^i) be $(\mathcal{F}_t)$-adapted processes such that the following holds.

i) $X_0^1 \le X_0^2$ a.s.

ii) $b^1(t,x) \le b^2(t,x) \quad \forall\, x$.

iii) Either b^1 or b^2 are Lipschitz.

iv) X^i is solution of $E(a, b^i)$, $i = 1, 2$.

Then

$$P\{X_t^1 \le X_t^2; \forall\, t \ge 0\} = 1.$$

In some cases $E(a, b)$ admits pathwise uniqueness even if b is only measurable, and sometimes for unbounded drifts b.

We conclude the section about strong existence and pathwise uniqueness with a celebrated theorem of A.Yu. Veretennikov.

Theorem 13.38 *We set $\sigma = aa^T$. Assume the following.*

(i) a, b are Borel bounded.

(ii) a is Lipschitz with linear growth.

(iii) The non-degeneracy Assumption 13.13. holds.

Then $E(a, b)$ admits strong existence and pathwise uniqueness.

Proof. The result follows from Theorem 1. in [37].

■

Remark 13.39 *(i) There exists still big activity in this field. For instance [21] has establishes pathwise uniqueness for the case $m = p$, a being the identity and b is locally in some suitable L^q spaces in time and space. For instance if b is time-independent then b is allowed to locally in $L^q(\mathbb{R}^m)$ when $q > m$.*

(ii) Other extensions on pathwise uniqueness verifying suitable properties on the flow appear in [8] where a, b do not depend on time. b is allowed to be suitably Hölder continuous, $aa^\top$ fulfills Assumption 13.13 and it is bounded of class C^3.

(iii) Concerning the study of SDEs in the multidimensional case, see also [9] and references therein.

14 Bessel processes

An application of previous considerations concerns the class of Bessel and square Bessel processes.

Example 14.1 *Let $a(x) = c\sqrt{|x|}$, where c is a positive constant. Let b be continuous and Lipschitz. Then $E(a,b)$ admits strong existence and pathwise uniqueness.*

In fact, first we observe that a, b are continuous with linear growth.

We have

$$|a(x) - a(y)| = c|\sqrt{|x|} - \sqrt{|y|}| \leq c\sqrt{|x-y|}$$

by (13.23). *Proposition 13.29 can be applied taking $h(z) = c\sqrt{|z|}$. By Corollary 13.34, the result follows.*

Let $x_0 \geq 0$, $\delta \geq 0$. Consider the SDE $E(a,b)$, with $a(x) = 2\sqrt{|x|}, b(t,x) = \delta$.

$$Z_t = x_0 + 2\int_0^t \sqrt{|Z_s|}dW_s + \delta t. \tag{14.26}$$

According to the comparison Theorem 13.37 the unique solution Z of $E(a,b)$ is non-negative and so the absolute value in (14.26) *can be raised.*

In fact for $x_0 = \delta = 0$, the unique solution is $Z \equiv 0$. So we have

$$Z_t = x_0 + 2\int_0^t \sqrt{Z_s}dW_s + \delta t, t \geq 0. \tag{14.27}$$

Definition 14.2 *The unique solution Z to* (14.26) *is called* **square δ-dimensional Bessel process** *starting at x_0 ; it is denoted by $BESQ(x_0, \delta)$. For fine properties of this process, see [31], ch. IX.3.*

Taking into account $Z \geq 0$, we call $\delta-$ **dimensional Bessel process starting at $\sqrt{x_0}$** *the process $X = \sqrt{Z}$. It is denoted by $BES(\sqrt{x_0}, \delta)$. .*

Proposition 14.3 *Let p be a positive integer. Let $W = (W^1, \cdots, W^p)$ be a classical p-dimensional Brownian motion, $y_0 \in \mathbb{R}^d$. We set $X_t = \|y_0 + W_t\|$. $(X_t)_{t\geq 0}$ is a p-dimensional Bessel process starting at $\|y_0\|$.*

Proof. We set $Z_t = \|y_0 + W_t\|^2$. By Itô formula, we have

$$\begin{aligned} Z_t &= \|y_0\|^2 + 2\sum_{i=1}^p \int_0^t (W_s^i + y_0^i)dW_s^i + \frac{1}{2}\sum_{i=1}^p 2t \\ &= \|y_0\|^2 + 2M_t + pt, \end{aligned}$$

where

$$M_t = \sum_{i=1}^{p} \int_0^t (W_s^i + y_0^i) dW_s^i.$$

Now, (M_t) is a continuous local martingale whose quadratic variation is

$$[M]_t = \sum_{i=1}^{p} \int_0^t (W_s^i + y_0^i)^2 ds = \int_0^t \|W_s + y_0\|^2 ds.$$

Therefore

$$B_t = \int_0^t \frac{dM_s}{\|W_s + y_0\|}$$

defines a classical Brownian motion because $[B]_t = t$ and because of Lévy characterization Theorem 13.11 of Brownian motion. Now,

$$2 \int_0^t \sqrt{Z_s} dB_s = 2 \int_0^t \frac{\|W_s + y_0\|}{\|W_s + y_0\|} dM_s = 2M_t,$$

and the final result follows. ■

Remark 14.4 *a) If $\delta > 1$, it is possible to see that the process $BES(x_0, \delta)$ fulfills*

$$X_t = x_0 + B_t + \frac{\delta - 1}{2} \int_0^t \frac{ds}{X_s},$$

where B is a standard Brownian motion.

b) For $0 \leq \delta \leq 1$, the integral $\int_0^t X_s^{-1} ds$ does not converge and $BES(x_0, \delta)$ is a non-semimartingale process, except for $\delta = 1$ and $\delta = 0$, see [39, 17], Chapter XI Section 1 and Section 6.1, respectively.

c) Let $\delta \in]0, 1[$. X is a Dirichlet process, see [39].

Exercise 14.5 *Show item a) supposing that $X_t > 0$ a.s. for every t.*

Exercise 14.6 *Let (S_t) be a $BESQ(x_0, \delta)$. For $N > 0$, we set $\tau_N = \inf\{s \geq 0 | S_s > N\}$.*

(i) Show that a.s. (τ_N) is an increasing sequence of stopping times diverging to infinity.

(ii) Show that $E(S_t^{\tau_N}) = x_0 + \delta E(\tau_N \wedge t)$.

(iii) *Calculate* $E(S_t), t \geq 0$.

(iv) *Show that* $E(\sup_{t\leq T} S_t^2) < \infty$.

(v) *Calculate* $Var(S_t)$ *for any* $t \geq 0$.

We aim here at establishing the marginal distribution law $\rho_t(dy)$ of S_t where S is a Square Bessel process. In fact we will determine the Laplace transform i.e.

$$E(e^{-\lambda S_t}) = \int_0^\infty e^{-\lambda y}\rho_t(dy). \tag{14.28}$$

We will establish a particular case of the celebrated **Pitman-Yor theorem** which allows to calculate an even more general quantity i.e.

$$E\left(e^{-\lambda\int_0^\infty S_s d\mu(s)}\right),$$

where μ is a Borel measure with compact support on $\mathbb{R}_+$ which will correspond to (14.28) taking $\mu = \delta_t$.

First we need a lemma.

Lemma 14.7 *Let* Z *(resp.* Z'*) be a* $BESQ(x,\delta)$ *(resp.* $BESQ(x',\delta')$*), driven by a Brownian motion* W *(resp.* W'*). We suppose* W, W' *independent and defined on the same probability space. Then* $X = Z + Z'$ *is a* $BESQ(x+x',\delta+\delta')$ *process.*

Proof. We set $X_t = Z_t + Z'_t$. By additivity,

$$\begin{aligned} X_t &= Z_t + Z'_t = x + x' + 2\int_0^t \sqrt{Z_s}dW_s + \delta t \\ &+ 2\int_0^t \sqrt{Z'_s}dW'_s + \delta' t. \end{aligned} \tag{14.29}$$

In order to conclude, we need to show the existence of some Brownian motion $\tilde{W}$ with

$$\int_0^t \sqrt{Z_s}dW_s + \int_0^t \sqrt{Z'_s}dW'_s = \int_0^t \sqrt{X_s}d\tilde{W}_s.$$

We introduce the process

$$\begin{aligned} M_t &= \int_0^t 1_{\{X_s>0\}} \frac{\sqrt{Z_s}dW_s + \sqrt{Z'_s}dW'_s}{\sqrt{X_s}} + \\ &+ \int_0^t 1_{\{X_s=0\}} dW''_s, \end{aligned}$$

where W'' is another Brownian motion independent of W, W', possibly by enlarging the probability space. We omit the details about this. We show that M is Brownian motion. Since M is a local martingale, according to Lévy's characterization theorem, it will be enough to show $[M]_t = t$.

Recalling that $\frac{Z_s+Z'_s}{X_s} = 1$, we obtain

$$[M]_t = \int_0^t 1_{\{X_s>0\}} ds + \int_0^t 1_{\{X_s=0\}} ds = t.$$

Coming back to (14.29), we get

$$\begin{aligned} \int_0^t \sqrt{X_s} dM_s &= \int_0^t 1_{\{X_s>0\}} (\sqrt{Z_s}dW_s + \sqrt{Z'_s}dW'_s) + \int_0^t \sqrt{X_s} 1_{\{X_s=0\}} dW''_s \\ &= \int_0^t 1_{\{X_s>0\}} (\sqrt{Z_s}dW_s + \sqrt{Z'_s}dW'_s) \\ &= \int_0^t (\sqrt{Z_s}dW_s + \sqrt{Z'_s}dW'_s). \end{aligned}$$

Indeed $X_t = 0$ implies that $Z_t = Z'_t = 0$ X_t being positive. So the result follows. ■

We go on with another significant partial result.

Theorem 14.8 *Let μ be a Borel finite measure on $[0, T]$. We set*

$$\varphi(x, \delta) := E\left(e^{-\lambda \int_0^\infty X_s d\mu(s)}\right),$$

where X be $BESQ(x, \delta)$. Then

$$\varphi(x, \delta) = \varphi(x, 0)\varphi(0, \delta).$$

Proof.

We remark that this quantity is finite since μ is a positive measure and Bessel process is always non-negative.

Let Z (resp. Z') respectively a $BESQ(x,\delta)$ (resp. $BESQ(x',\delta')$) for $x,x'\in\mathbb{R}_+,\delta,\delta'\geq 0$. Using Lemma 14.7 we have

$$\begin{aligned}\varphi(x+x',\delta+\delta') &= E\left(e^{-\lambda\int_0^\infty(Z_s+Z'_s)d\mu(s)}\right)=E\left(e^{-\lambda\int_0^\infty Z_s d\mu(s)}\right)E\left(e^{-\lambda\int_0^\infty Z'_s d\mu(s)}\right)\\ &= \varphi(x,\delta)\varphi(x',\delta').\end{aligned}$$

So we have proved that

$$\varphi(x+x',\delta+\delta')=\varphi(x,\delta)\varphi(x',\delta'). \tag{14.30}$$

In particular, for $x,\delta\geq 0$,

$$\varphi(x,\delta)=\varphi(x,0)\varphi(0,\delta).$$

■

Next lemma is technical.

Lemma 14.9 *(i)* $\varphi(x,\delta)=\varphi(\frac{x}{\delta},1)^\delta$, *if* $x,\delta>0$.

(ii) $\varphi(x,\delta)=\varphi(1,\frac{\delta}{x})^x$.

Proof. It is enough to prove (i) since (ii) follows similarly. If δ is rational, then (i) is the consequence of the multiplicative property (14.30). The general case follow because φ is continuous, see Lemma 14.10. ■

Lemma 14.10 *The function* $\varphi:\mathbb{R}_+\times\mathbb{R}_+\to\mathbb{R}$ *is continuous.*

Proof. We omit the details of proof. However the exercise below allows to show that if there are two sequences (x_n) and (δ_n) which converge increasingly (or decreasingly) respectively to x_0 and δ_0, then $\varphi(x_n,\delta_n)\to\varphi(x_0,\delta_0)$.

■

Exercise 14.11 *Let* $S^i, i=1,2$, *be the solution (on some filtered probability space, equipped with a Brownian motion* W*) of*

$$S^i_t=x_i+2\int_0^t\sqrt{S^i_s}dW_s+\delta_i t, i=1,2,$$

where $0\leq x_1\leq x_2, 0\leq\delta_1\leq\delta_2$.

(i) Show that $S_t^1 \leq S_t^2, \forall t \in [0,T]$.

(ii) Express $E(|S_t^1 - S_t^2|)$.

(iii) Show that

$$\mathrm{Var}(S_t^2 - S_t^1) \leq (x_2 - x_1)t + \frac{(\delta_2 - \delta_1)t^2}{2}.$$

(iv) Using Doob's inequality find an upper bound for

$$E\left(\sup_{t \leq T} |S_t^1 - S_t^2|^2\right).$$

(v) Let (x_n) *(resp.* δ_n*) be an increasing sequence of non-negative numbers coverging to* x_0 *(resp.* δ_0*). Let* $S = S^n$ *the solution to*

$$S_t = x_n + \int_0^t \sqrt{S_s} dW_s + \delta_n t, \quad n \in \mathbb{N}.$$

Deduce that the sequence S^n *converges ucp to* S^0.

(vi) Deduce that $\varphi(x_n, \delta_n) \to \varphi(x, \delta)$ *when* μ *has its support on a compact interval* $[0,T]$.

Next theorem allows to calculate the Laplace transform of the law of a square Bessel process.

Theorem 14.12 *Let* S *be a* $BESQ(x,\delta)$*. Then*

$$E\left(\exp(-\lambda S_t)\right) = \left(\frac{1}{2t\lambda + 1}\right)^{\frac{\delta}{2}} \exp\left(-\frac{\lambda x}{2t\lambda + 1}\right).$$

Proof. We apply previous results with μ being the positive measure δ_t. Let $\lambda > 0$. Using the notation φ as before we have

$$E(\exp(-\lambda S_t)) = \varphi(x, \delta) = \varphi\left(\frac{x}{\delta}, 1\right)^{\delta},$$

by Lemma 14.9. So by Proposition 14.3, it is enough to evaluate previous quantity for $S_t = (\sqrt{\frac{x}{\delta}} + W_t)^2$, where (W_t) is a classical Wiener process. We denote $x_0 = \frac{x}{\delta}$. We get

$$\begin{aligned}
\varphi(x_0,1) &= E\left(e^{-\lambda(\sqrt{x_0}+W_t)^2}\right) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-\lambda(\sqrt{x_0}+y\sqrt{t})^2-\frac{y^2}{2}}dy \\
&= \frac{1}{\sqrt{2\pi}}e^{-\lambda x_0}\int_{\mathbb{R}} e^{-(\lambda t+\frac{1}{2})y^2-2\lambda y\sqrt{x_0 t}}dy \\
&= \frac{1}{\sqrt{2\pi}}e^{-\lambda x_0}\int_{\mathbb{R}} e^{-\frac{1}{2}\{(y\sqrt{2\lambda t+1})^2+4\lambda y\sqrt{x_0 t}\}}dy \\
&= \frac{1}{\sqrt{2\pi}}e^{-\lambda x_0}e^{\frac{1}{2}\left(2\lambda\sqrt{\frac{x_0 t}{2\lambda t+1}}\right)^2}\int_{\mathbb{R}} e^{-\frac{1}{2}\left(y\sqrt{2\lambda t+1}+2\lambda\sqrt{\frac{x_0 t}{2\lambda t+1}}\right)^2}dy \\
&= \exp\left(-\lambda x_0 + 2\lambda^2\frac{x_0 t}{2\lambda t+1}\right)\frac{1}{\sqrt{2\lambda t+1}}\int_{\mathbb{R}} q(y)dy,
\end{aligned}$$

where q is some law density. Finally

$$\varphi(x_0,1) = \frac{1}{(1+2\lambda t)^{\frac{1}{2}}}\exp\left(\frac{-\lambda x_0}{2\lambda t+1}\right).$$

Taking into account the definition of x_0 the result follows.

■

Exercise 14.13 *Deduce $E(S_t)$ from previous formula.*

Remark 14.14 *(i) Suppose $x = 0$ and δ be an integer n. Then the law of $\frac{S_t}{t}$ is a Chi-square of parameter n.*

(ii) If S is a $BESQ(x_0,\delta)$, for every $t > 0$, the law of S_t is absolutely continuous and its density is denoted by $x \mapsto p^\delta(x_0,\cdot)$, see Chapter IX of [31].

15 Cox-Ingersoll-Ross equation

A succesful model in finance is provided by Cox-Ingersoll-Ross model which is often used to model the evolution of interest rates.

We consider the SDE

$$\left\{\begin{array}{rcl} dX_t &=& (a+bX_t)dt + c\sqrt{|X_t|}dW_t \\ X_0 &=& x_0, \end{array}\right. \tag{15.31}$$

where $b, c \in \mathbb{R}$ and $a, x_0 \geq 0$.

Proposition 15.1 *Equation (15.31) admits strong existence and pathwise uniqueness. Moreover* $X_t \geq 0$ *a.s.*

Proof.

The function $x \mapsto a + bx$ is Lipschitz and $x \mapsto \sqrt{|x|}$ fulfills the assumption of Proposition 13.29. Moreover those functions are continuous with linear growth. The result follows by Corollary 13.34. ■

Remark 15.2 (i) *The unique solution* X *of Proposition 15.1 is said* **Cox-Ingersoll-Ross** *process.*

(ii) *Again, as for the case of square Bessel process, We apply comparison theorem for SDEs, setting*

$$b_1(x) = a + bx, b_2(x) = bx.$$

Since $b_1(x) \geq b_2(x), x_0 \geq 0$*, then the unique solution* X *of (15.33) is greater or equal than the unique solution* Y *of*

$$\begin{cases} dY_t &= bY_t dt + c\sqrt{|Y_t|}dW_t \\ X_0 &= 0. \end{cases} \tag{15.32}$$

Since $Y \equiv 0$*, then* $X_t \geq 0$ *and we can remove the absolute value to (15.33).*

(iii) *We can therefore affirm that the Cox-Ingersoll-Ross process is (the unique) solution of*

$$\begin{cases} dX_t &= (a + bX_t)dt + c\sqrt{X_t}dW_t \\ X_0 &= x_0. \end{cases} \tag{15.33}$$

Next step allows to associate the solution of (15.31) to square Bessel process. Before this we need a lemma that can be found in [18].

Lemma 15.3 *Let* (M_t) *be an* $(\mathcal{F}_t)$*-local martingale with quadratic variation* $[M]_t = \int_0^t \Phi_s^2 ds$*, for* Φ *being a measurable* $(\mathcal{F}_t)$*-progressively measurable non-negative process such that* $\Phi_s(\omega) \neq 0$*,* $dPds$ *a.e.*

Then, there is a Brownian motion $(\bar{W}_t)$ *such that*

$$M_t = M_0 + \int_0^t \Phi_s d\bar{W}_s.$$

Proposition 15.4 *Suppose $b > 0$. Let (X_t) be the solution of (15.33) and (T_t) be a BESQ(x_0, δ) where $\delta = \frac{4a}{c^2}, x_0 = \frac{c^2}{b}$. Then $X_t = e^{bt} T_{\varphi(t)}$ in law where*

$$\varphi(t) = \frac{c^2}{4b}(1 - e^{-bt}).$$

Proof. We set $R_t = T_{\varphi(t)}$. We determine the SDE that it fulfills. Let $(Z_t)_{t \geq 0}$ be a classical Wiener process such that

$$T_t = T_0 + 2\int_0^t \sqrt{T_s} dZ_s + \frac{4a}{c^2} t.$$

So, setting $M_t := 2\int_0^t \sqrt{T_s} dZ_s, t \geq 0$, we have

$$\begin{aligned} R_t &= R_0 + \int_0^{\varphi(t)} \frac{4a}{c^2} ds + M_{\varphi(t)} \\ &= R_0 + a\int_0^t e^{-b\tilde{s}} d\tilde{s} + M_{\varphi(t)}. \end{aligned} \tag{15.34}$$

Indeed the change of variable in the second line comes from

$$\tilde{s} = \varphi^{-1}(s), ds = \varphi'(\tilde{s}) d\tilde{s} = \frac{c^2}{4} e^{-b\tilde{s}} d\tilde{s}, \varphi : \mathbb{R}_+ \to [0, \frac{c^2}{4b}[.$$

Since $[M]_t = 4\int_0^t T_s ds$ we have

$$\begin{aligned} [(M)_{\varphi(t)}]_t &= [M]_{\varphi(t)} = 4\int_0^{\varphi(t)} T_s ds = 4\int_0^t T_{\varphi(u)} \varphi'(u) du = 4\int_0^t R_u \varphi'(u) du \\ &= c^2 \int_0^t R_u e^{-bu} du. \end{aligned}$$

$(M_{\varphi(t)})$ is an $(\mathcal{F}_{\varphi(t)})$-local martingale.

By Chapter IX of [31], if (T_t) is a square Bessel process then for every $t > 0$, the law of T_t has a density so $P\{T_t = 0\} = 0$. The same holds for R and so $R_s(\omega) \neq 0$, $dPds$ a.e.

According to Lemma 15.3, there is a classical Brownian motion W with

$$M_{\varphi(t)} = \int_0^t c e^{-\frac{bu}{2}} \sqrt{R_u} dW_u.$$

Consequently by (15.34) we have

$$R_t = R_0 + a\int_0^t e^{-bs} ds + \int_0^t c e^{-\frac{bs}{2}} \sqrt{R_s} dW_s.$$

We evaluate now $Y_t = e^{bt}R_t$ integrating by parts. We get

$$\begin{aligned}
Y_t &= Y_0 + \int_0^t e^{bs} dR_s + b\int_0^t R_s e^{bs} ds \\
&= Y_0 + \int_0^t e^{bs}(ae^{-bs}ds + ce^{-\frac{bs}{2}}\sqrt{R_s}dW_s + bR_s ds) \\
&= Y_0 + at + c\int_0^t \sqrt{Y_s}dW_s + b\int_0^t Y_s ds.
\end{aligned}$$

So Y solves equation (15.33). Since pathwise uniqueness implies uniqueness in law then X and Y have the same law.

■

Exercise 15.5 *Evaluate the Laplace transform of the Cox-Ingersoll-Ross process X_t, i.e.*

$$E(e^{-\lambda X_t}).$$

16 Backward stochastic differential equations

16.1 Preliminaries and motivations.

These equations were motivated by applications to stochastic control. They represent an efficient tool to represent probabilistically a non-linear parabolic or elliptic PDEs. Today, there are several applications to mathematical finance, see [6]. A nice survey is provided by [28] and a recent reference is given in [29]. The theorems we expose here are written with relatively heavy assumptions, so they are not optimal.

Let (W_t) be classical Wiener process equipped with its canonical filtration $(\mathcal{F}_t)$.

We recall the notation $L^2(W)_T$ (see Definition 7.3) indicating the linear space of progressively measurable processes $(X_t)_{t\in[0,T]}$ such that

$$E\left(\int_0^T |X_s|^2 ds\right) = E\left(\int_0^T |X_s|^2 d[W]_s\right) < \infty.$$

Assumption 16.1 *a) We are given a square integrable random variable ξ which plays the role of final condition.*

b) A coefficient $f : \Omega \times [0,T] \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, for which there is a real number K with $K > 0$ with the following features.

(i) $f(\cdot, y, z)$ is progressively measurable for every y, z;

(ii)

$$E\left(\int_0^T f(t,0,0)^2 dt\right) < \infty.$$

(iii) $|f(t,y,z) - f(t,y',z')| \le K(|y-y'| + |z-z'|)$, for every t, y, y', z, z'.

Remark 16.2 *(i) In this first section, Assumption (iii) on f will not operate.*

(ii) The Lipschitz conditions with respect to y given in item (iii) are not necessary. They can be replaced by monotonicity conditions.

(iii) We write everything in the one-dimensional case. The extension to the multi-dimensional case holds with similar arguments.

Definition 16.3 *A pair of progressively measurable real valued processes (Y, Z) indexed on $[0,T]$ will be said* **solution** *to BSDE (ξ, f) if*

a) $E\left(\int_0^T |Z_s|^2 ds\right) < \infty$,

b) $Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s, 0 \le t \le T$ a.s.

Remark 16.4 *(i) We remark that the adaptedness of Y implies in particular that Y_0 is deterministic.*

(ii) Since the right-hand side in item b) is continuous then every solution is a continuous process.

We discuss now some motivations in the introduction to the study of backward SDEs.

Remark 16.5 *(i) $f = 0$. It constitutes the representation theorem of martingales and hedging problem in finance.*

In this case the equation BSDE(ξ, f) becomes

$$\begin{cases} Y_t & = & \xi - \int_t^T Z_s dW_s \\ E(\int_0^T Z_s^2 ds) & < & \infty. \end{cases}$$

We set $Y_t = E(\xi|\mathcal{F}_t)$. By the representation theorem of Brownian martingales, there is $Z \in L^2(\Omega \times [0,T])$, progressively measurable that

$$Y_t = Y_0 + \int_0^t Z_s dW_s, t \in [0,T].$$

So

$$Y_t = Y_T - \int_t^T Z_s dW_s = \xi - \int_t^T Z_s dW_s,$$

which shows that $BSDE(\xi, 0)$ has a solution.

We go on with **uniqueness**.

Let $(Y^1, Z^1), (Y^2, Z^2)$ be two solutions. We set

$$Y = Y^1 - Y^2, Z = Z^1 - Z^2.$$

We have

$$Y_t = -\int_t^T Z_s dW_s, t \in [0,T]. \tag{16.1}$$

Since $E\left(\int_0^T Z_s^2 ds\right) < \infty$ then $\left(\int_0^\cdot Z_s dW_s\right)$ is a (square integrable) martingale. Taking the conditional expectation in (16.1) with respect to $(\mathcal{F}_t)$, we get $Y_t = 0$ and so $Y \equiv 0$.

Then

$$0 = \int_0^t Z_s dW_s, t \geq 0.$$

Taking the quadratic variation we obtain

$$0 = \int_0^t Z_s^2 ds,$$

so $Z_s(\omega) = 0$ $dsdP$ a.e.

(ii) Suppose the existence of $g : \mathbb{R}_+ \times \mathbb{R}^3 \to \mathbb{R}$ such that

$$f(\omega, t, y, z) = g(t, X_t(\omega), y, z),$$

where (X_t) is the solution of a classical SDE

$$X_t = x_0 + \int_0^t a(s, X_s) dW_s + \int_0^t b(s, X_s) ds, \quad E(a,b), \tag{16.2}$$

where a, b are with linear growth. We suppose g to be bounded and continuous. Let $\phi : \mathbb{R} \to \mathbb{R}$ continuous with linear growth such that $\xi = \phi(X_T)$.

Exercise. *Show that ξ is square integrable.*

In this case the system

$$\left\{\begin{array}{lcl} X_t & = & x_0 + \int_0^t a(s, X_s)dW_s + \int_0^t b(s, X_s)ds \\ Y_t & = & \xi + \int_t^T g(s, X_s, Y_s, Z_s)ds - \int_t^T Z_s dW_s \end{array}\right. \qquad \text{FBSDE}(g, \phi, a, b, x_0),$$

is called **forward-backward** *SDE or also* **Markovian** *SDE.*

We suppose the existence of $u \in C^{1,2}([0, T[\times\mathbb{R})$, continuous on $[0, T] \times \mathbb{R}$ solving

$$\text{(PDE)} \quad \left\{\begin{array}{lcl} (\partial_t u & + & \frac{a^2}{2}\partial^2_{xx}u + b\partial_x u)(t, x) \\ & + & g(t, x, u(t, x), (a\partial_x u)(t, x)) = 0, \\ u(T, x) & = & \phi(x). \end{array}\right.$$

We also suppose that $\partial_x u$ is bounded on $[0, T[\times\mathbb{R}$.

Proposition 16.6 *The system FBSDE(g, ϕ, a, b, x_0) admits a solution (Y, Z) where*

$$\left\{\begin{array}{lcl} Y_t & = & u(t, X_t) \\ Z_t & = & \partial_x u(t, X_t)a(t, X_t), \end{array}\right. \tag{16.3}$$

where X is a solution of $E(a, b)$.

Proof. We apply Itô formula to $Y_t = u(t, X_t)$. Let $\tau < T$. We obtain

$$\begin{aligned} Y_\tau &= Y_t + \int_t^\tau \partial_s u(s, X_s)ds + \int_t^\tau \partial_x u(s, X_s)dX_s + \frac{1}{2}\int_t^\tau \partial^2_{xx}u(s, X_s)d[X]_s \\ &= Y_t + \int_t^\tau \partial_x u(s, X_s)a(s, X_s)dW_s \\ &+ \int_t^\tau \left(\partial_s u(s, X_s) + b(s, X_s)\partial_x u(s, X_s) + \frac{1}{2}\partial^2_{xx}u(s, X_s)a^2(s, X_s)\right) ds \\ &= Y_t + \int_t^\tau Z_s dW_s - \int_t^\tau g(s, X_s, Y_s, Z_s)ds, \end{aligned}$$

so

$$Y_t = Y_\tau - \int_t^\tau Z_s dW_s + \int_t^\tau g(s, X_s, Y_s, Z_s)ds.$$

Finally, letting τ converge to T we get

$$Y_t = \xi - \int_t^T Z_s dW_s + \int_t^T g(s, X_s, Y_s, Z_s) ds,$$

which constitutes a probabilistic representation of the non-linear PDE. ■

In conclusion, the system FBSDE(ξ, g, a, b, x_0) constitutes a probabilistic representation of the non-linear PDE.

16.2 Existence and uniqueness of the BSDE

We start with a simple basic lemma often used here.

Lemma 16.7 *Let Y be an adapted measurable process such that $E(\sup_{t\leq T} Y_t^2) < \infty$. Let $Z \in L^2(W)_T$. Then $M_t := \int_0^t Y_s Z_s dW_s, t \geq 0$, is a martingale.*

Proof. Since M is a local martingale vanishing at zero, by Proposition 6.5 it is enough to prove that $E([M,M]_T^{\frac{1}{2}}) < \infty$. Indeed, by Cauchy-Schwarz, we get

$$\begin{aligned} E([M,M]_T^{\frac{1}{2}}) &= E\left(\sqrt{\int_0^T Y_s^2 Z_s^2 ds}\right) \leq E\left(\sqrt{\sup_{t\leq T} Y_t^2 \int_0^T Z_s^2 ds}\right) \\ &\leq \left\{E\left(\sup_{t\leq T} Y_t^2\right)\right\}^{\frac{1}{2}} \left\{E\left(\int_0^T Z_s^2 ds\right)\right\}^{\frac{1}{2}}, \end{aligned}$$

which is finite.

■

From now on, we will make Assumption (iii) (Lipschitz on (y,z)) on f.

Proposition 16.8 *Under the above conditions, if (Y,Z) is a solution of the BSDE(ξ, f), then there exists a constant c, which depends only on T and K, such that*

$$E\left(\sup_{0\leq t\leq T} |Y_t|^2 + \int_0^T |Z_t|^2 dt\right) \leq cE\left(|\xi|^2 + \int_0^T |f(t,0,0)|^2 dt\right).$$

Proof. If (Y,Z) is a solution, then

$$Y_t = Y_0 - \int_0^t f(s,Y_s,Z_s)ds + \int_0^t Z_s dW_s. \qquad (16.4)$$

For each $n \in \mathbb{N}$, we introduce the stopping time

$$\tau_n = \inf\{0 \le t \le T : |Y_t| \ge n\},$$

(with the convention that the infimum over an empty set is infinite) and the stopped process

$$Y_t^n = Y_{t\wedge\tau_n}.$$

Since Y is continuous and therefore almost all paths are bounded on $[0,T]$ we have $\mathcal{T}_n \to +\infty$ when $n \to \infty$. Then $|Y_t^n| \le n \vee |Y_0|$ and

$$Y_t^n = Y_0^n - \int_0^t 1_{[0,\tau_n]}(s) f(s,Y_s^n,Z_s)ds + \int_0^t 1_{[0,\tau_n]}(s) Z_s dW_s.$$

For $n \ge |Y_0|$ can write

$$E(|Y_t^n|^2) \quad \le \quad 3\left(|Y_0|^2 + E\left(\int_0^t |f(s,Y_s^n,Z_s)|ds\right)^2 + \int_0^t E(|Z_s|^2)ds\right), \quad t\in[0,T].$$

Taking into account Assumption 16.1 b) (ii), (iii) there is a constant C depending on K and (which in the sequel can change from one line to the other) such that previous expression is inferior or equal to

$$C\left(|Y_0^2| + \int_0^t E(f(s,0,0)^2 + |Y_s^n|^2 + |Z_s|^2)ds\right).$$

So there exists a quantity $\mathcal{C}$, depending on $K, Y_0, E(\int_0^T f(s,0,0)^2 ds), E(\int_0^T Z_s^2 ds)$ but not on n such that

$$E|Y_t^n|^2 \le \mathcal{C}\left(1 + \int_0^t E(|Y_s^n|^2)ds\right).$$

Gronwall's Lemma 13.10 implies

$$E((Y_t^n)^2) \le \mathcal{C}e^{\mathcal{C}t}, t\in[0,T].$$

On the other hand

$$\lim_{n\to\infty} Y_t^n = Y_t \quad \text{a.s.}$$

Therefore, by Fatou's lemma

$$E(Y_t^2) = E(\liminf_{n\to\infty}(Y_t^n)^2) \leq \liminf_{n\to\infty} E(Y_t^n)^2 \leq \mathcal{C}e^{\mathcal{C}T}, t \in |0,T]. \tag{16.5}$$

From this, from (16.4) and b) (iii) in Assumption 16.1 we have

$$\begin{aligned} E(\sup_{t\leq T}|Y_t|^2) &\leq c\left(Y_0^2 + K^2 E\left(\int_0^T (|Y_s|^2 + |Z_s|^2)ds\right) + E\left(\int_0^T f^2(s,0,0)ds\right)\right) \\ &+ E\left(\sup_{t\leq T}|\int_0^t Z_s dW_s|^2\right), \end{aligned} \tag{16.6}$$

where c is a constant which may change from line to line. Therefore, by Doob's inequality,

$$E(\sup_{t\leq T}|Y_t|^2) \leq c\left(Y_0^2 + E\left(\int_0^T |Y_s|^2 ds\right) + E\left(\int_0^T |Z_s|^2 ds)\right) + E\left(\int_0^T f(s,0,0)^2 ds\right)\right), \tag{16.7}$$

where c only depends on K and T. This quantity is finite. Finally by (16.5)

$$E(\sup_{t\leq T}|Y_t|^2) < \infty. \tag{16.8}$$

We go on with the proof of the proposition. Let (Y,Z) be a solution. It follows that

$$E\left(\int_t^T Y_s Z_s dW_s\right) = 0 \tag{16.9}$$

This will follow from the fact that the local martingale

$$N_t = \int_0^t Y_s Z_s dW_s$$

is a true martingale, because of Lemma 16.7.

We apply Itô formula and (16.4) to get

$$\begin{aligned} Y_T^2 &= Y_t^2 - 2\int_t^T Y_s dY_s + [Y]_T - [Y]_t \\ &= Y_t^2 - 2\int_t^T Y_s f(s,Y_s,Z_s)ds + 2\int_t^T Y_s Z_s dW_s \\ &+ \int_t^T Z_s^2 ds. \end{aligned}$$

Taking the expectation in previous expression and using (16.9) we get

$$
\begin{aligned}
E(\xi^2) &= E(Y_t^2) - 2\int_t^T E(Y_s f(s, Y_s, Z_s))ds \\
&+ E\left(\int_t^T Z_s^2 ds\right).
\end{aligned}
$$

Therefore

$$
E(Y_t^2) + E\left(\int_t^T Z_s^2 ds\right) = E(\xi^2) + 2\int_t^T E(Y_s f(s, Y_s, Z_s))ds. \quad (16.10)
$$

On the other hand, taking into account the initial assumptions on f we get

$$
\begin{aligned}
2E(|Y_s f(s, Y_s, Z_s)|) &\leq 2E\left|Y_s(f(s, Y_s, Z_s) - f(s, 0, 0))\right| + 2E|Y_s f(s, 0, 0)| \\
&\leq 2K\left(E(|Y_s|(|Y_s| + |Z_s|) + 2E(|Y_s||f(s, 0, 0)|)\right)) \\
&\leq c\left(E(|Y_s|^2) + E(|Y_s Z_s|) + E(|Y_s||f(s, 0, 0)|)\right),
\end{aligned}
$$

for a certain constant c only depending on T and K. Using

$$
ab \leq \frac{a^2 + b^2}{2}, \quad a^2 = c|Y_s|^2, \quad b^2 = \frac{|Z_s|^2}{c},
$$

it yields

$$
E(|Y_s Z_s|) \leq \frac{1}{2}E\left(cY_s^2 + \frac{Z_s^2}{c}\right).
$$

So

$$
\begin{aligned}
2E(|Y_s f(s, Y_s, Z_s)|) &\leq c\Bigg(E(|Y_s|^2) + \frac{c}{2}E(|Y_s|^2) + \frac{1}{2c}E(|Z_s|^2) \\
&+ \frac{1}{2}E(f(s, 0, 0))^2 + \frac{1}{2}E(|Y_s|^2)\Bigg) \\
&= C_0\left((E(|Y_s|^2) + E(|f(s, 0, 0)^2))\right) + \frac{1}{2}E(|Z_s|^2),
\end{aligned}
$$

where C_0 is a constant only depending on K, T. Coming back to (16.10), we obtain

$$
\begin{aligned}
E(Y_t^2) &+ E\left(\int_t^T Z_s^2 ds\right) \leq E(\xi^2) + C_0 E\left(\int_t^T Y_s^2 ds + f(s, 0, 0)^2 ds\right) \\
&+ \frac{1}{2}E\left(\int_t^T Z_s^2 ds\right).
\end{aligned}
$$

Therefore, since $E(Y_t^2) < \infty, t \in [0,T]$

$$\begin{aligned} E(Y_t^2) &+ \frac{1}{2}E\left(\int_t^T Z_s^2 ds\right) \leq E(\xi^2) + C_0\left(\int_t^T E(Y_s^2)ds\right) \\ &+ C_0 E\left(\int_0^T f(s,0,0)^2 ds\right). \end{aligned} \tag{16.11}$$

So, by Gronwall's lemma (applied backwardly with $g(t) = E(Y_{T-t}^2)$) there is a constant C_1 such that

$$E(Y_t^2) \leq C_1\left(E(\xi^2) + E\left(\int_0^T f(s,0,0)^2 ds\right)\right), t \in [0,T]. \tag{16.12}$$

Combining (16.11) and (16.12) provides the final statement but with the supremum outside the expectation, i. e.

$$\sup_{0\leq t\leq T} E\left(|Y_t|^2 + \int_0^T |Z_t|^2 dt\right) \leq cE\left(|\xi|^2 + \int_0^T |f(t,0,0)|^2 dt\right). \tag{16.13}$$

The result follows by (16.7) Fubini's and (16.13).

■

We go on preparing the main theorem of existence and uniqueness.

We denote by $\mathcal{B}^2 := L^2(W)_T \times L^2(W)_T$. We introduce a map $\Phi : \mathcal{B}^2 \to \mathcal{C}_{\mathcal{F}}([0,T]) \times L^2(W)_T$: to each couple $(U,V) \in \mathcal{B}^2$ we associate the following couple (Y,Z) of processes. We first define the continuous version of the square integrable martingale

$$M_t := E\left(\xi + \int_0^T f(s,U_s,V_s)ds|\mathcal{F}_t\right), t \in [0,T].$$

The r.v. $\xi + \int_0^T f(s,U_s,V_s)ds$ is square integrable since ξ is and $(U,V) \in \mathcal{B}^2$. Observe in particular

$$\int_0^T f(s,U_s,V_s)^2 ds \leq 2\left(\int_0^T f(s,0,0)^2 ds + 2K^2\int_0^T (U_s^2 + V_s^2)ds\right). \tag{16.14}$$

We recall that by the Brownian martingale representation theorem there is $Z \in L^2(W)_T$ such that

$$M_t = E\left(\xi + \int_0^T f(s, U_s, V_s)ds\right) + \int_0^t Z_r dW_r, t \in [0, T]. \qquad (16.15)$$

We also set

$$Y_t = M_t - \int_0^t f(s, U_s, V_s)ds, t \in [0, T], \qquad (16.16)$$

which is a continuous process. In particular $Y_T = \xi$.

Lemma 16.9 *(i)* Φ *maps a couple of process* $(U, V) \in \mathcal{B}^2$ *into a couple* $(Y, Z) \in \mathcal{B}^2$.

(ii) $E(\sup_{t \le T} Y_t^2) < \infty$ *for every* $(U, V) \in \mathcal{B}^2$.

(iii) $(Y, Z) = \Phi(U, V)$ *if and only if*

$$Y_t = \xi - \int_t^T Z_r dW_r + \int_t^T f(s, U_s, V_s)ds, t \in [0, T]. \qquad (16.17)$$

(iv) (Y, Z) *is a fixed point of* Φ *if and only if* (Y, Z) *is a solution of BSDE*(f, ξ).

Proof. Let $(U, V) \in \mathcal{B}^2$. $Z \in L^2(W)_T$ by construction. About Y, item (i) comes from (ii). Concerning (ii), since M is a square integrable martingale and taking into account Doob's inequality and (16.14) we have

$$E(\sup_{t \le T} Y_t^2) \le 8E(M_T^2) + 2TE\left(\int_0^T f(s, U_s, V_s)^2 ds\right) < \infty.$$

Concerning item (iii), if $(Y, Z) = \Phi(U, V)$ we have

$$\begin{aligned} Y_t &= Y_T - \int_t^T Z_r dW_r + \int_t^T f(r, U_r, V_r)dr \\ &= \xi - \int_t^T Z_r dW_r + \int_t^T f(r, U_r, V_r)dr, t \in [0, T]. \end{aligned}$$

If viceversa (Y, Z) is an element of $\mathcal{B}^2$ fulfilling previous equality then by taking the expectation with respect to $\mathcal{F}_t$ we get

$$\begin{aligned} Y_t &= E\left(\xi + \int_0^T f(r, U_r, V_r)dr | \mathcal{F}_t\right) - \int_0^t f(r, U_r, V_r)dr \\ &= M_t - \int_0^t f(r, U_r, V_r)dr, \end{aligned}$$

where

$$M_t = E\left(\xi + \int_0^T f(r, U_r, V_r)dr|\mathcal{F}_t\right) \text{ a.s.},$$

which says that $\Phi(U,V) = (Y,Z)$. This shows (iii). (iv) is a direct consequence of (iii).

■

Theorem 16.10 *Under Assumption 16.1 BSDE(ξ, f) admits a unique solution (Y,Z) in $\mathcal{B}^2$.*

Proof. According to Lemma 16.9 we need to show the existence of a unique fixed point for the map Φ. Let $\gamma \geq 0$; Let $(U,V) \in \mathcal{B}^2$ and $(Y,Z) = \Phi(U,V)$. Let $(U,V),(U',V') \in \mathcal{B}^2$ and $(Y,Z) = \Phi(U,V),(Y',Z') = \Phi(U',V')$. We set $\bar{U} = U - U', \bar{V} = V - V', \bar{Y} = Y - Y', \bar{Z} = Z - Z'$. We apply Itô's formula from t to T we get

$$\begin{aligned}
-e^{\gamma t}\bar{Y}_t^2 &= 2\int_t^T e^{\gamma s}\bar{Y}_s d\bar{Y}_s + \int_t^T \gamma e^{\gamma s}\bar{Y}_s^2 ds + \int_t^T e^{\gamma s} d[\bar{Y}]_s \\
&= 2\int_t^T e^{\gamma s}\bar{Y}_s\bar{Z}_s dW_s + \int_t^T \bar{Y}_s(f(s,U_s,V_s) - f(s,U_s',V_s'))e^{\gamma s} ds \\
&+ \int_t^T \gamma e^{\gamma s}\bar{Y}_s^2 ds + \int_t^T e^{\gamma s}\bar{Z}_s^2 ds.
\end{aligned}$$

By Lemma 16.9 (ii) we have $E(\sup_{t\leq T}\bar{Y}_t^2) < \infty$. By Lemma 16.7 $(\int_0^t e^{\gamma s}\bar{Y}_s\bar{Z}_s dW_s)$ is a martingale. So taking the expectation in (16.18), taking into account the Lipschitz condition we get

$$\begin{aligned}
e^{\gamma t}E(|\bar{Y}|_t^2) &+ E\left(\int_t^T e^{\gamma s}(\gamma\bar{Y}_s^2 + \bar{Z}_s^2)ds\right) \\
&\leq 2K\int_t^T e^{\gamma s}|\bar{Y}_s|(|\bar{U}_s| + |\bar{V}_s|)ds \\
&\leq 4K^2 E\left(\int_t^T e^{\gamma s}|\bar{Y}_s|^2 ds\right) + \frac{1}{4}E\left(\int_t^T e^{\gamma s}(|\bar{U}_s| + |\bar{V}_s|)^2 ds\right),
\end{aligned}$$

making use of the inequality

$$2ab \leq a^2(4K) + \frac{b^2}{4K}$$

with

$$a = |\bar{Y}_s|, b = |\bar{U}_s| + |\bar{V}_s|.$$

We choose $\gamma = 1 + 4K^2$ hence

$$E\left(\int_0^T e^{\gamma s}(|\bar{Y}_s|^2 + |\bar{Z}_s|^2)ds\right) \leq \frac{1}{2}E\left(\int_0^T e^{\gamma s}(|\bar{U}_s|^2 + |\bar{V}_s|^2)ds\right),$$

from which it follows that Φ is a strict contraction on $\mathcal{B}^2$ equipped with the norm $\|\cdot\|_\gamma$, equivalent to the initial norm, where

$$\|(Y,Z)\|_\gamma^2 = E\left(\int_0^T e^{\gamma s}(|Y_s|^2 + |Z_s|^2)ds\right),$$

for $\gamma = 1+4K^2$. Then, by Banach fixed point theorem, Φ has a unique fixed point and the theorem is proved. ■

We now want to estimate the difference between two solutions in terms of the difference between the data. Given two final conditions $\xi, \xi' \in L^2(\Omega)$ and the two drivers f, f' where f' verifies the Assumption 16.1 and (Y,Z) (resp. (Y',Z')) be a solution of BSDE(f,ξ) (resp. BSDE(f',ξ'). We also suppose that $E(\sup_{t\leq T} Y_t^2) + E(\sup_{t\leq T}(Y')_t^2) < \infty$.

Theorem 16.11 *There exists a constant c which depends on the Lipschitz conditions K' of f' with*

$$E\left(\sup_{t\leq T}|Y_t - Y'_t|^2 + \int_0^T |Z_s - Z'_s|^2)ds\right) \leq cE\left(|\xi - \xi'|^2 + \int_0^T |f(s,Y_s,Z_s) - f'(s,Y_s,Z_s)|^2 ds\right).$$

Proof. We use Itô's formula to develop the increment of $|Y_s - Y'_s|^2$ between $s = t$ and T, yielding

$$\begin{aligned} |Y_t - Y'_t|^2 &+ \int_t^T |Z_s - Z'_s|^2 ds = |\xi - \xi'|^2 \\ &- 2\int_t^T (Y_s - Y'_s)(f(s,Y_s,Z_s) - f'(s,Y'_s,Z'_s))ds \quad (16.18) \\ &- 2(M_T - M_t), \end{aligned}$$

where $M_t = \int_0^t (Y_s - Y_s')(Z_s - Z_s')dW_s$, is a martingale by Lemma 16.7. We remark that

$$\begin{aligned} 2|(Y_s \quad &- \quad Y_s')(f(s,Y_s,Z_s) - f'(s,Y_s',Z_s'))| \qquad (16.19)\\ &\leq \quad 2|Y_s - Y_s'| \left(|(f(s,Y_s,Z_s) - f'(s,Y_s,Z_s)| + K'|Z_s - Z_s'|\right) + 2K'|Y_s - Y_s'|^2 \\ &\leq \quad |Y_s - Y_s'|^2 + |f(s,Y_s,Z_s) - f'(s,Y_s,Z_s)|^2 \\ &+ \quad 2K'^2|Y_s - Y_s'|^2 + \frac{1}{2}|Z_s - Z_s'|^2 + 2K'|Y_s - Y_s'|^2, \end{aligned}$$

where we have used successively the inequalities $2ab \leq a^2 + b^2$ and $2ab \leq 2a^2K' + \frac{b^2}{2K'}$ with

$$a = |Y_s - Y_s'|, b = f(s,Y_s,Z_s) - f'(s,Y_s,Z_s),$$

and

$$a = |Y_s - Y_s'|, b = |Z_s - Z_s'|.$$

Hence taking the expectation (16.18) and taking (16.19) into account we obtain

$$\begin{aligned} &E\left(|Y_t - Y_t'|^2 + \frac{1}{2}\int_t^T |Z_s - Z_s'|^2 ds\right) \leq E(|\xi - \xi'|^2) \qquad (16.20)\\ + \quad &E\left(\int_t^T (f(s,Y_s,Z_s) - f'(s,Y_s,Z_s))^2 ds\right) + (1 + 2K' + 2K'^2)E\left(\int_t^T |Y_s - Y_s'|^2 ds\right). \end{aligned}$$

So, by Gronwall's lemma (applied backwardly with $g(t) = E(|Y - Y'|^2_{T-t})$ there is a constant C_1 such that

$$E(|Y_t - Y_t'|^2) \leq C_1 \left(E(|\xi - \xi'|^2 + \int_0^T |f(s,Y_s,Z_s) - f'(s,Y_s,Z_s)|^2 ds\right), t \in [0,T]. \qquad (16.21)$$

Combining (16.20) and previous equality provides the final statement but with the supremum outside the expectation, i. e.

$$\begin{aligned} &\sup_{0 \leq t \leq T} E\left(|Y_t - Y_t'|^2 + \int_0^T |Z_t - Z_t'|^2 dt\right) \qquad (16.22)\\ \leq \quad &C_2 E\left(|\xi - \xi'|^2 + \int_0^T |f(s,Y_s,Z_s) - f'(s,Y_s,Z_s)|^2 ds\right), t \in [0,T]. \end{aligned}$$

On the other hand

$$Y_t - Y_t' = Y_0 - Y_0' - \int_0^t \ell(s)ds - \int_0^t (f'(s,Y_s,Z_s) - f'(s,Y_s',Z_s'))ds + \int_0^t (Z_s - Z_s')dW_s, \qquad (16.23)$$

where

$$\ell(s) = f(s, Y_s, Z_s) - f'(s, Y_s, Z_s).$$

By Doob's we get

$$\begin{aligned}
E(\sup_{t\le T}|Y_t - Y'_t|^2) &\le 4|Y_0 - Y'_0|^2 + 4T\int_0^T \ell(s)^2 ds + 8K'^2 E\left(\int_0^T (Z_s - Z'_s)^2 ds\right) \\
&+ 8K'^2\int_0^T E(Y_s - Y'_s)^2 ds + 16E\left(\int_0^T (Z_s - Z'_s)^2 ds\right).
\end{aligned}$$

Previous expression together with (16.22) and Fubini's provide the final statement.

■

We continue the same set-up and prove a comparison theorem.

Theorem 16.12 *Let ξ, ξ' be two square integrable r.v. such that $\xi \le \xi'$ a.s. Let f, f' fulfilling Assumption 16.1 and $f(t,y,z) \le f'(t,y,z)$ $dt \otimes dP$ a.e. Let Y (resp. Y') be the solution of BSDE(ξ, f) (resp. BSDE(ξ', f'). We have the following.*

(i) *$Y_s \le Y'_s$ a.s. for all $s \in [0,T]$.*

(ii) *If moreover $Y_0 = Y'_0$ then $Y_t = Y'_t$ a.s. for all $t \in [0,T]$.*

(iii) *Whenever $P\{\xi < \xi'\} > 0$ or $f(t, Y'_t, Z'_t) < f'(t, Y'_t, Z'_t)$ on a set of positive $dt \otimes dP$ measure, then $Y_0 < Y'_0$.*

Proof. We define

$$\alpha_t = \begin{cases} \frac{f(t,Y'_t,Z'_t) - f(t,Y_t,Z'_t)}{Y'_t - Y_t}, & \text{if} \quad Y_t \ne Y'_t \\ 0 & \text{if} \quad Y_t = Y'_t. \end{cases}$$

We also define the process $(\beta_t)_{0\le t\le T}$ as follows. We denote by

$$\beta_t = \begin{cases} \frac{f(t,Y_t,Z'_t) - f(t,Y_t,Z_t)}{Z'_t - Z_t}, & \text{if} \quad Z_t \ne Z'_t \\ 0 & \text{if} \quad Z_t = Z'_t. \end{cases}$$

We note that $(\alpha_t)_{0\le t\le T}$ and $(\beta_t)_{0\le t\le T}$ are measurable adapted processes, $|\alpha| \le K$, $|\beta| \le K$.

For $0 \le s \le t \le T$, let

$$\Gamma_{s,t} := \exp\left(\int_s^t \left(\alpha_r - \frac{|\beta_r|^2}{2}\right) dr + \int_s^t \beta_r dW_r\right).$$

Define

$$(\bar{Y}_t, \bar{Z}_t) = (Y'_t - Y_t, Z'_t - Z_t), \bar{\xi} = \xi' - \xi, U_t = f'(t, Y'_t, Z'_t) - f(t, Y'_t, Z'_t).$$

Then $(\bar{Y}, \bar{Z})$ solves the linear BSDE

$$\bar{Y}_t = \xi + \int_t^T (\alpha_r \bar{Y}_r + \beta_r \bar{Z}_r) dr + \int_t^T U_r dr - \int_t^T \bar{Z}_r dW_r. \tag{16.24}$$

We state a lemma.

Lemma 16.13 *For $0 \le s < t \le T$ we have*

$$\begin{aligned} \bar{Y}_s &= \Gamma_{s,t}\bar{Y}_t + \int_s^t \Gamma_{s,r} U_r dr - \int_s^t \Gamma_{s,r}(\bar{Z}_r + \bar{Y}_r \beta_r) dW_r \\ \bar{Y}_s &= E\left(\Gamma_{s,t}\bar{Y}_t + \int_s^t \Gamma_{s,r} U_r dr | \mathcal{F}_s\right). \end{aligned}$$

Proof (of Lemma 16.13).
By an easy application of Itô formula, we can show

$$\Gamma_{s,t} = \int_s^t \Gamma_{s,r}(\beta_r dW_r + \alpha_r dr). \tag{16.25}$$

(16.24) gives

$$\bar{Y}_t = \bar{Y}_s - \int_s^t (\alpha_r \bar{Y}_r + \beta_r \bar{Z}_r) dr - \int_s^t U_r dr + \int_s^t \bar{Z}_r dW_r. \tag{16.26}$$

Then we apply integration by parts to $\Gamma_{s,t}\bar{Y}_t$ using (16.26) and (16.25).

Consequently

$$
\begin{aligned}
\Gamma_{s,t}\bar{Y}_t &= \underbrace{\Gamma_{s,s}}_{=1}\bar{Y}_s + \int_s^t \Gamma_{s,r} d\bar{Y}_r \\
&+ \int_s^t \bar{Y}_r d\bar{\Gamma}_{s,r} + [\bar{Y}, \Gamma_{s,\cdot}]_t \\
&= \bar{Y}_s + \int_s^t \Gamma_{s,r}(-\alpha_r \bar{Y}_r - \beta_r \bar{Z}_r) dr \\
&- \int_s^t \Gamma_{s,r} U_r dr + \int_s^t \Gamma_{s,r} \bar{Z}_r dW_r \\
&+ \int_s^t \bar{Y}_r \Gamma_{s,r}(\beta_r dW_r + \alpha_r dr) \\
&+ \int_s^t \Gamma_{s,r}\beta_r \bar{Z}_r dr \\
&= \bar{Y}_s - \int_s^t \Gamma_{s,r} U_r dr + \int_s^t \Gamma_{s,r}(\bar{Z}_r + \beta_r \bar{Y}_r) dW_r,
\end{aligned}
$$

which implies the first equality. The second equality of the statement follows taking the conditional expectation of the first expression with respect to $\mathcal{F}_s$. Indeed the local martingale

$$
M_t := \int_s^t \Gamma_{s,r}(\bar{Z}_r + \bar{Y}_r \beta_r) dW_r,
$$

is a martingale by Lemma 16.7. Indeed first β is bounded and so $E\left(\int_0^T (\bar{Z}_r + \bar{Y}_r\beta_r)^2 dr\right) < \infty$. Moreover

$$
\sup_{s \le t \le T} \Gamma_{s,t} \le \exp(T\|\alpha\|_\infty)\tilde{\Gamma}_{s,t}, \tag{16.27}
$$

where

$$
\tilde{\Gamma}_{s,t} = \exp\left(\int_s^t -\frac{|\beta_r|^2}{2} dr + \int_s^t \beta_r dW_r\right), s \le t \le T,
$$

which is a martingale because of the Novikov criterion. Indeed

$$
E\left(\exp\left(\int_0^T \frac{|\beta_r|^2}{2} dr\right)\right) < \infty,
$$

holds because β is bounded. Now $(\tilde{\Gamma}_{s,t})_t$ is also square integrable. Indeed

$$
\tilde{\Gamma}^2_{s,t} = \exp\left(\int_s^t \beta_r^2 dr\right) L_{s,t},
$$

with

$$L_{s,t} = \exp\left(\int_s^t -\frac{|2\beta_r|^2}{2}dr + \int_s^t 2\beta_r dW_r\right),$$

which is a martingale again by Novikov criterion. So $(\tilde{\Gamma}_{s,t})$ is a square integrable martingale and

$$E(\sup_{s\le t\le T} \Gamma_{s,t}^2) \le \exp(2T\|\alpha\|_\infty)E(\sup_{s\le t\le T} \tilde{\Gamma}_{s,t}^2),$$

which is finite by Doob inequality. ■

Proof of Theorem 16.12.

ξ and U are by assumption non-negative. Item (i) of Theorem 16.12 follows setting $t = T$ in the second equality in Lemma 16.13 statement. Item (ii) follows setting $s = 0$ in the same mentioned equality and by the strict positivity of $\Gamma_{s,t}$. Item (iii) is a consequence of the strict positivity of $\Gamma_{s,t}$ and Lemma 16.13 for $t = T$ and $s = 0$. ■

The theory of forward-backward SDEs is an important chapter of stochastic analysis connecting deterministic PDEs and world of stochastic processes. In the introduction of the topic we have shown that classical solutions of parabolic PDEs produce solutions of a FBSDE. Previous comparison theorem is useful to show that whenever a forward-backward SDE is well-posed for every initial time and initial conditions, those produce solutions of PDEs in generalized sense, called *viscosity solutions*.

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